

UMC 0.11um LOGIC/MIXED_MODE AI Advanced Enhancement Standard Performance 1.2V MOSFET SPICE Model Document

**Version 1.0 Phase 1
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1 Introduction

1.1 Revision history

Version	Date	Owner	Comment
0.2_p1	2010/06/30	Kelly Pao	1. Original. 2. Copy from G-05-LOGIC/MIXED_MODE11- MMC/L110E-SPICE and revise the Idlin corner.
0.2_p2	2010/08/11	Shih-Hsin Yeh	1. Copy Igate model from L110E model.
0.3_p1	2010/10/22	Kelly Pao	1. Removed the m in igate area formula. (area='w*0.9*1*0.9*m*1e4*tcig' -- > area='w*0.9*1*0.9*1e4*tcig')
0.4_p1	2010/10/28	Yealing Lin	1. Add Twell NMOS model (copy from Twin well NMOS). Modify Igate model: (a) m=m to m=1. (b) area='w*0.9*1*0.9*1e4*tcig'to area='w*0.9*(1*0.9+1.2e- 8)*1e4*tcig'. (c) 'rsh*nrd/m' to 'rsh*nrd'. (d) 'rsh*nrs/m' to 'rsh*nrd'.
0.5_p1	2010/12/24	Shih-Hsin Yeh	1. Recenter the model to align EDR target. 2. Revise Cgc parameters.
0.6_p1	2011/06/07	Shih-Hsin Yeh	1. Revise Cgc parameters.
1.0_p1	2012/09/14	Laney Fan	1. Version up to 1.0 based on 6 pcs Si data.

1.2 General information

This document serves as a reference to the design works of products that will be manufactured by UMC using the following process.

UMC 0.11um LOGIC/MIXED_MODE AI Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-Logic/Mixed_Mode11-1P8M-MMC/FSG/L110E-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

1.2V n-ch MOSFET

1.2V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.24

For detailed descriptions of this compact model equations and parameters, please refer to this material.

BSIM3V3.24 MOSFET Model User Manual

In BSIM3V3.24, there are two types of models that could be used. One is binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is global model, which means one scalable parameter set is provided and it could cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for its many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the hardware data, which can be found in the following sections.

1.3 Model usage

The compact model is provided in various formats. It's guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2005.3
Cadence Spectre version 5.10.41
Mentor Graphics Eldo version 6.6_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNFP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
- Add the following statement in the input file to the simulator.
.lib "/ModelFileName.lib" CaseName
, where CaseName could be TT, FF, SS, FNFP, or SNFP.

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
SetOption1 options scale=0.9
- Add the following statement in the input file to the simulator.
include "/ModelFileName.lib.scs" section=CaseName
, where CaseName could be TT, FF, SS, FNFP, or SNFP.

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

.Option scale=0.9

- Add the following statement in the input file to the simulator.

.lib "/ModelFileName.lib.eldo" CaseName

, where CaseName could be TT, FF, SS, FNFP, or SNFP.

For 1.2V n-ch MOSFET: n_12_spl110e_ig

For 1.2V p-ch MOSFET: p_12_spl110e_ig

For 1.2V n-ch MOSFET: n_12bpw_spl110e_ig (for Twell model)

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range(after 90% shrink):

For 1.2V n-ch MOSFET:

$$0.108 \mu\text{m} \leq L_{\text{DES}} \leq 45 \mu\text{m}$$

$$0.144 \mu\text{m} \leq W_{\text{DES}} \leq 90 \mu\text{m}$$

For 1.2V p-ch MOSFET:

$$0.108 \mu\text{m} \leq L_{\text{DES}} \leq 45 \mu\text{m}$$

$$0.144 \mu\text{m} \leq W_{\text{DES}} \leq 90 \mu\text{m}$$

Adopt multi-finger structures if $W_{\text{DES}} > 9\text{mm}$.

$$0.306 \mu\text{m} \leq S_A, S_B \leq 2.2 \mu\text{m}, S_D > 0.36 \mu\text{m}$$

Note: $S_A, S_B = 2.2 \mu\text{m}$ is the reference dimension of the STI effect.

Voltage range:

For 1.2V n-ch MOSFET:

$$0\text{V} \leq V_{\text{GS}} \leq 1.2\text{V} (*1.1)$$

$$0\text{V} \leq V_{\text{DS}} \leq 1.2\text{V} (*1.1)$$

$$-1.2\text{V} \leq V_{\text{BS}} \leq 0\text{V}$$

For 1.2V p-ch MOSFET:

$$0\text{V} \geq V_{\text{GS}} \geq -1.2\text{V} (*1.1)$$

$$0\text{V} \geq V_{\text{DS}} \geq -1.2\text{V} (*1.1)$$

$$1.2\text{V} \geq V_{\text{BS}} \geq 0\text{V}$$

Temperature range:

For both 1.2V n-ch and p-ch MOSFET:

$$-50^\circ\text{C} \sim +125^\circ\text{C}$$

UMC L110E is UMC L130E 90% direct shrunk option.The following table give users (including designers) a dimension correlation between Drawing vs. SPICE/Resistor dimensions.

Feature\ GDSII	Drawing in L130E GDSII database (unit : um)	L110 GDSII =0.9*(L130E GDSII) Dimension (unit : um)	L110 SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	(L sizing back 12 nm) 9/0.12
Salicide poly resistor	(Width/Length)	0.9*(Width/Length)	
3.3V I/O (W/L)	10/0.34	9/0.306	(L sizing back 12 nm) 9/0.318
contact size	0.16*0.16	0.144*0.144	
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	
Metal size	Width/Space	0.9*(Width/Space)	
Sloting inserting	metal line width>5.556um	mwatal line width>5um	
N-well,salicide and non-salicide resistor	(Width/Length)	0.9*(Width/Length)	

Note:

- 1.Because process bias make drawn and physical dimension are different , please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.
- 2.Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

2 1.2 V n-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	D1A0BBF3	D1A0BBF3	D1A0BBF3	D1A0BBF3
Lot ID	QGSML	QGSML	QGSML	QGSML
Wafer Number	12	12	12	12

2.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300\text{nA} * (W_{DES} * 0.9) / (L_{DES} * 0.9 + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

Table 2-2 Electrical Parameters for NMOS

All parameters are given for T=25 °C ,V_{BS}=0 V unless specified otherwise.

Target & Centered Value are given for SA=2.2u SB=2.2u

Measured & Extracted Value are given for SA=2.2u SB=2.2u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
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Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.319	0.318	0.324	0.324
I _{ON}	V _{DS} = 1.2 V	μA/μm	11.16	11.27	11.43	11.16

Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.438	0.428	0.434	0.438
V _{TSAT}	V _{DS} = 1.2 V	V	0.350	0.347	0.346	0.350
Body Effect V _T shift	V _{DS} = 1.2 V V _{BS} = 0 ~ -1.2 V	V	--	0.093	0.110	0.110
DIBL V _T shift	V _{DS} = 0.05~1.2 V	V	0.080	0.081	0.088	0.088
I _{ON}	V _{DS} =V _{GS} = 1.2 V	μA/μm	550	548.33	552.57	550.00
I _{OFF}	V _{DS} = 1.2 V, V _{GS} =0 V	A/μm	2E-10	1.6E-10	1.7E-10	1.6E-10

Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.290	0.287	0.302	0.302
V _{TSAT}	V _{DS} = 1.2 V	V	--	0.276	0.280	0.280
I _{ON}	V _{DS} =V _{GS} = 1.2 V	μA/μm	11.44	11.50	11.65	11.35

Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = 0.05 V	V	0.373	0.391	0.390	0.394
V _{TSAT}	V _{DS} = 1.2 V	V	0.317	0.327	0.335	0.339
I _{ON}	V _{DS} =V _{GS} = 1.2 V	μA/μm	778.57	765.64	784.07	777.86
I _{OFF}	V _{DS} = 1.2 V, V _{GS} =0 V	A/μm	--	2.4E-10	1.9E-10	3.8E-10

Capacitance

CJ	V _J = 0 V	F/m ²	7.13e-4	7.15E-04	7.20E-04	7.20E-04
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CJSW	$V_J = 0 \text{ V}$	F/m	1.01E-10	9.90E-11	9.94E-11	9.94E-11
CJSWG	$V_J = 0 \text{ V}$	F/m	3.47E-10	3.15E-10	3.76E-10	3.76E-10
C _{OVL}	$V_{DS} = V_{GS} = -0.5 \text{ V}$	F/m	2.68E-10	2.44E-10	2.93E-10	2.93E-10

2.3 MOSFET DC model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

WL	10	1	0.5	0.24	0.18	0.15	0.12
10	X	X	X	X	X	X	X
1	X			X			X
0.8				X			
0.65				X			X
0.43	X		X	X	X	X	X
0.25				X			X
0.16	X			X			
0.14	X		X	X	X	X	X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	0.05	1.2	1.15
V_{GS} (V)	-0.5	1.35	0.025
V_{BS} (V)	0	-1.2	-0.3

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	1.35	0.025
V_{GS} (V)	0.6	1.2	0.15
V_{BS} (V)	0	-1.2	-1.2

2.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension ($L_{DES} * 0.9 + 12nm$) and ($W_{DES} * 0.9$) respectively. (All the device dimensions in these plots are after 90% shrink and 12nm sizing channel length back.)

I_D vs. V_{DS} Plots

The following plots show measured and simulated output I-V curves for selected devices at 25 °C.

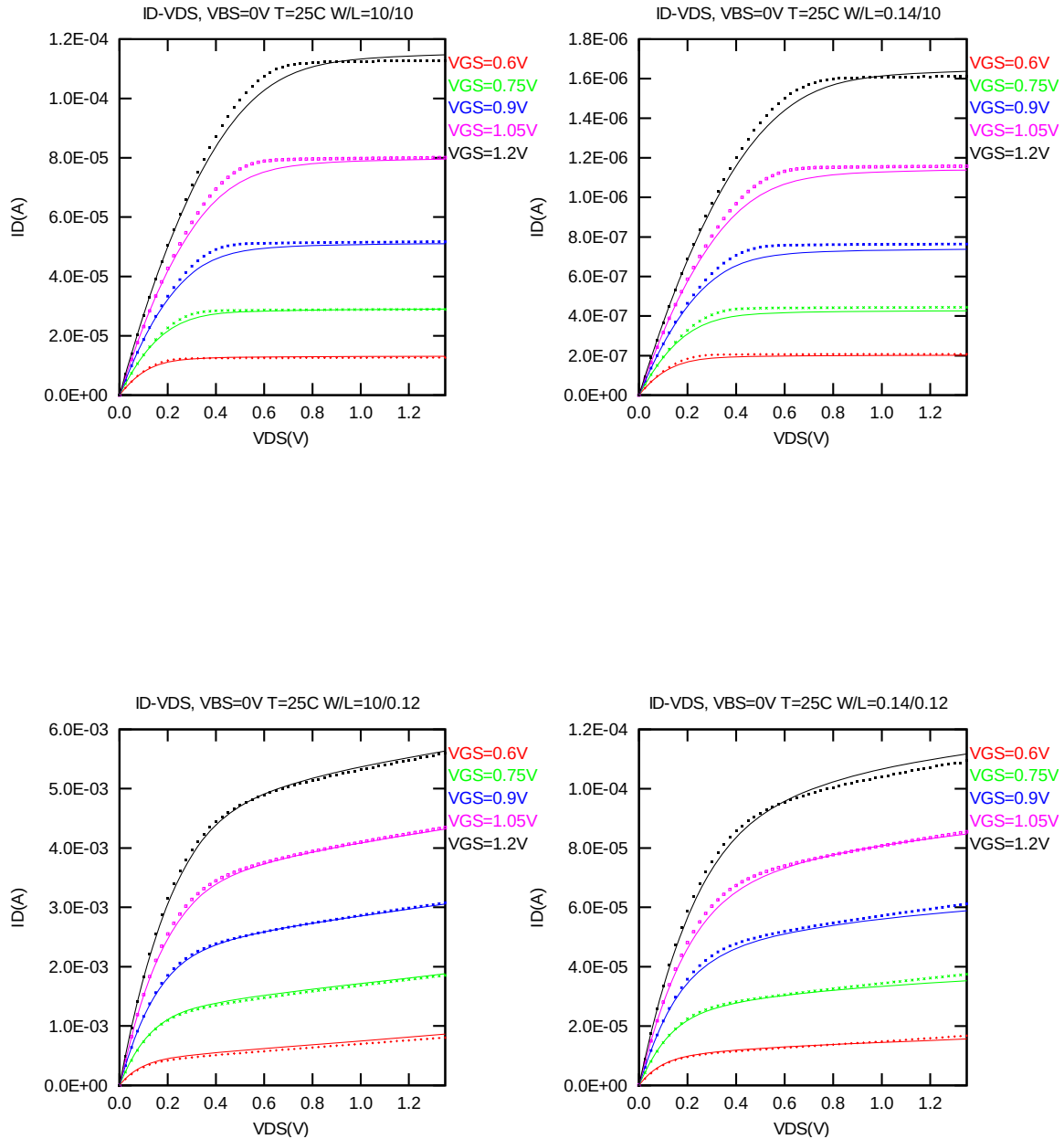


Figure 2-1 I_D vs. V_{DS} at $V_{BS} = 0V$ for NMOS

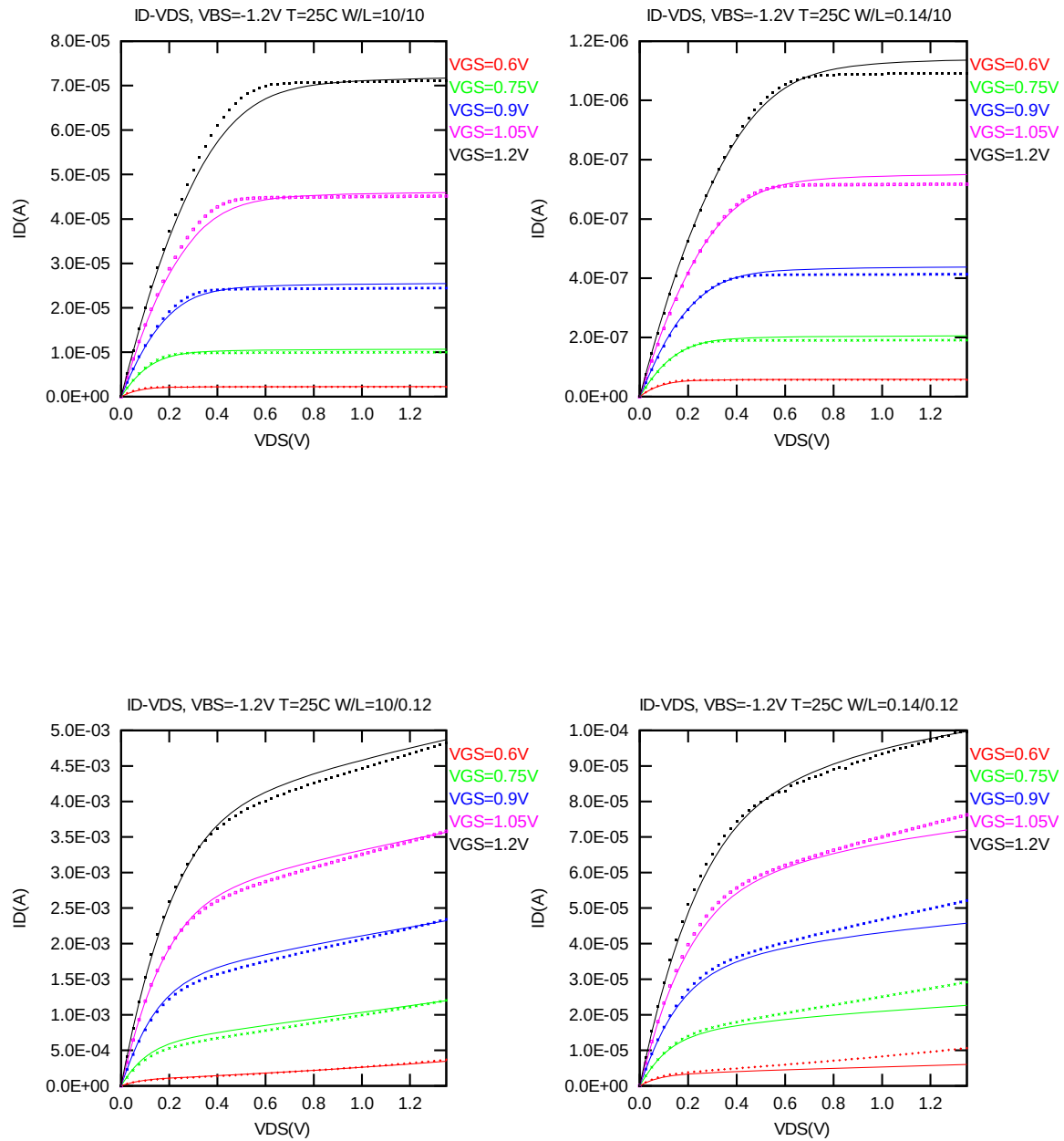


Figure 2-2 I_D vs. V_{DS} at $V_{BS} = -1.2$ V for NMOS

G_{DS} vs. V_{DS} Plots

The following plots show measured and simulated output conductance curves for selected devices at 25 °C.

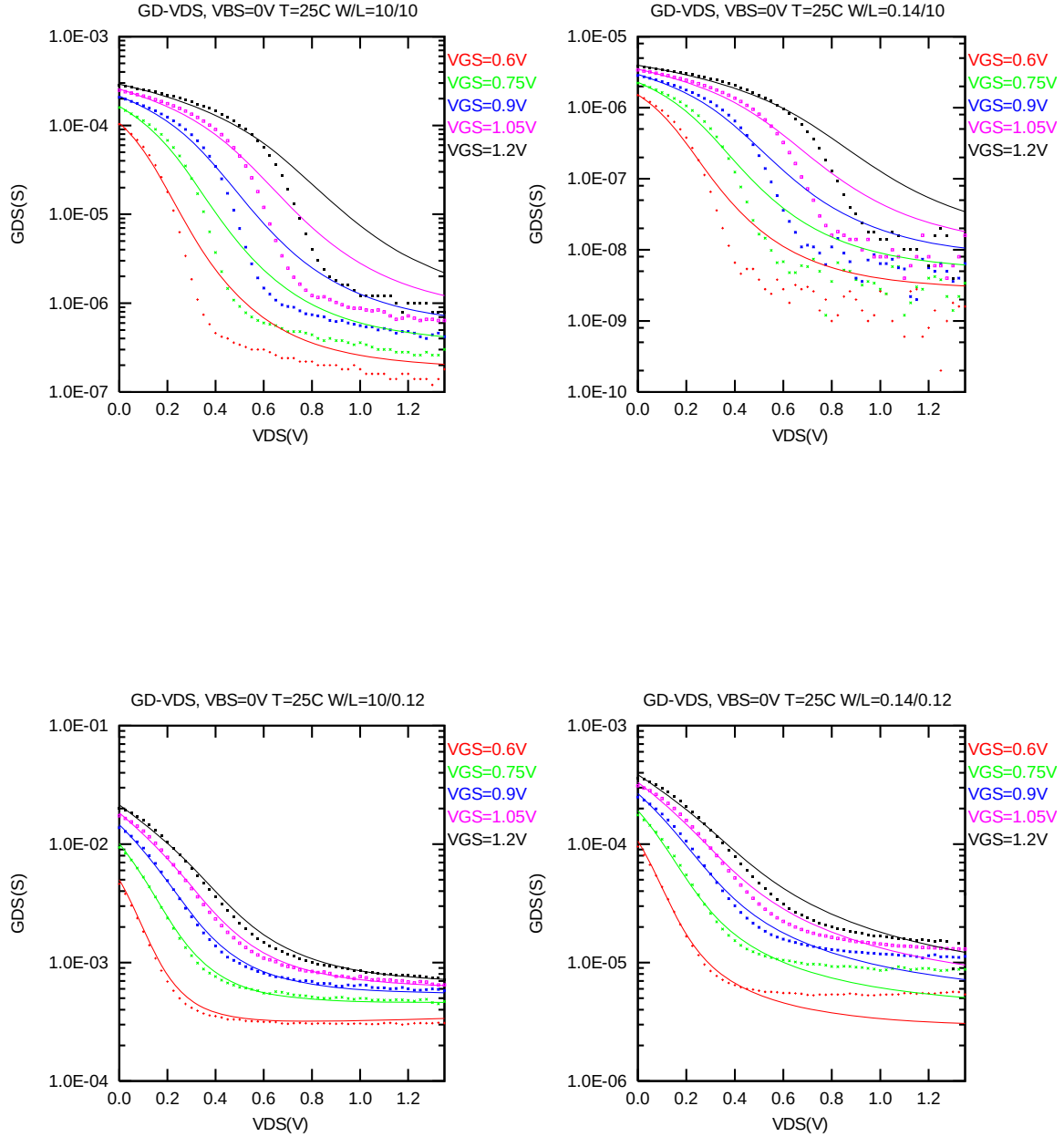


Figure 2-3 G_{DS} vs. V_{DS} at $V_{BS} = 0V$ for NMOS

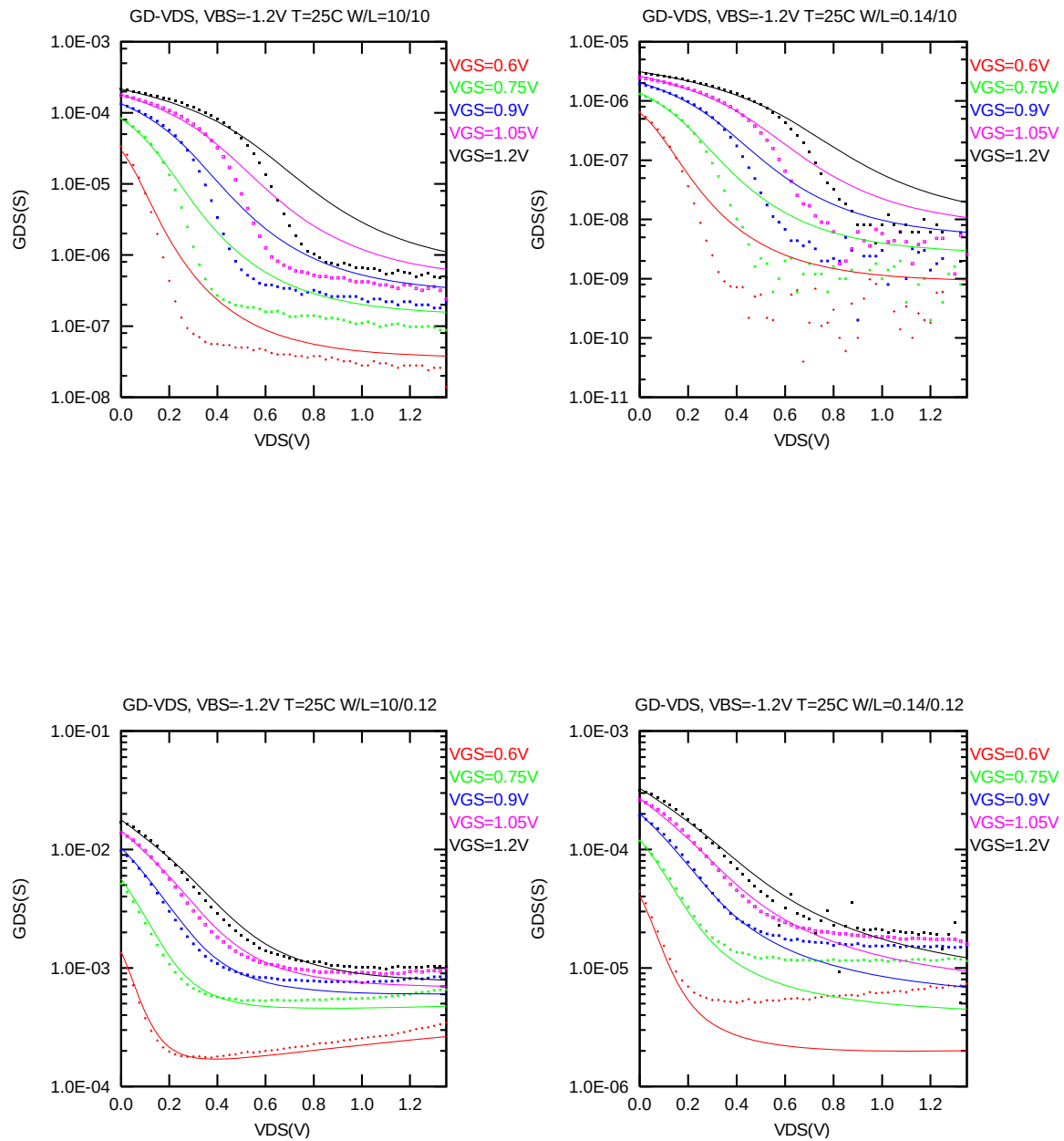


Figure 2-4 G_{DS} vs. V_{DS} at $V_{BS} = -1.2V$ for NMOS

I_D vs. V_{GS} Plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 °C.

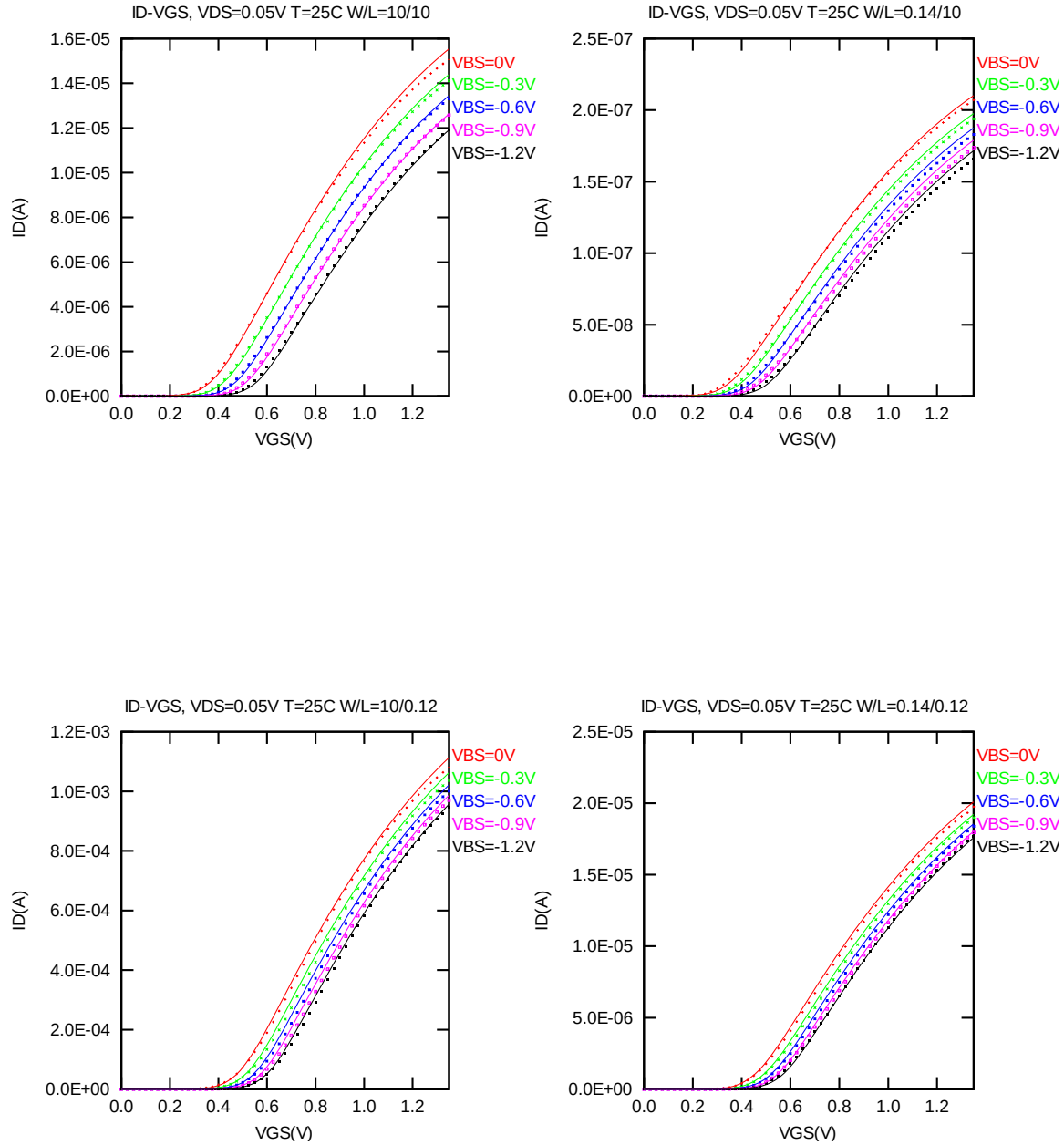


Figure 2-5 I_D vs. V_{GS} at $V_{DS} = 0.05 V$ for NMOS

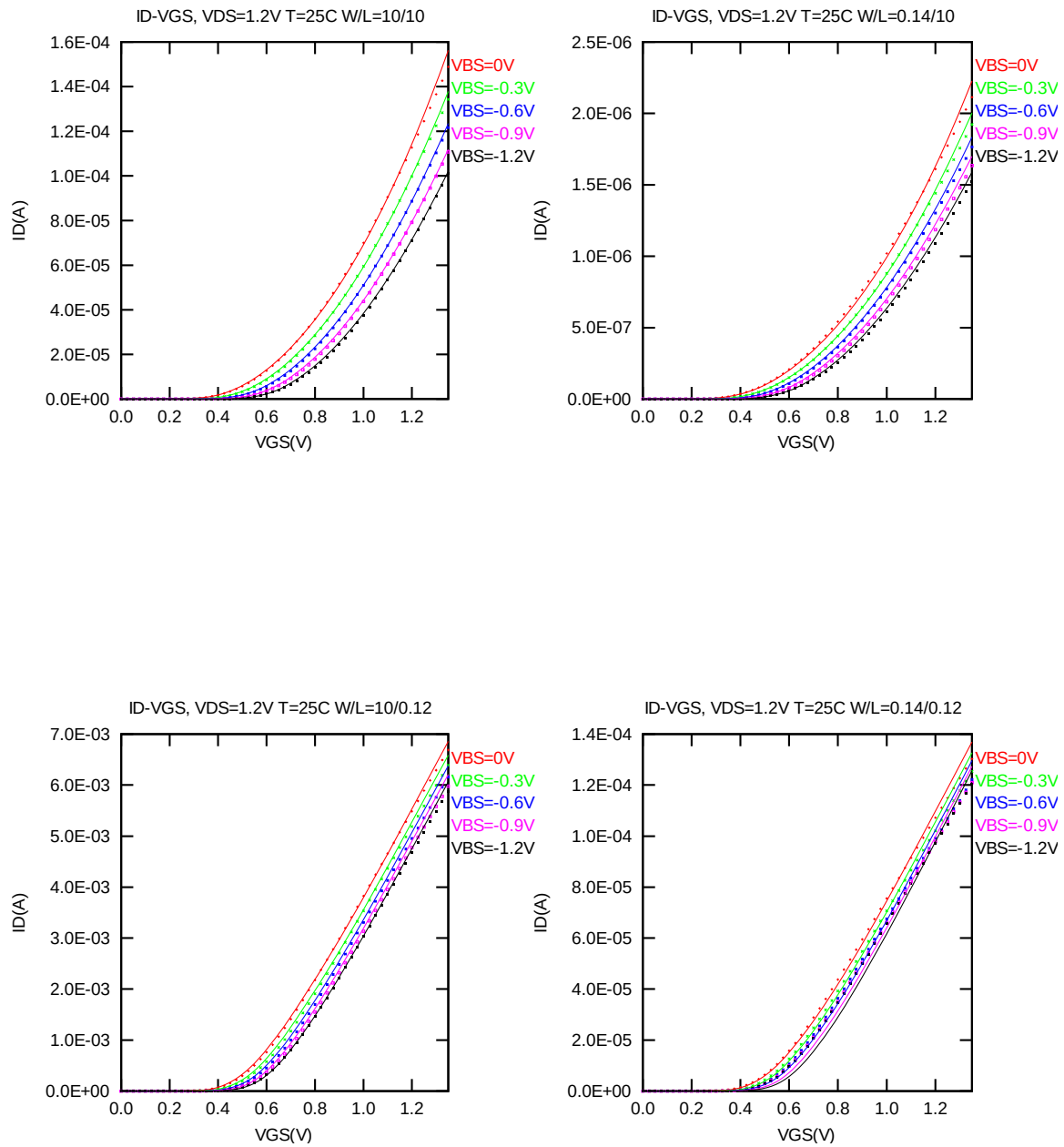


Figure 2-6 I_D vs. V_{GS} at $V_{DS} = 1.2V$ for NMOS

I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 °C.

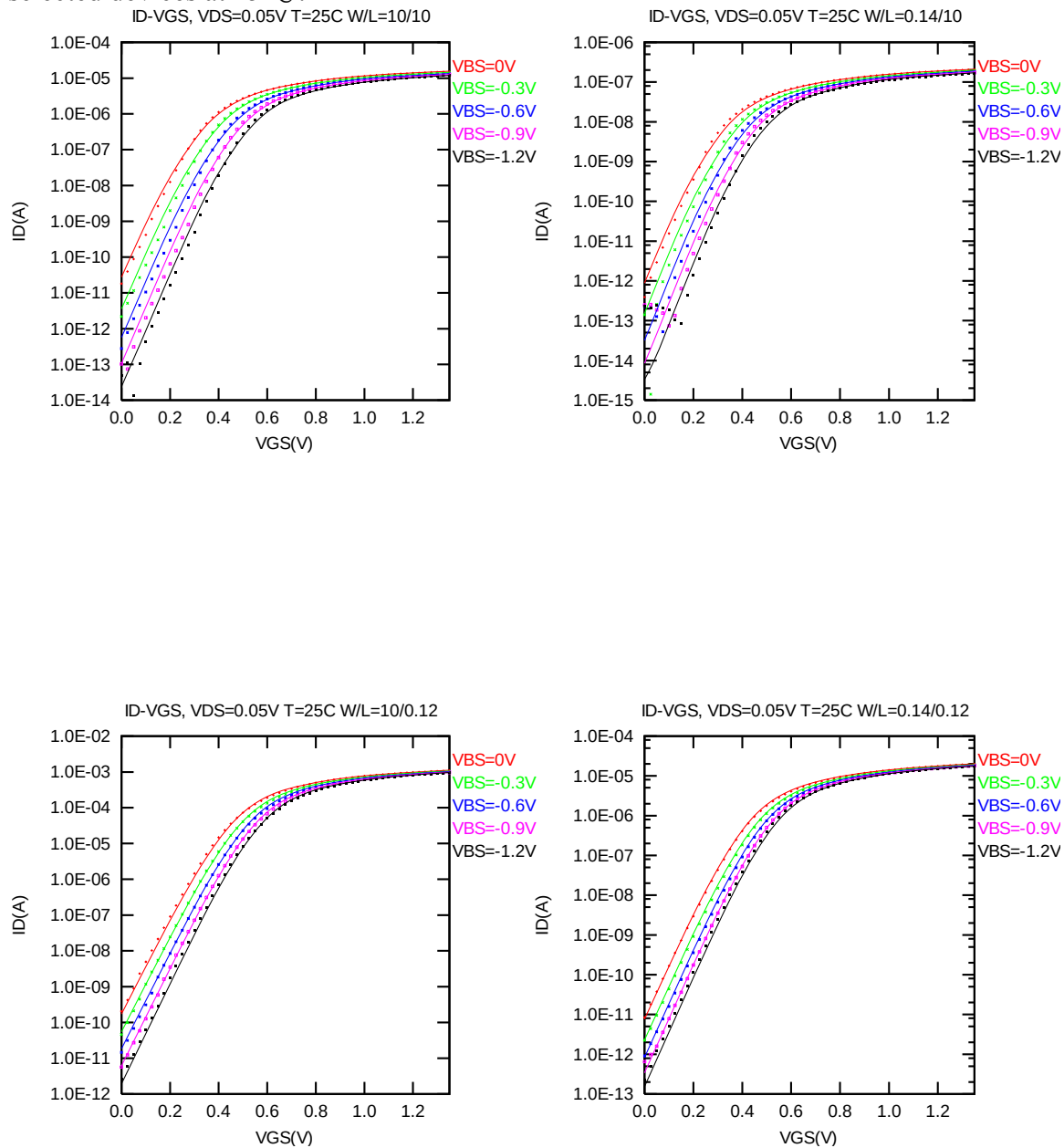


Figure 2-7 Log scaled I_D vs. V_{GS} at $V_{DS} = 0.05 V$ for NMOS

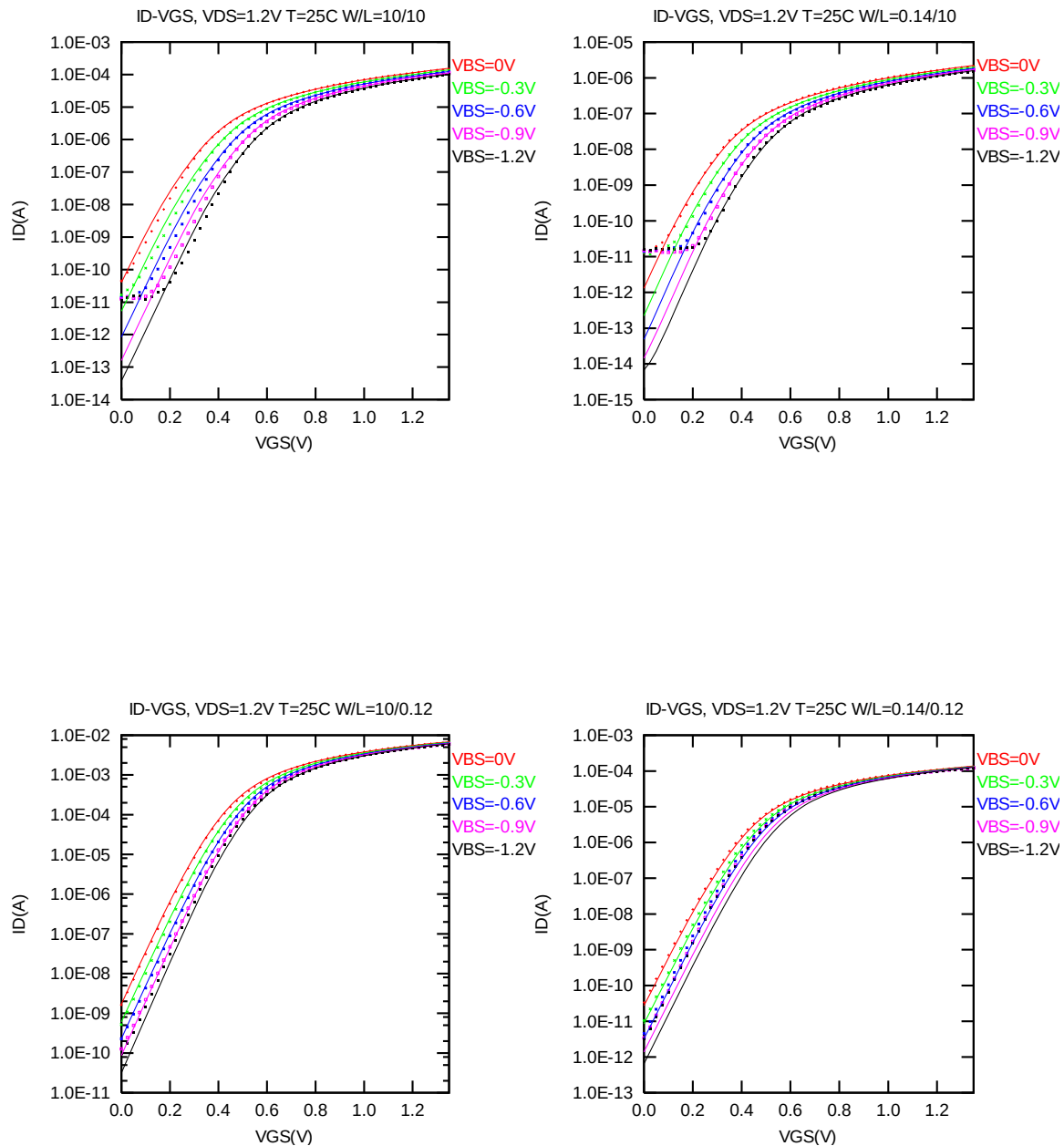


Figure 2-8 Log scaled I_D vs. V_{GS} at $V_{DS} = 1.2V$ for NMOS

G_M vs. V_{GS} Plots

The following plots show measured and simulated transfer conductance curves in linear scale for selected devices at 25 °C .

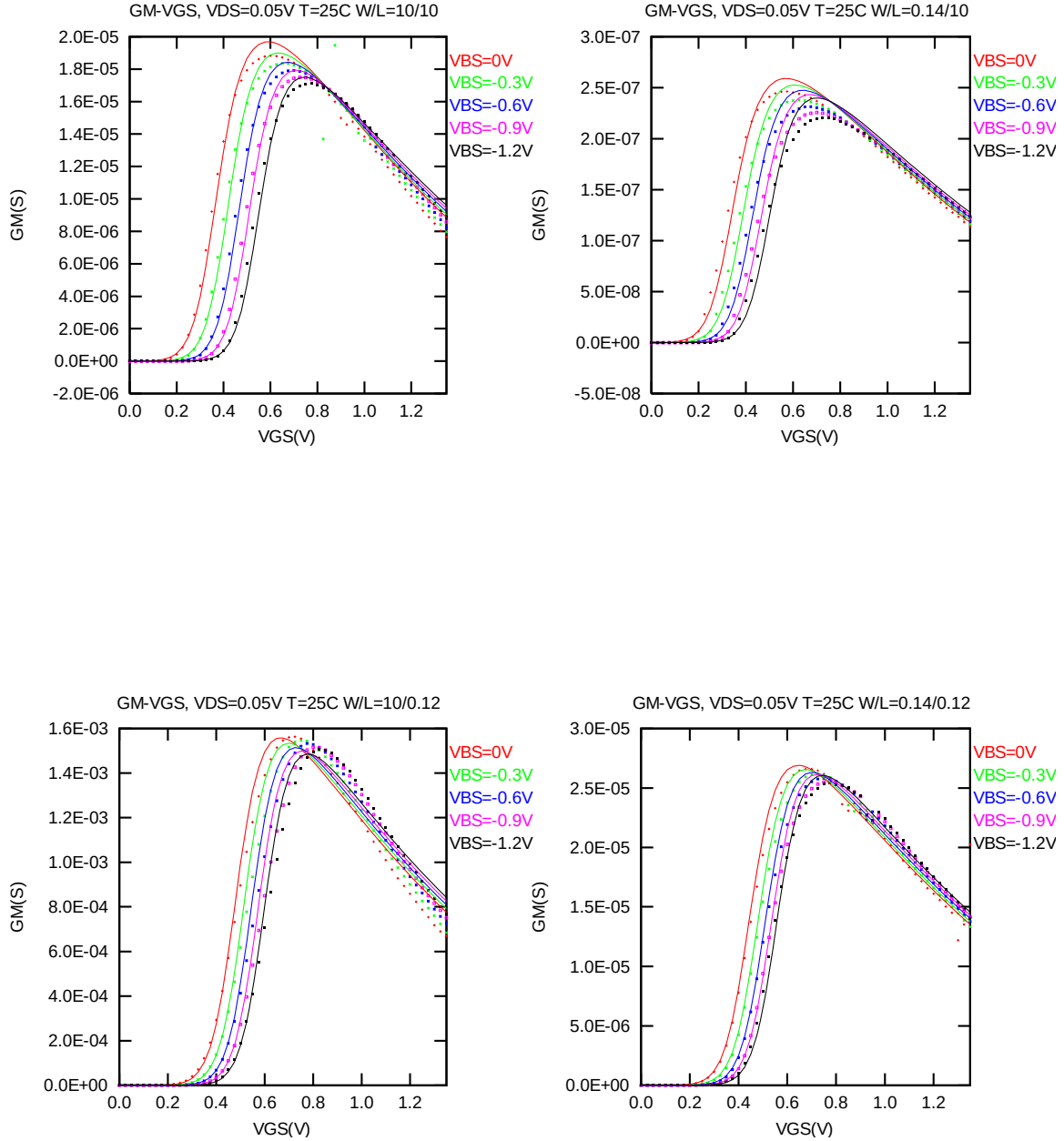


Figure 2-9 G_M vs. V_{GS} at $V_{DS} = 0.05$ V for NMOS

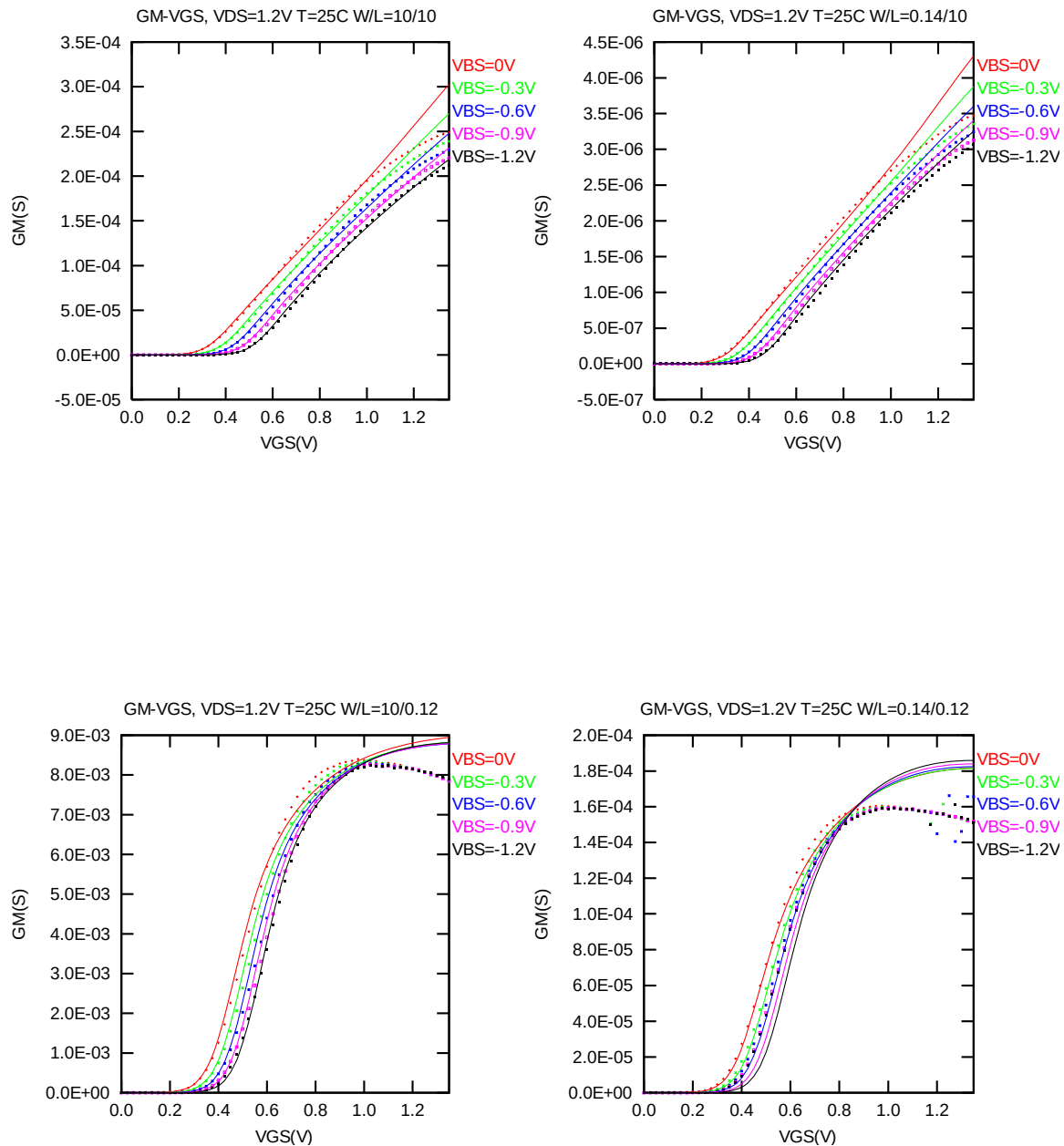


Figure 2-10 G_M vs. V_{GS} at $V_{DS} = 1.2V$ for NMOS

V_T vs. L_{DES} & V_T vs. W_{DES}

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012 \mu m)$ at $25^\circ C$.

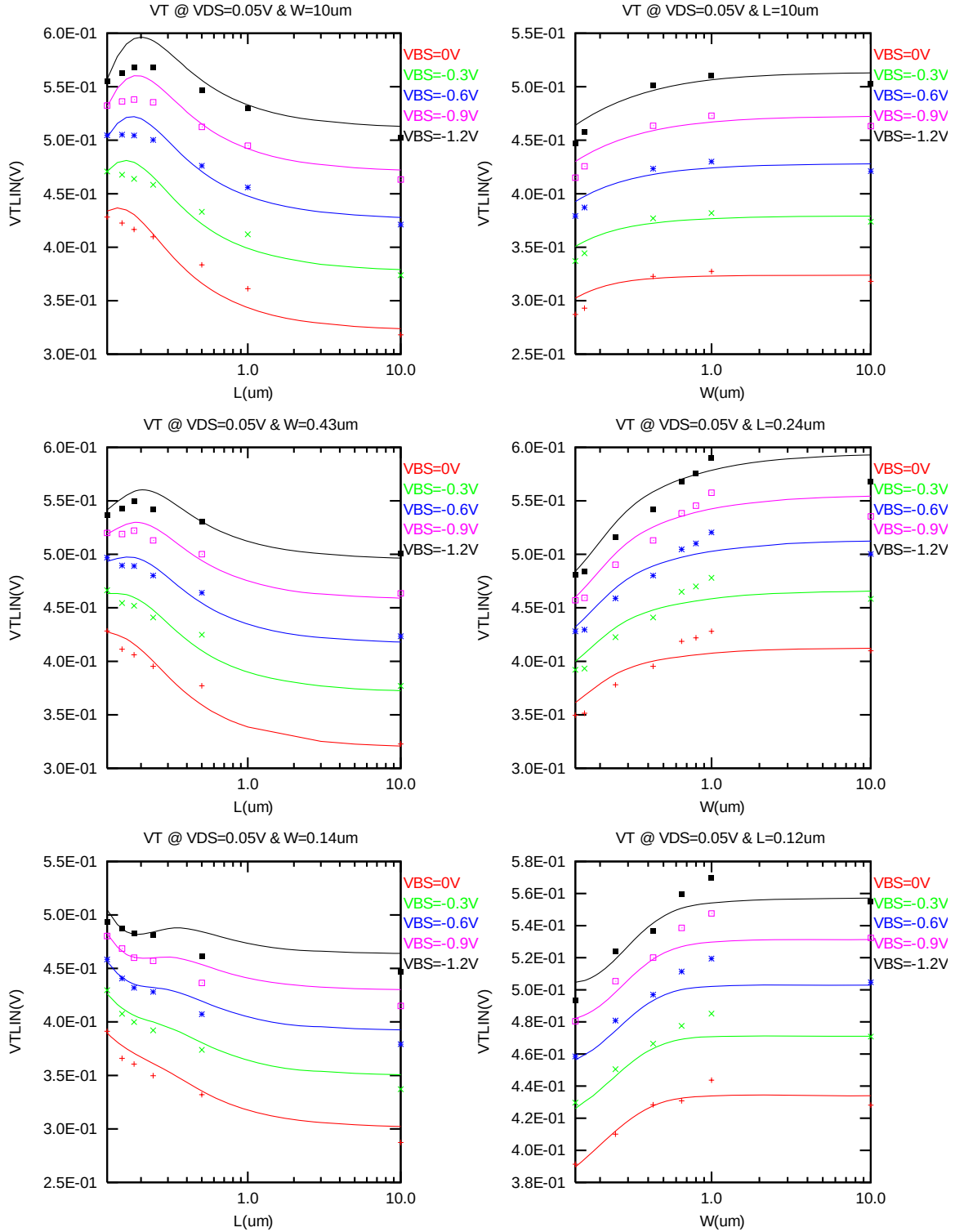


Figure 2-11 V_T vs. channel length & channel width for low drain bias ($V_{DS} = 0.05 V$) for NMOS

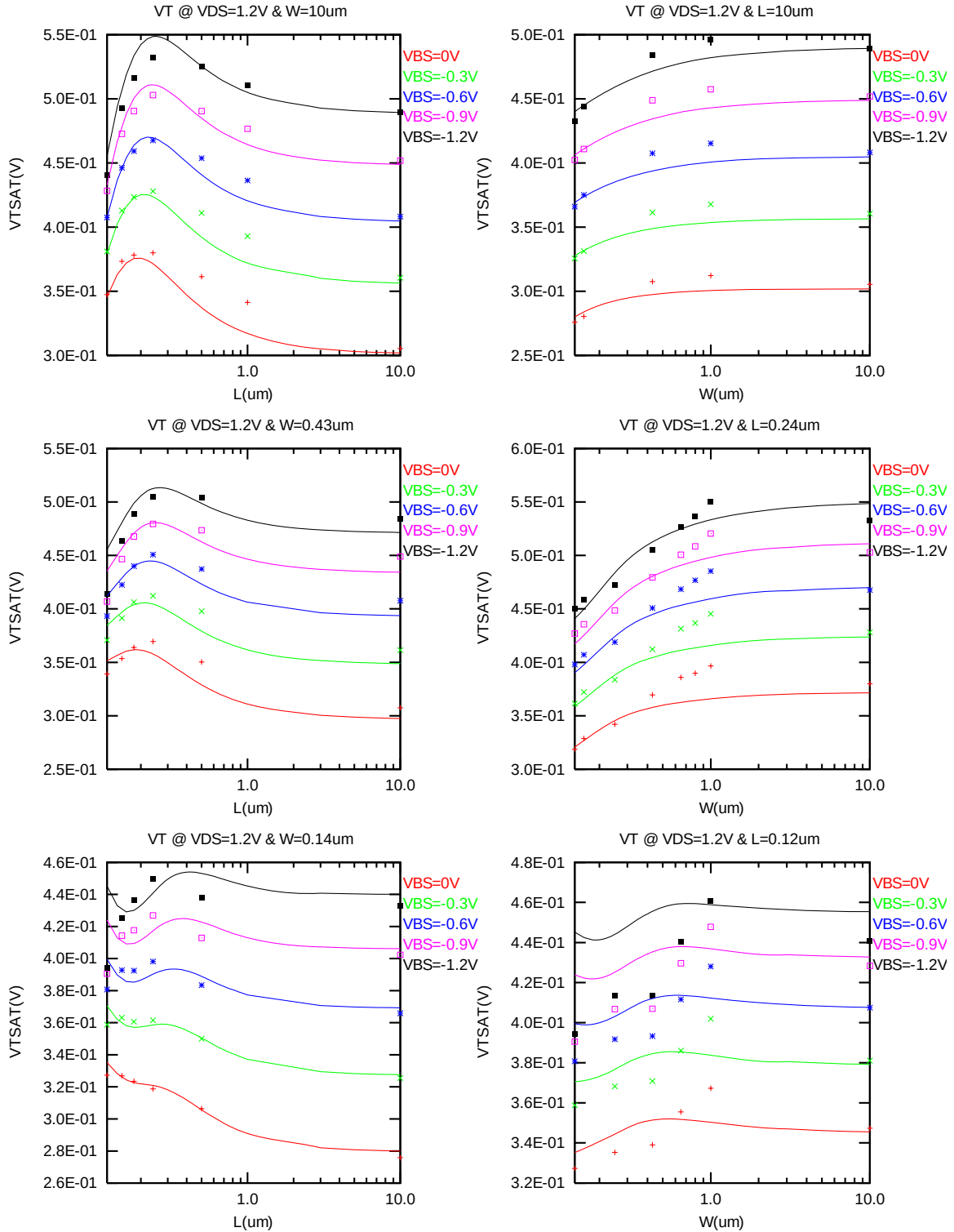


Figure 2-12 V_T vs. channel length & channel width for low drain bias ($V_{DS} = 1.2V$) for NMOS

I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = 1.2V$ for $V_{BS} = 0V$ and $V_{BS} = -1.2V$ at $25^\circ C$.

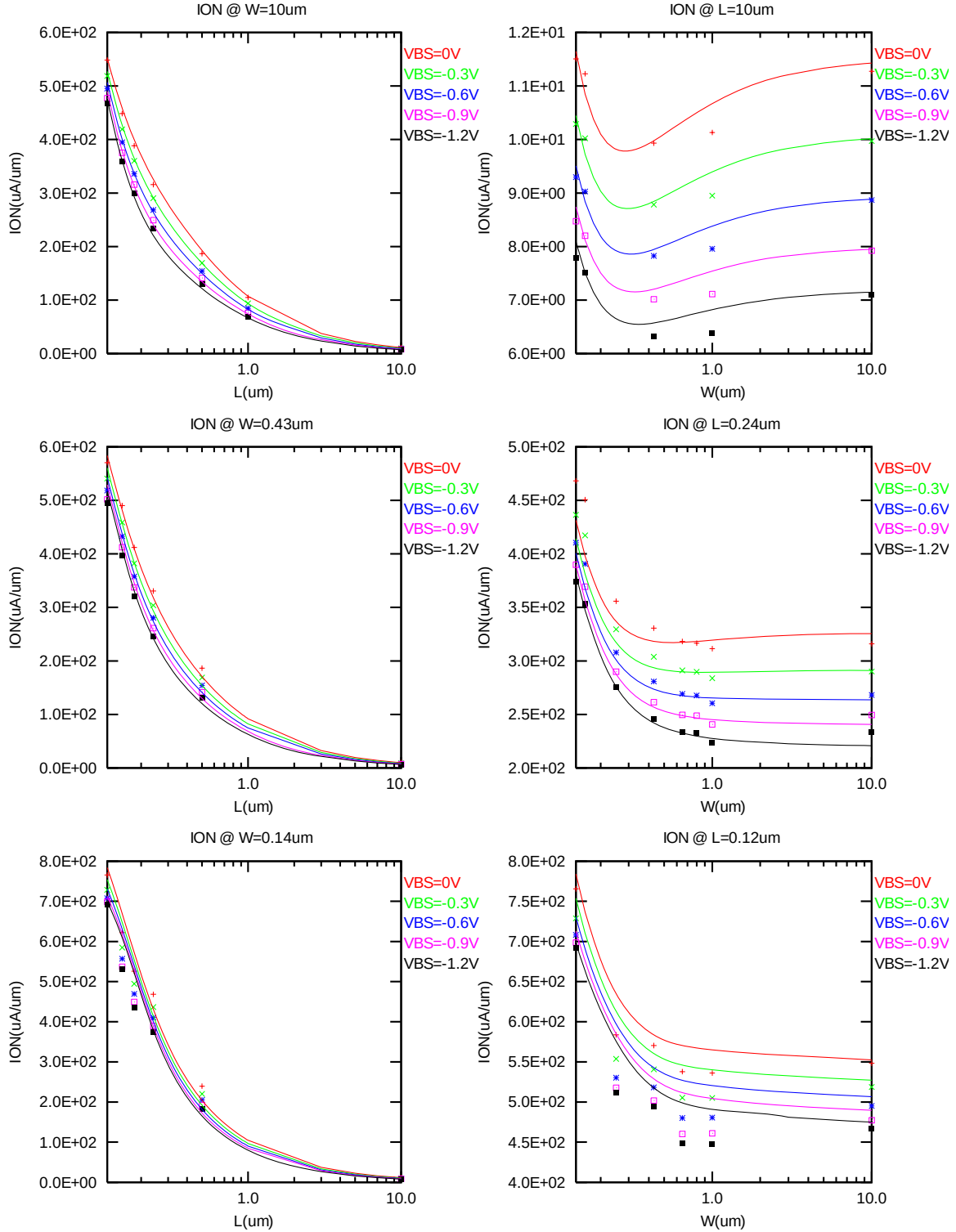


Figure 2-13 Saturation current I_{ON} vs channel length and channel width for NMOS

V_T vs. Temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $I_D = 300\text{nA} \cdot (W_{\text{DES}} \cdot 0.9) / (L_{\text{DES}} \cdot 0.9 + 0.012 \mu\text{m})$.

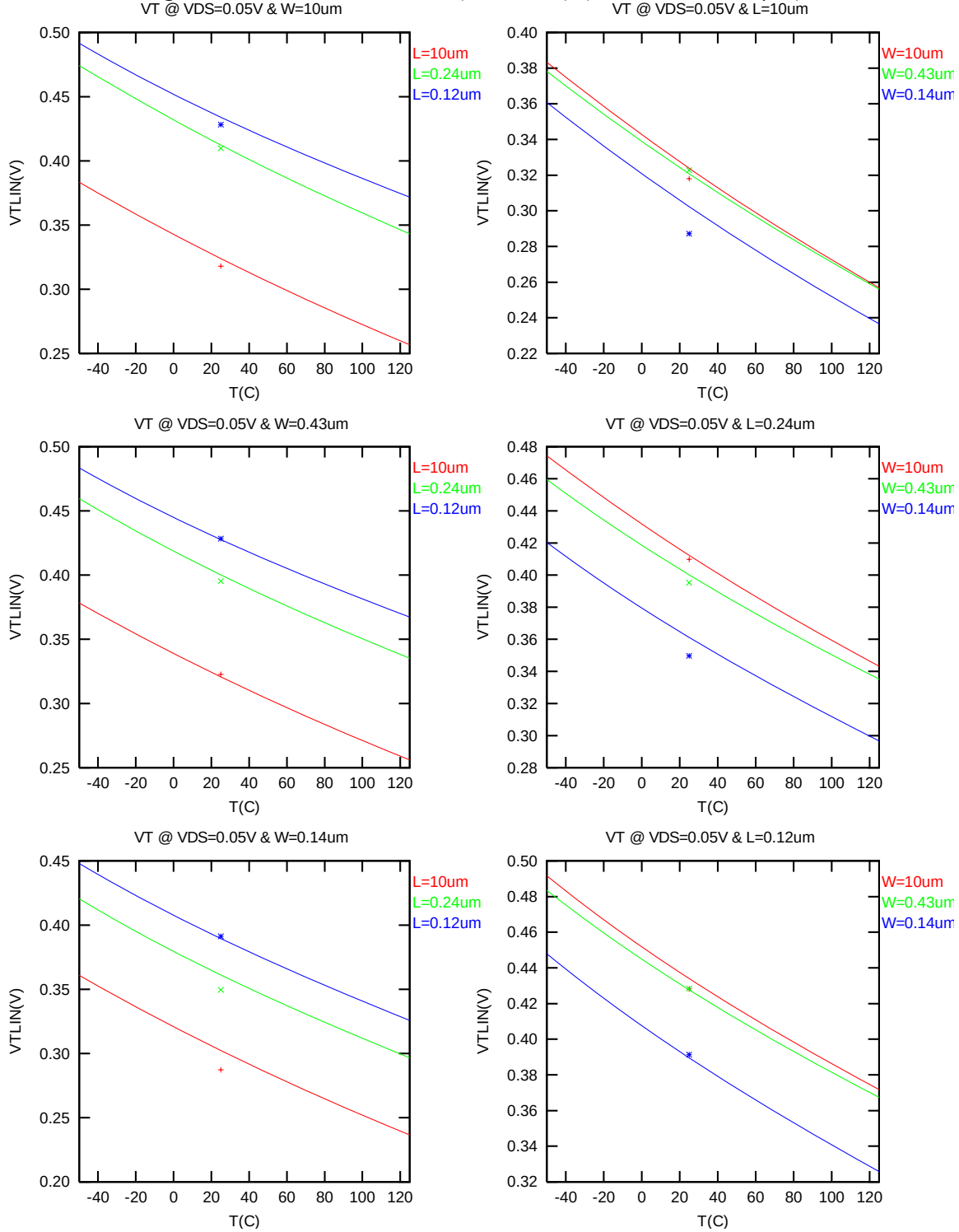


Figure 2-14 $V_{T\text{LIN}}$ vs temperature for NMOS

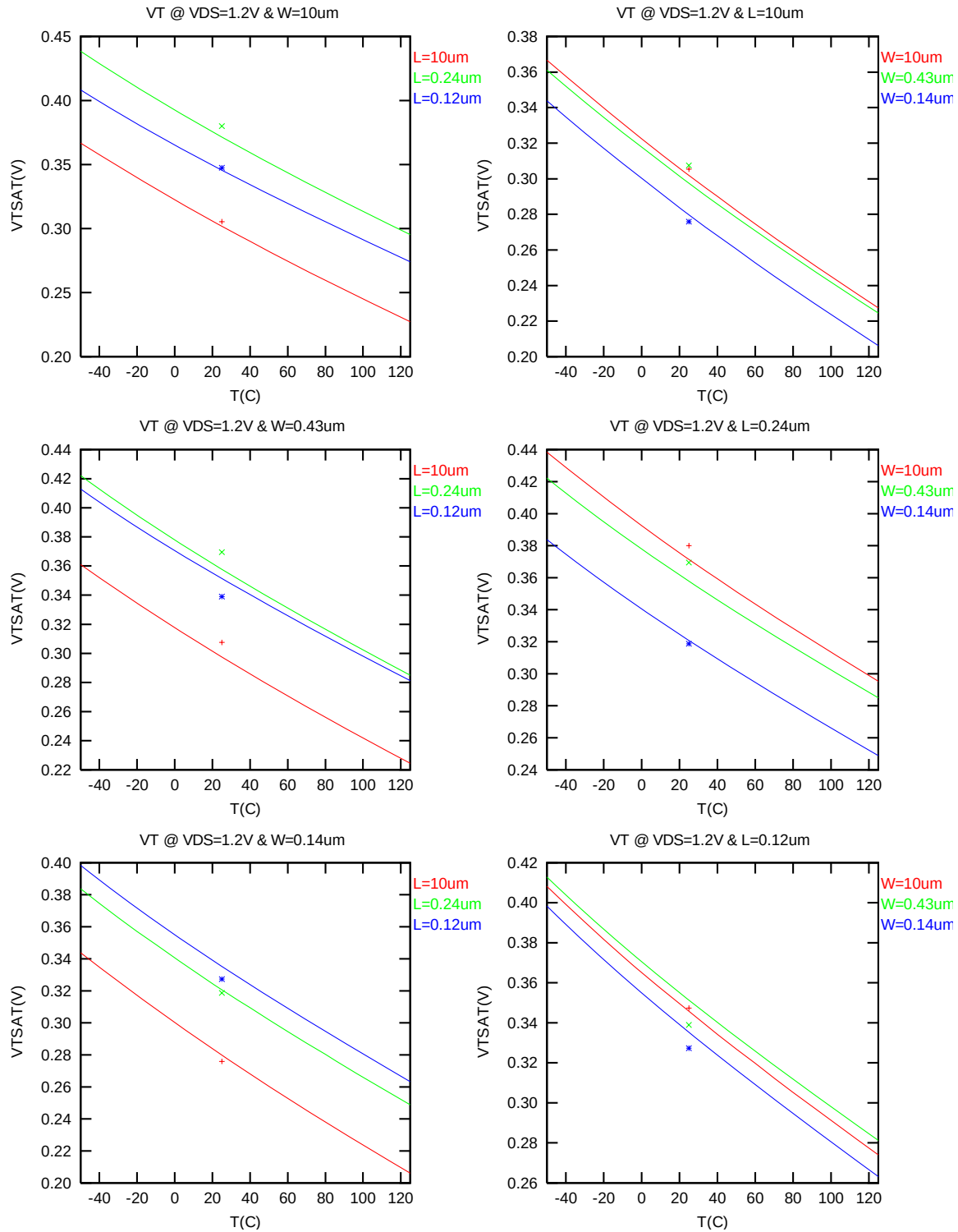


Figure 2-15 V_{TSAT} vs temperature for NMOS

I_{ON} vs Temperature

The following plots show the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = 1.2$ V, $V_{BS} = 0$ V.

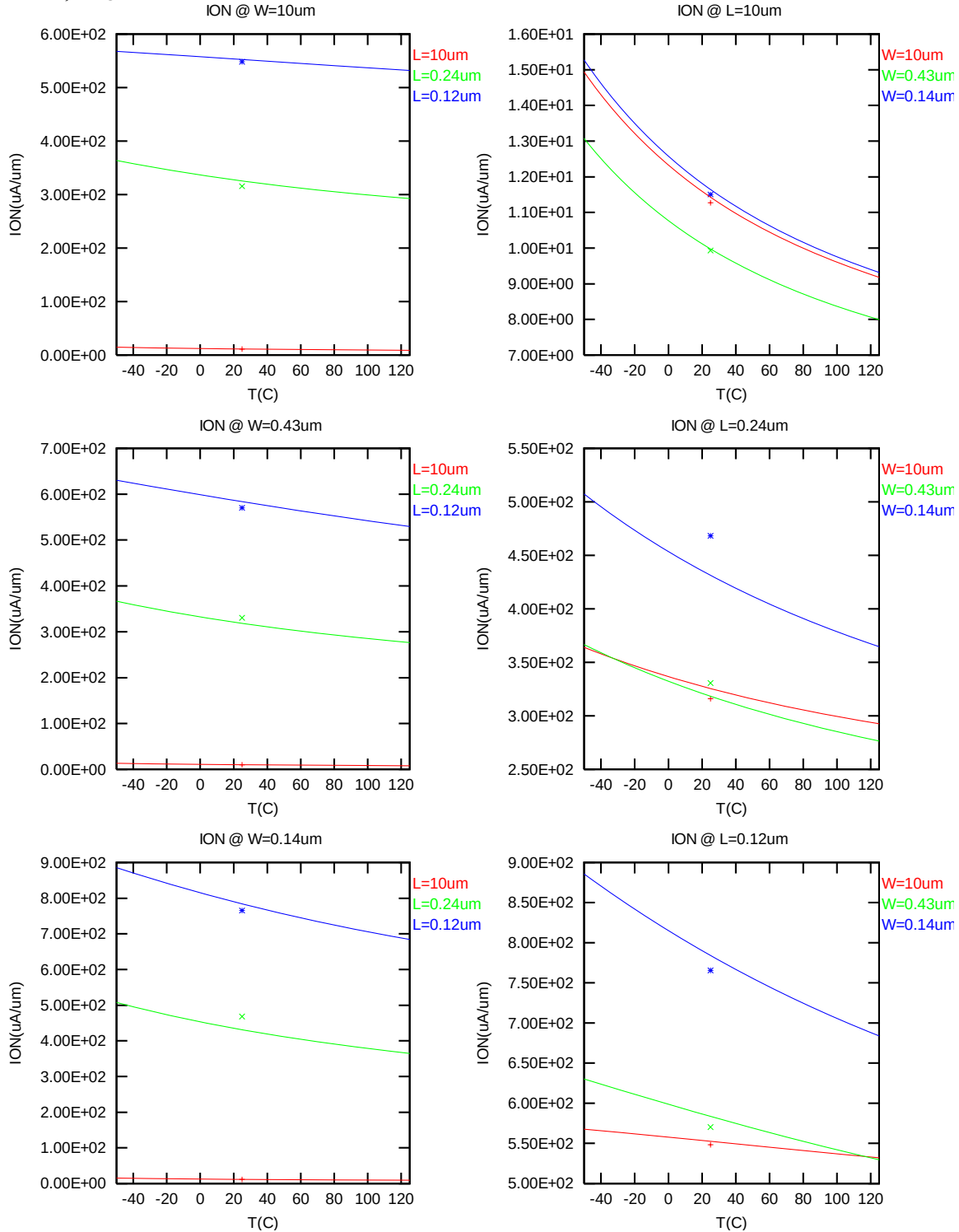


Figure 2-16 Saturation Current vs temperature for NMOS

2.3.4 I_{GATE} Extraction plot

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ 25C

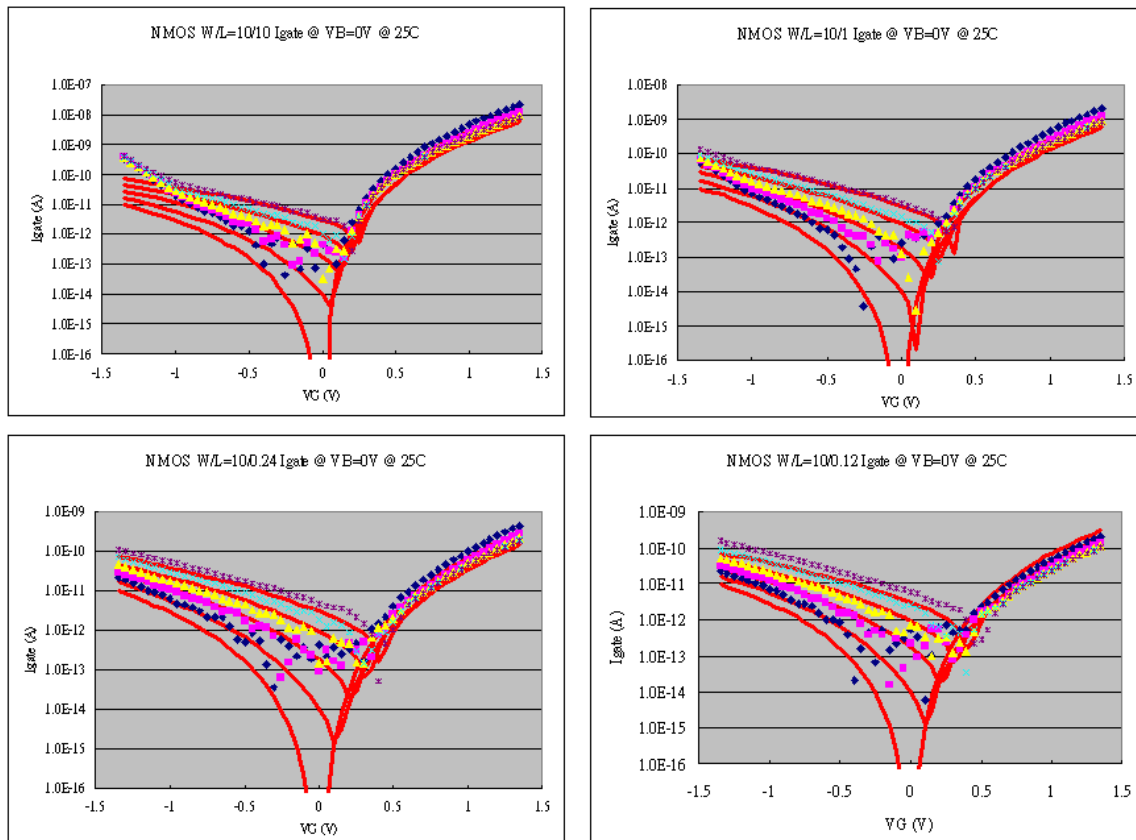


Figure 2-17 Measured and Simulated I_{GATE} for NMOS @ 25C

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ 125C

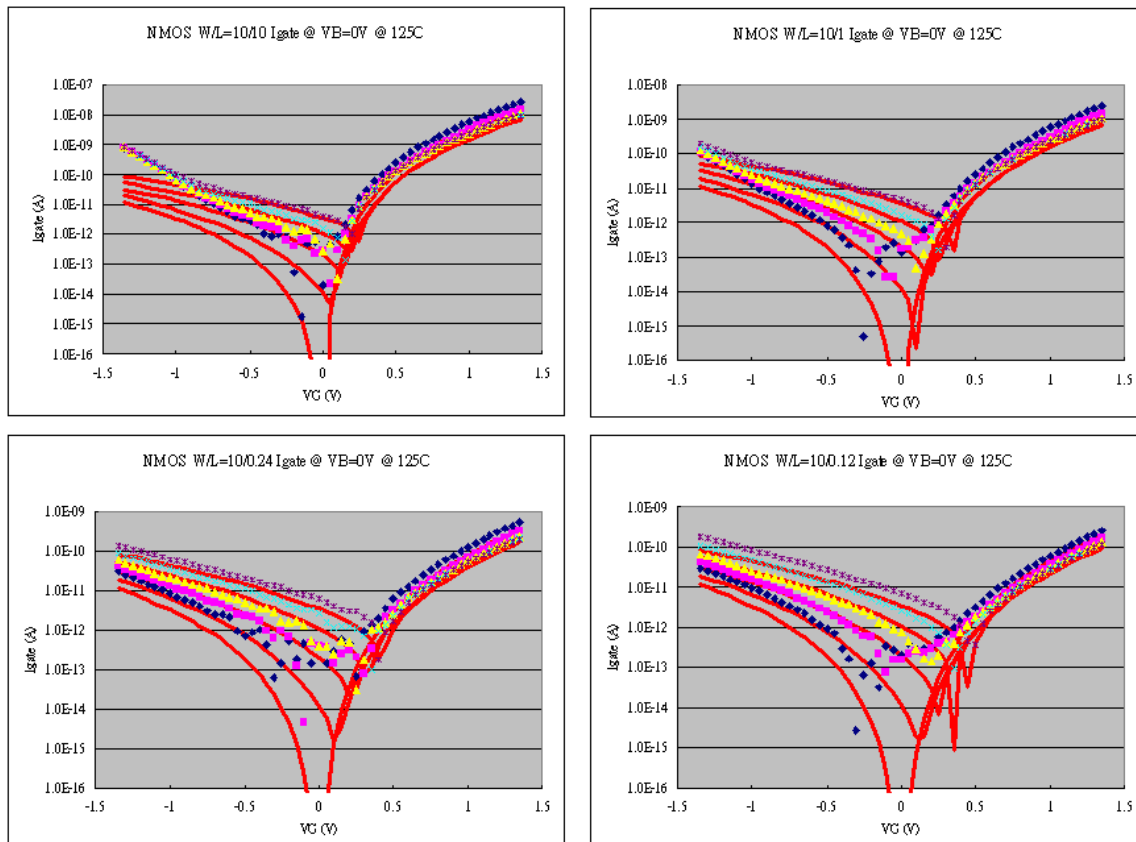


Figure 2-18 Measured and Simulated I_{GATE} for NMOS @ 125C

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ $-50C$

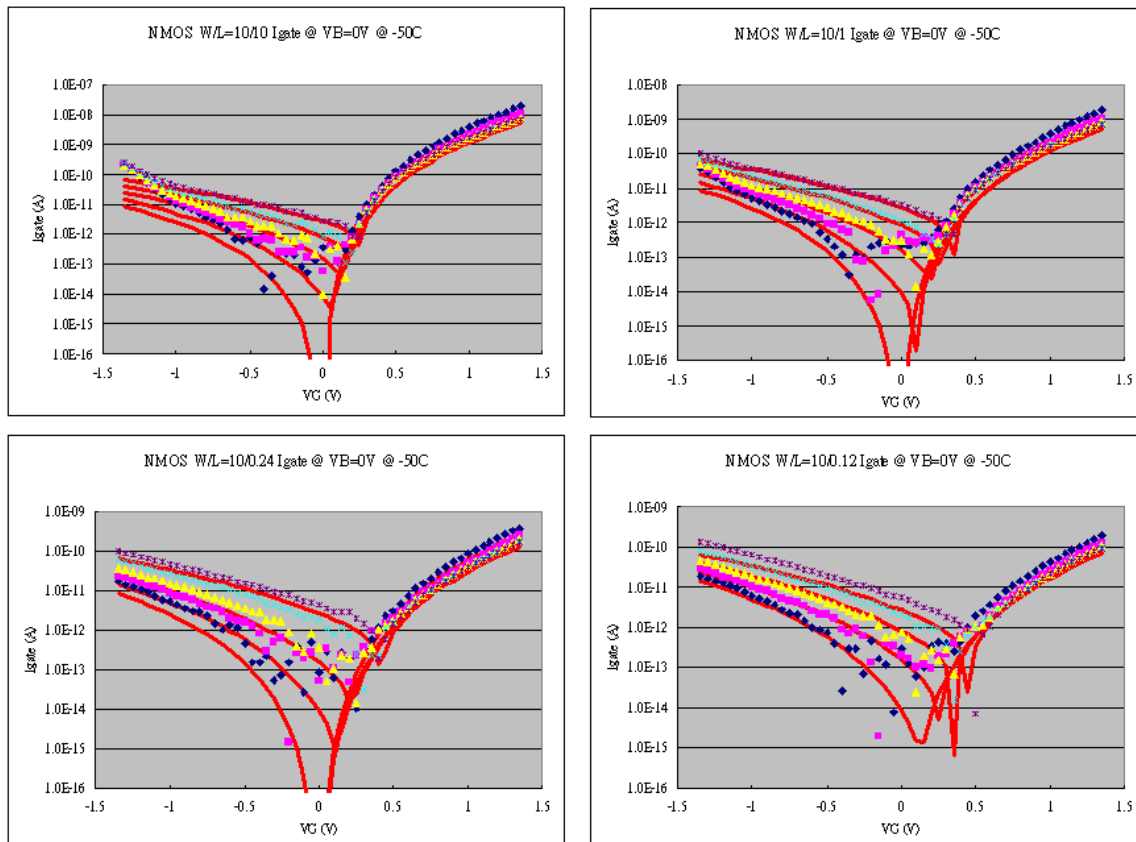


Figure 2-19 Measured and Simulated I_{GATE} for NMOS @ $-50C$

2.3.5 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

- I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

- G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 2-5 Requirement for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D Accuracy

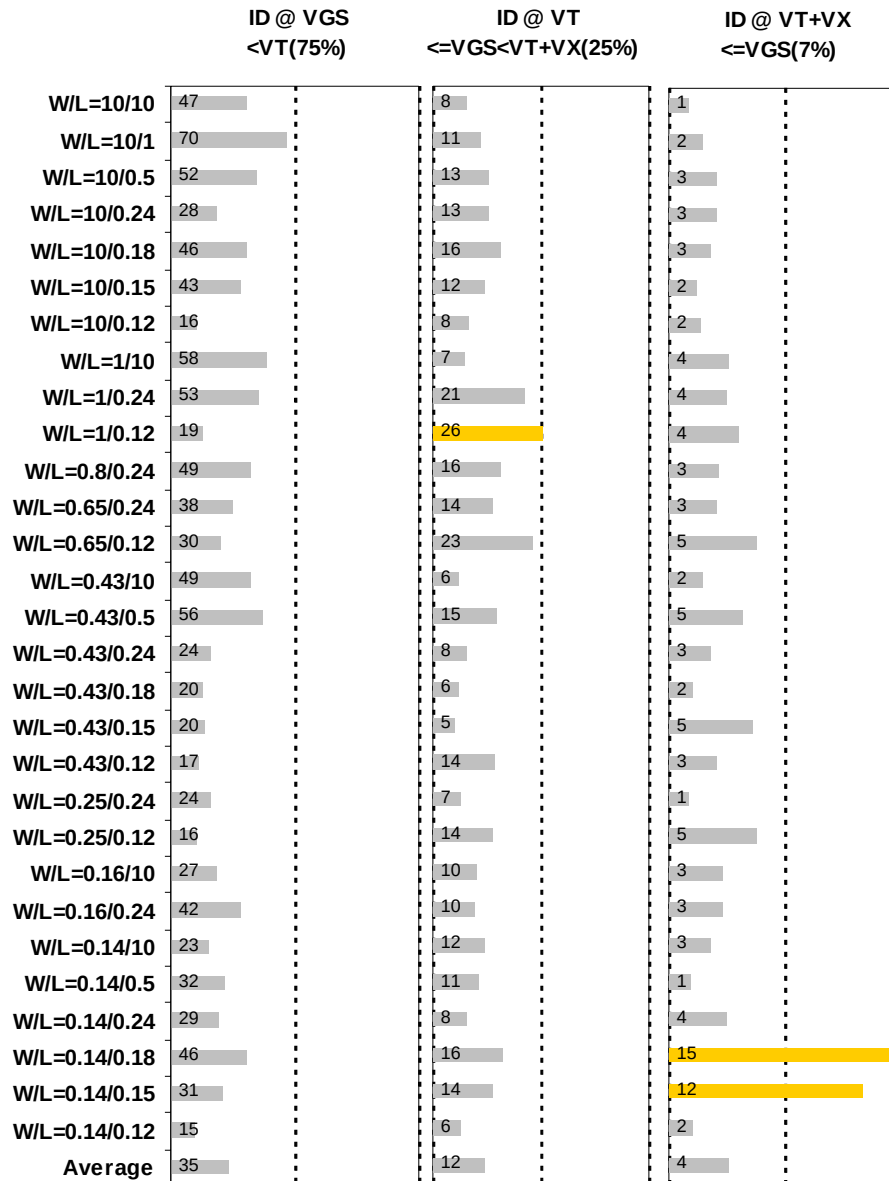


Figure 2-20 I_D accuracy at 25 °C for NMOS

G_M and G_{DS} Accuracy

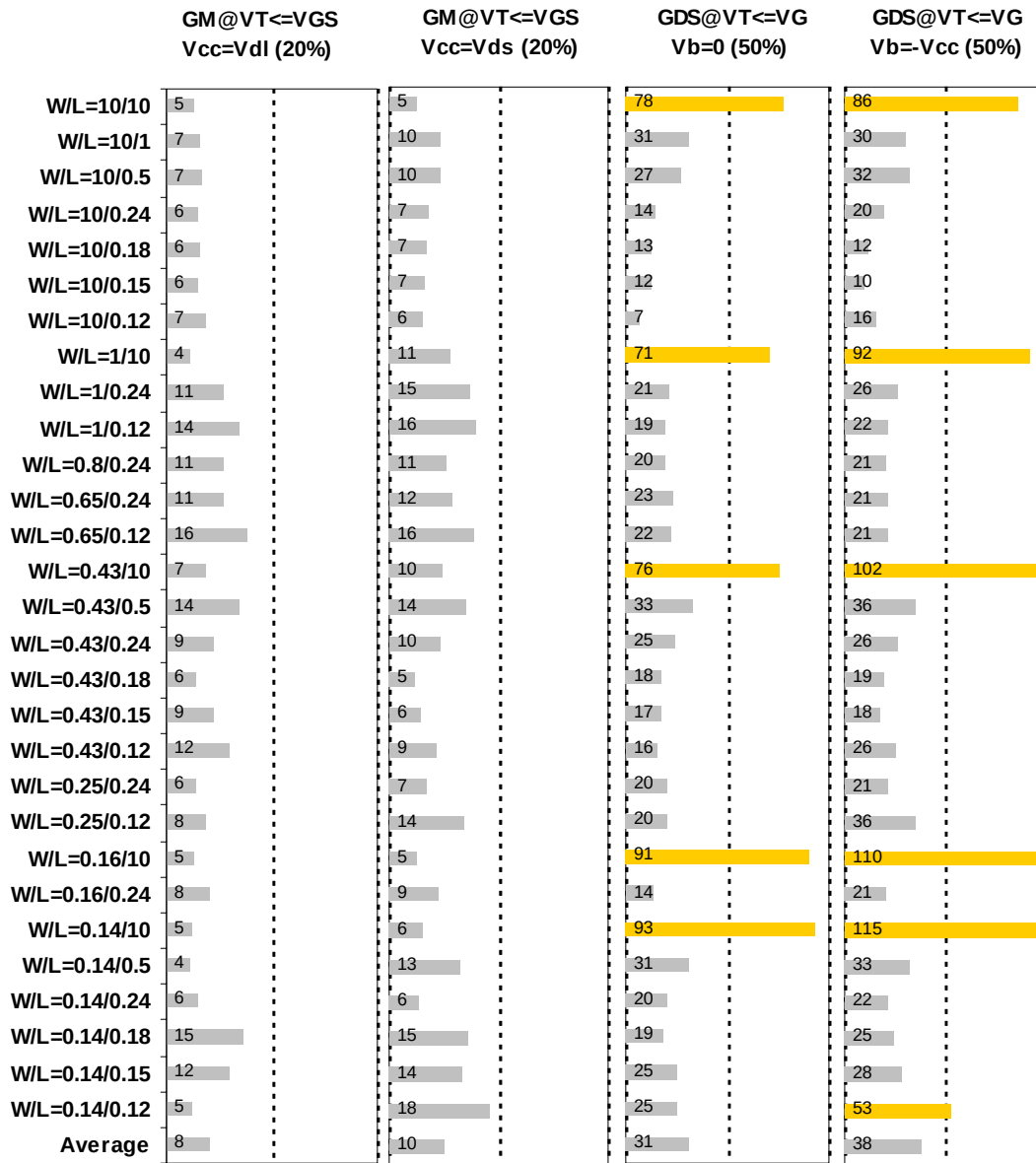


Figure 2-21 G_M and G_{DS} accuracy at 25 °C for NMOS

2.4 Parasitic source / drain junction capacitance model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Junction capacitor structures for NMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	3.60E-08	7.60E-04	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.70E-08	1.83E-02	Poly finger structure

2.4.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-7 Junction capacitor measurement conditions for NMOS

V _{BIAS} [V]	VH: pwell & VL: n+ From 0 to 1.3 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-50,25,125

2.4.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 2-8 Extracted parameters from junction capacitor measurement for NMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	Unit	Value	Parameter	Unit	Value	Parameter	Unit	Value
CJ	F/m ²	7.20E-04	CJSW	F/m	9.94E-11	CJSWG	F/m	3.76E-10
PB	V	5.60E-01	PBSW	V	5.40E-01	PBSWG	V	6.00E-01
MJ	--	2.58E-01	MJSW	--	1.30E-01	MJSWG	--	7.80E-01
TCJ	1/K	9.56E-04	TCJSW	1/K	7.32E-04	TCJSWG	1/K	3.30E-03
TPB	V/K	1.63E-03	TPBSW	V/K	3.10E-04	TPBSWG	V/K	2.05E-03

CGG vs. VBIAS Plots

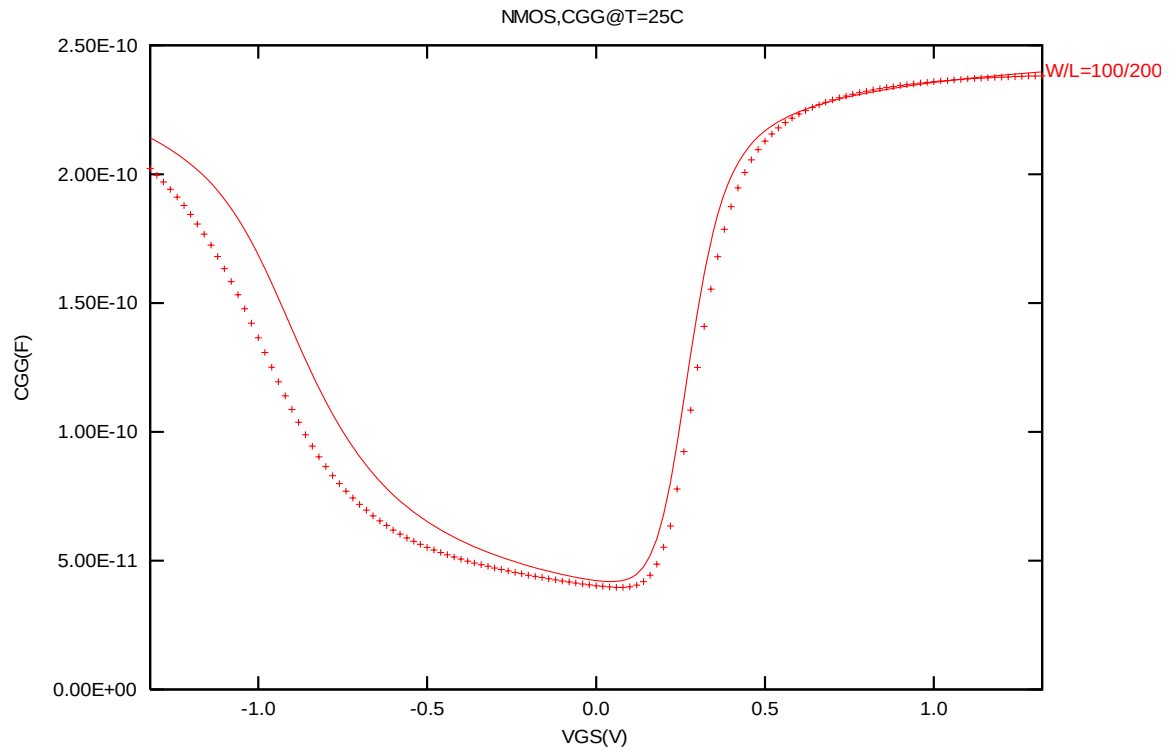


Figure 2-22 Measured and Simulated CGG curve for NMOS

CGC vs. VBIAS Plots

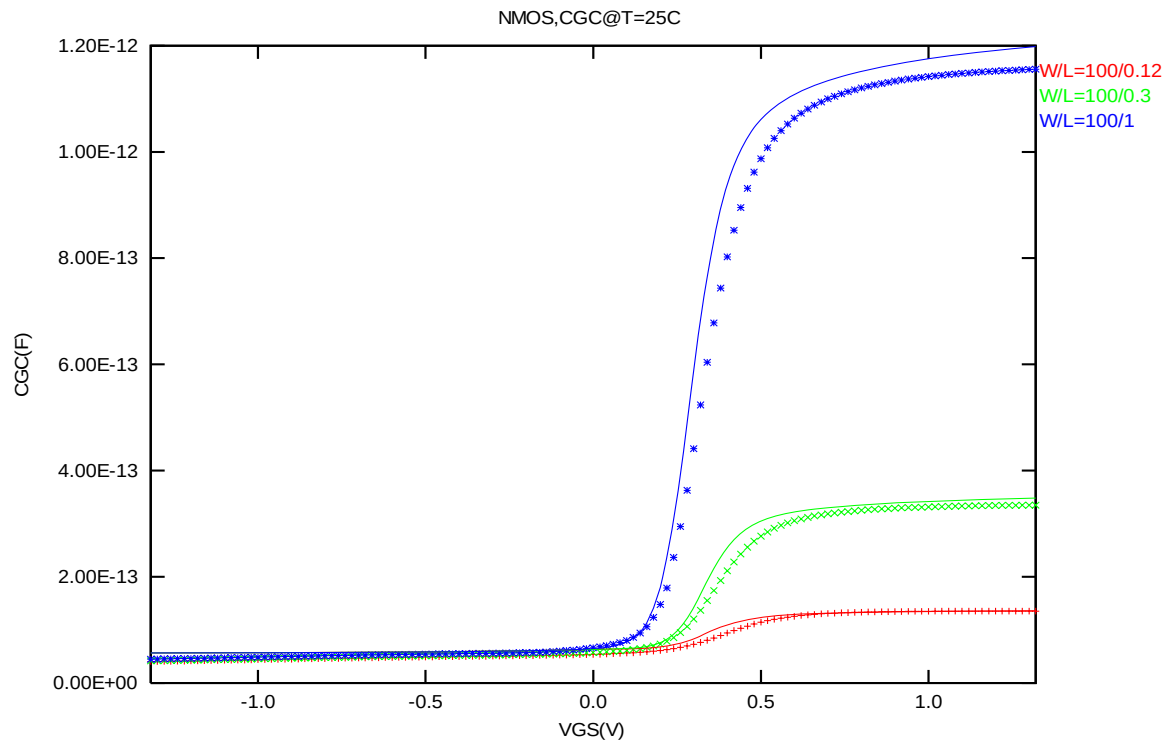


Figure 2-23 Measured and Simulated CGC curve for NMOS

CJ, CJSW vs. VBIAS Plots

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

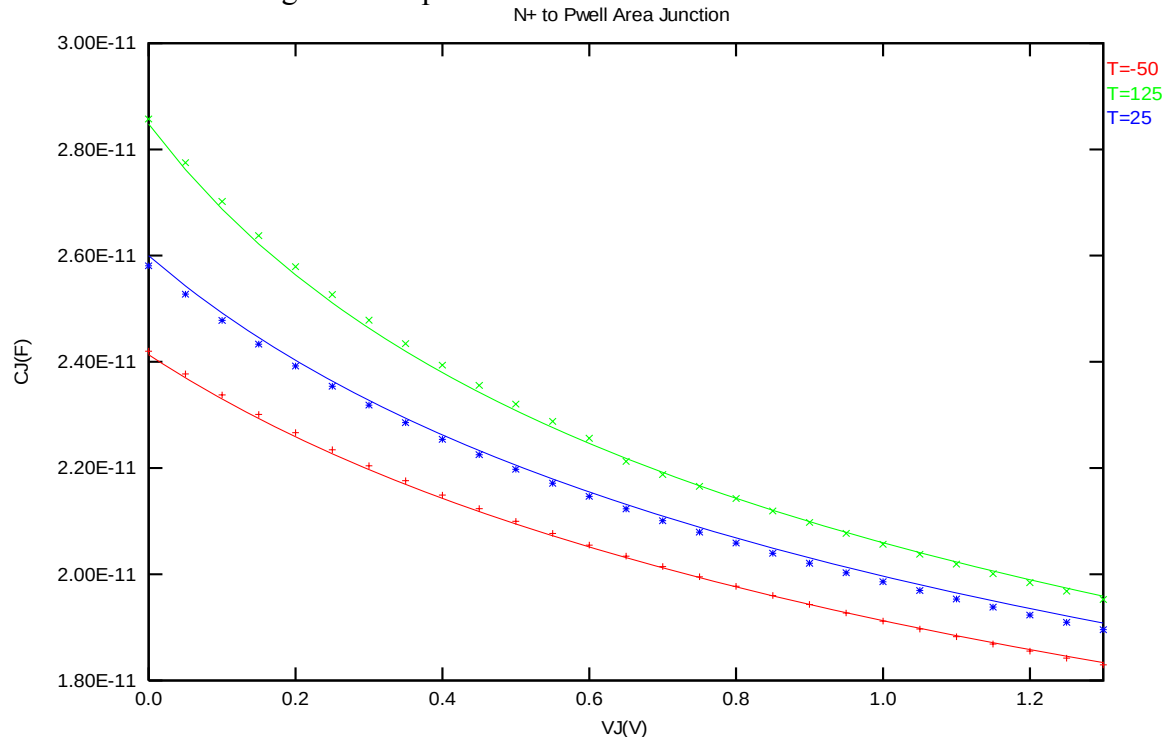


Figure 2-24 Bottom area junction capacitance at -50 °C, 25 °C and 125 °C for NMOS

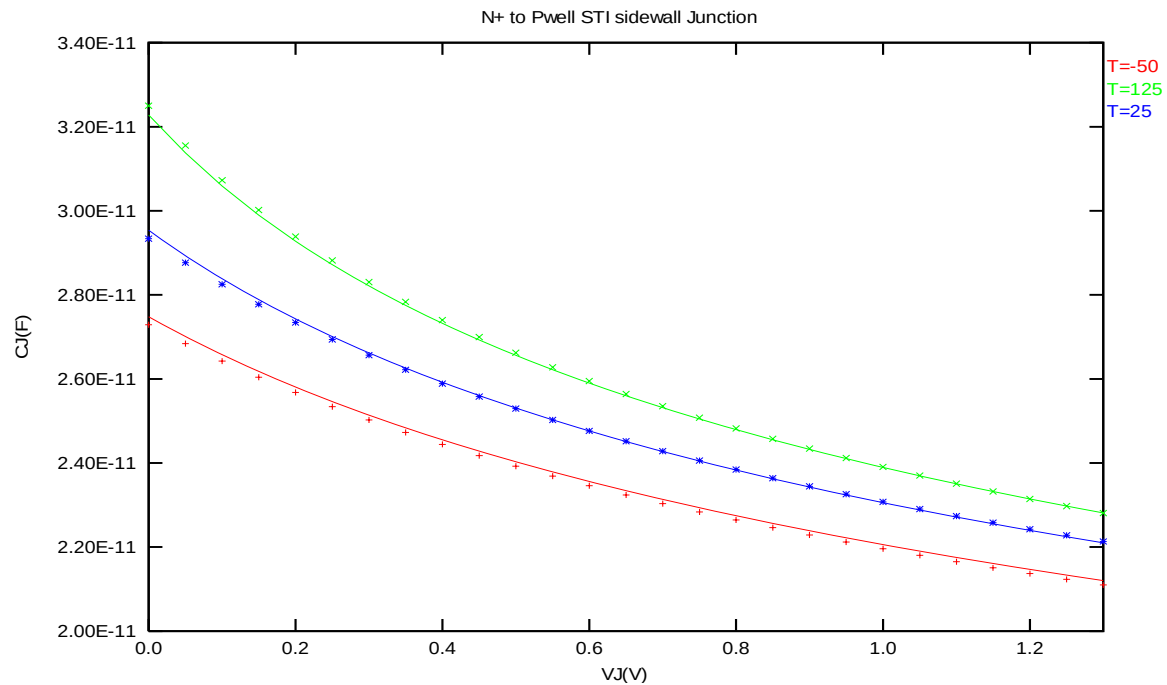


Figure 2-25 STI bounded sidewall junction capacitance at -50 °C, 25 °C and 125 °C for NMOS

3 1.2 V p-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	D1A0BBF3	D1A0BBF3	D1A0BBF3	D1A0BBF3
Lot ID	QGSML	QGSML	QGSML	QGSML
Wafer Number	12	12	12	12

3.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

Table 3-2 Electrical Parameters for PMOS

All parameters are given for T=25 °C ,V_{BS}=0 V unless specified otherwise.

Target & Centered Value are given for SA=2.2u SB=2.2u

Measured & Extracted Value are given for SA=2.2u SB=2.2u

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
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Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)

V _{TLIN}	V _{DS} = -0.05 V	V	-0.384	-0.381	-0.385	-0.386
I _{ON}	V _{DS} = -1.2 V	μA/ μm	-1.90	-1.91	-1.91	-1.89

Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = -0.05 V	V	-0.407	-0.409	-0.417	-0.407
V _{TSAT}	V _{DS} = -1.2 V	V	-0.330	-0.331	-0.340	-0.330
Body Effect V _T shift	V _{DS} = -1.2 V V _{BS} = 0 ~ -1.2 V	V	--	-0.190	-0.202	-0.202
DIBL V _T shift	V _{DS} = -0.05~-1.2 V	V	-0.080	-0.078	-0.077	-0.077
I _{ON}	V _{DS} =V _{GS} = -1.2 V	μA/μm	-195.0	-193.23	-193.67	-195.00
I _{OFF}	V _{DS} = -1.2 V, V _{GS} =0 V	A/μm	-2E-10	-9.0E-11	-1.1E-10	-1.5E-10

Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)

V _{TLIN}	V _{DS} = -0.05 V	V	-0.355	-0.353	-0.357	-0.357
V _{TSAT}	V _{DS} = -1.2 V	V	--	-0.340	-0.345	-0.345
I _{ON}	V _{DS} =V _{GS} = -1.2 V	μA/μm	-2.686	-2.70	-2.75	-2.72

Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μ m / 0.12 μ m)

V _{TLIN}	V _{DS} = -0.05 V	V	-0.407	-0.403	-0.399	-0.390
V _{TSAT}	V _{DS} = -1.2 V	V	-0.332	-0.337	-0.358	-0.341
I _{ON}	V _{DS} =V _{GS} = -1.2 V	μA/μm	-232.86	-223.63	-230.90	-234.61
I _{OFF}	V _{DS} = -1.2 V, V _{GS} =0 V	A/μm	--	-5.2E-11	-3.2E-11	-4.8E-11

Capacitance

C _J	V _J = 0 V	F/m ²	9.65E-04	9.65E-04	9.69E-04	9.69E-04
C _{JSW}	V _J = 0 V	F/m	1.24E-10	1.15E-10	1.18E-10	1.18E-10

CJSWG	$V_j = 0 \text{ V}$	F/m	5.34E-10	5.16E-10	5.43E-10	5.43E-10
C _{OVL}	$V_{DS} = V_{GS} = 0.5 \text{ V}$	F/m	3.07E-10	2.66E-10	2.99E-10	2.99E-10

3.3 MOSFET DC model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below.

Table 3-3 Geometry of PMOS devices measured for parameter extraction

WL	10	1	0.5	0.24	0.18	0.15	0.12
10	X	X	X	X		X	X
1	X			X			X
0.8				X			
0.65				X			X
0.43	X		X	X	X	X	X
0.25				X			X
0.16	X			X			
0.14	X		X	X		X	X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	-0.05	-1.2	-1.15
V_{GS} (V)	0.5	-1.35	-0.025
V_{BS} (V)	0	1.2	0.3

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	-1.35	-0.025
V_{GS} (V)	-0.6	-1.2	-0.15
V_{BS} (V)	0	1.2	1.2

3.3.3 Model verification plots

The simulation results obtained from the extracted model were plotted together with the measurement results for comparison. The L and W in the plots refer to physical dimension ($L_{DES} * 0.9 + 12nm$) and ($W_{DES} * 0.9$) respectively. (All the device dimensions in these plots are after 90% shrink and 12nm sizing channel length back.)

I_D vs. V_{DS} Plots

The following plots show measured and simulated output I-V curves for selected devices at 25 °C.

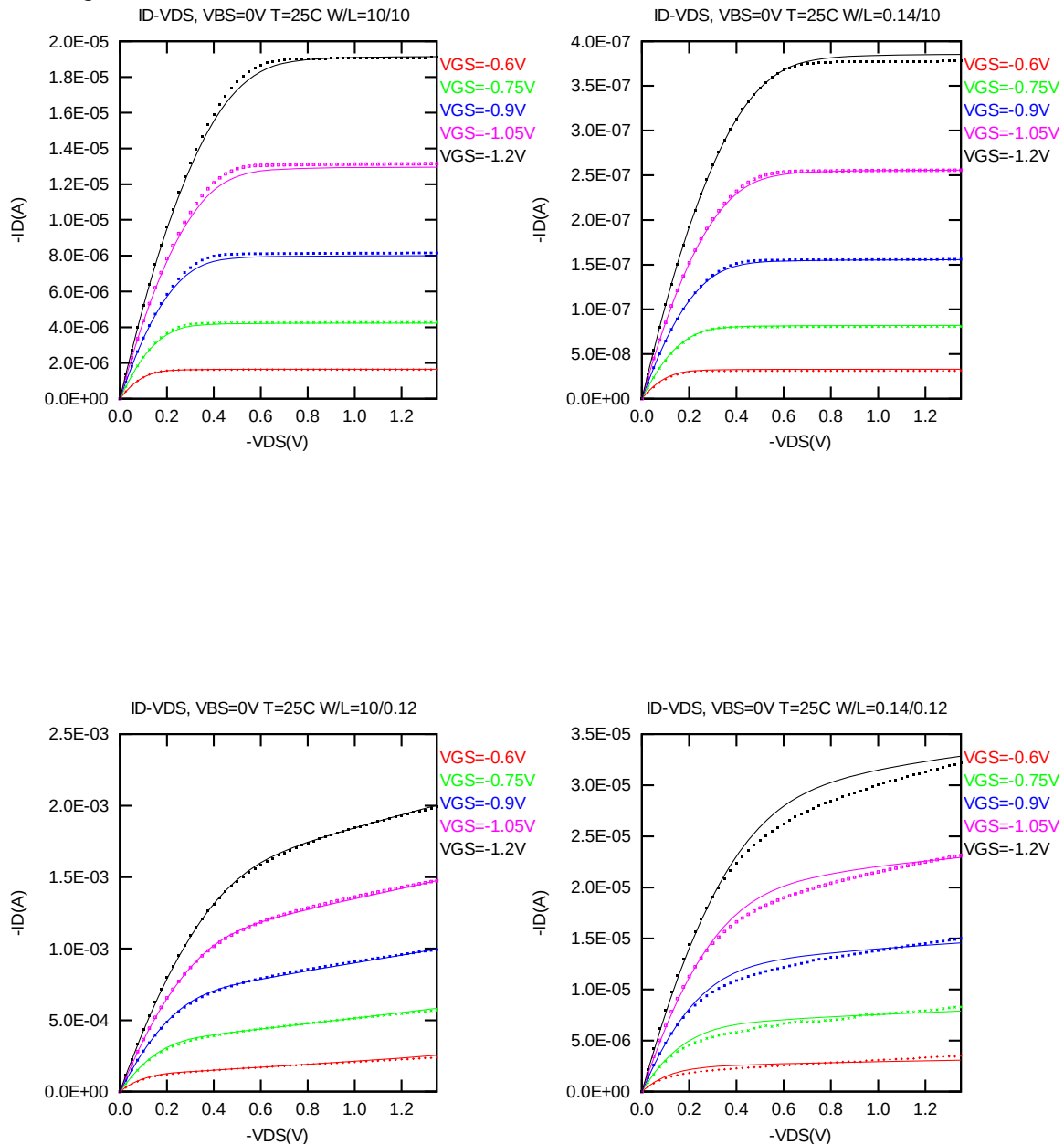


Figure 3-1 I_D vs. V_{DS} at $V_{BS} = 0V$ for PMOS

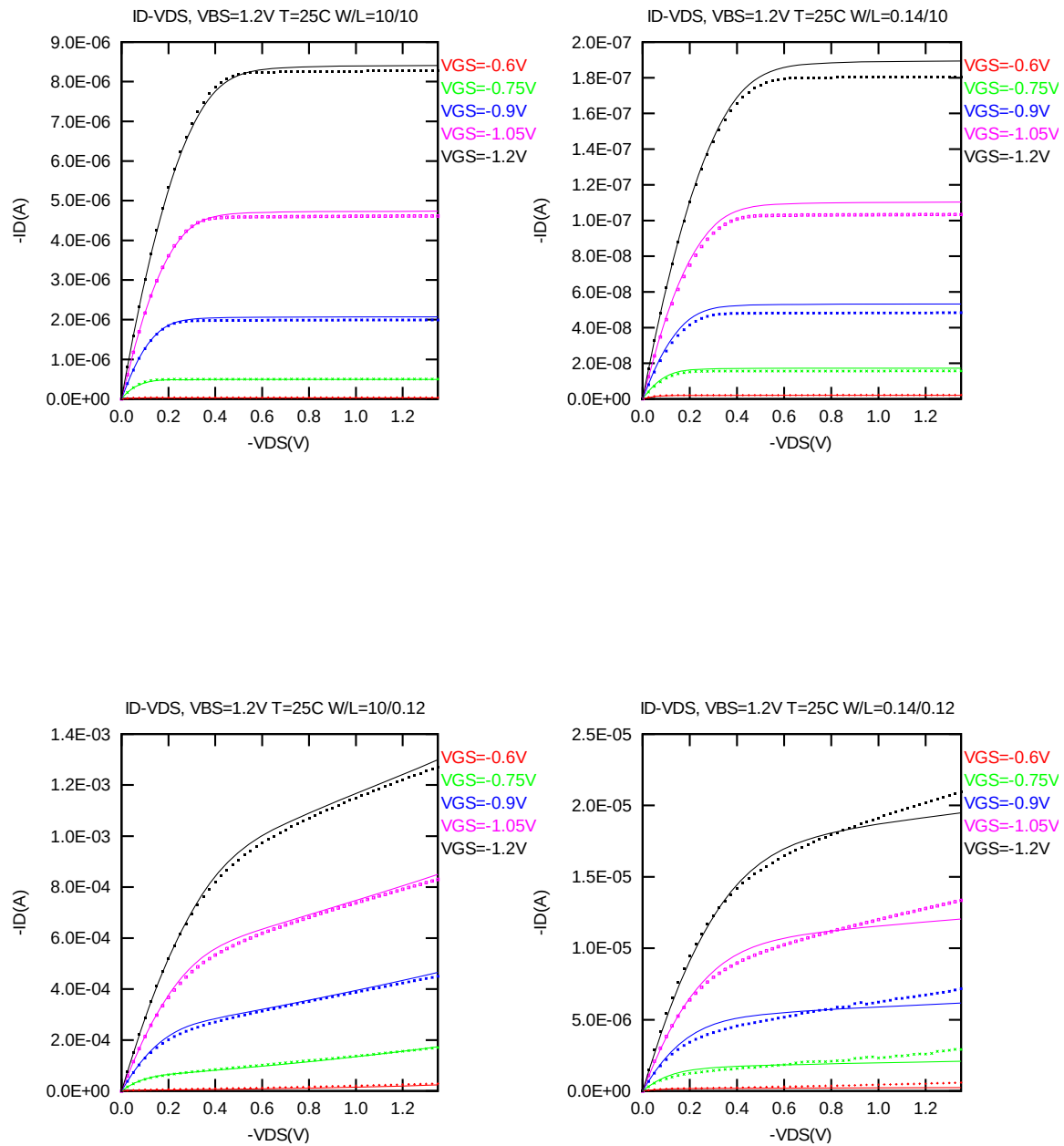


Figure 3-2 I_D vs. V_{DS} at $V_{BS}=1.2$ V for PMOS

G_{DS} vs. V_{DS} Plots

The following plots show measured and simulated output conductance curves for selected devices at 25 °C.

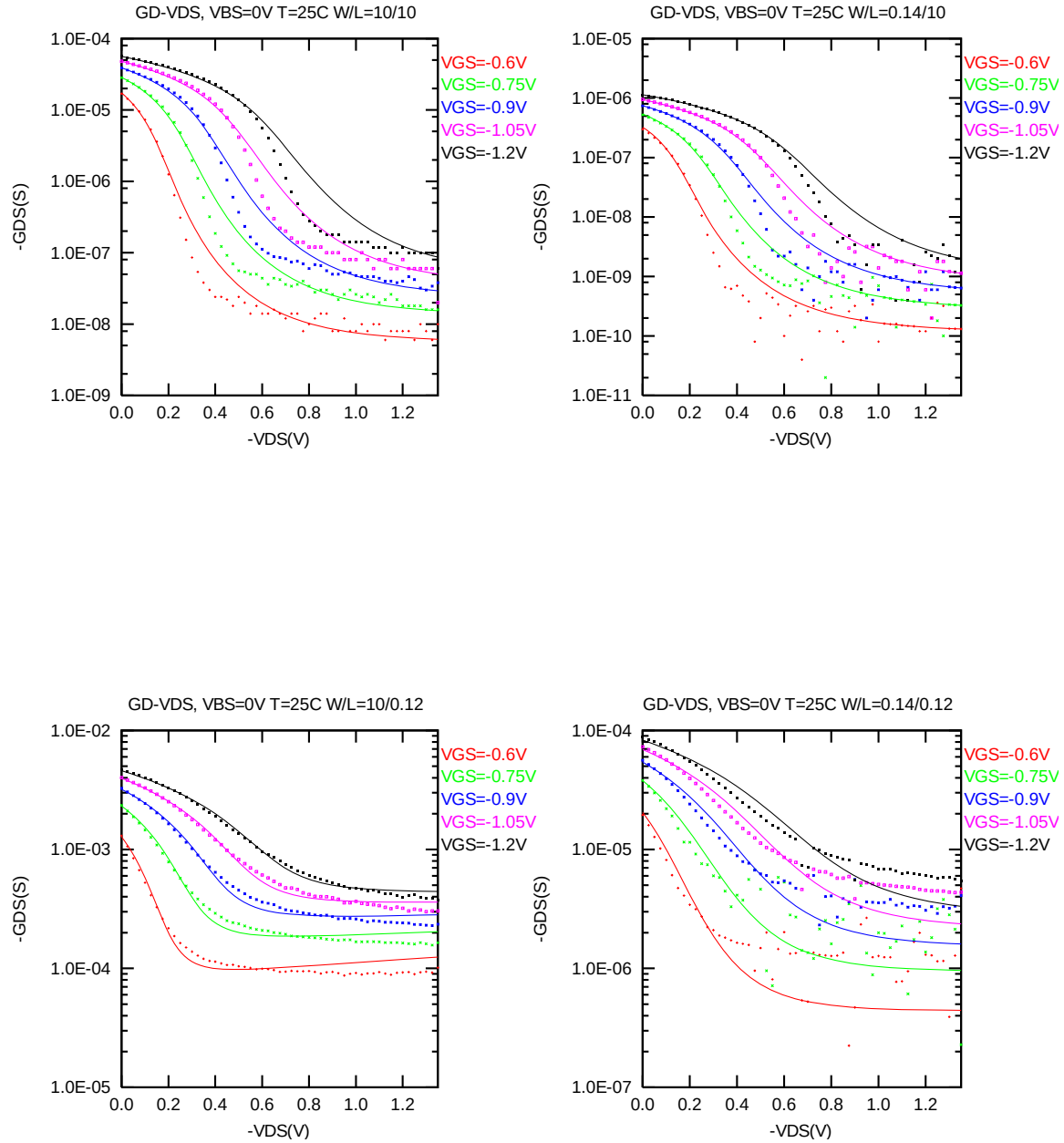


Figure 3-3 G_{DS} vs. V_{DS} at $V_{BS} = 0V$ for PMOS

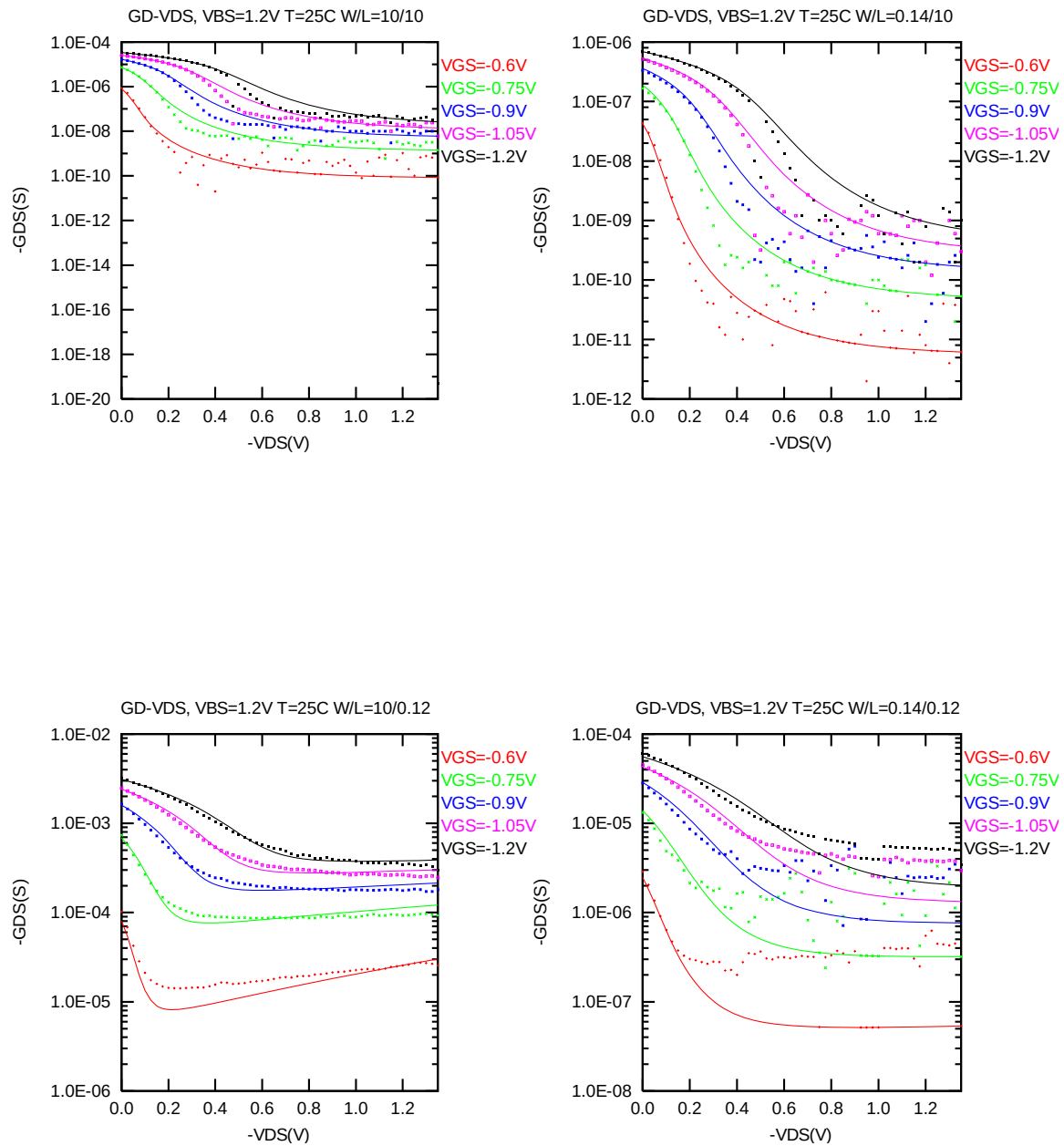


Figure 3-4 G_{DS} vs. V_{DS} at $V_{BS} = 1.2V$ for PMOS

I_D vs. V_{GS} Plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 °C.

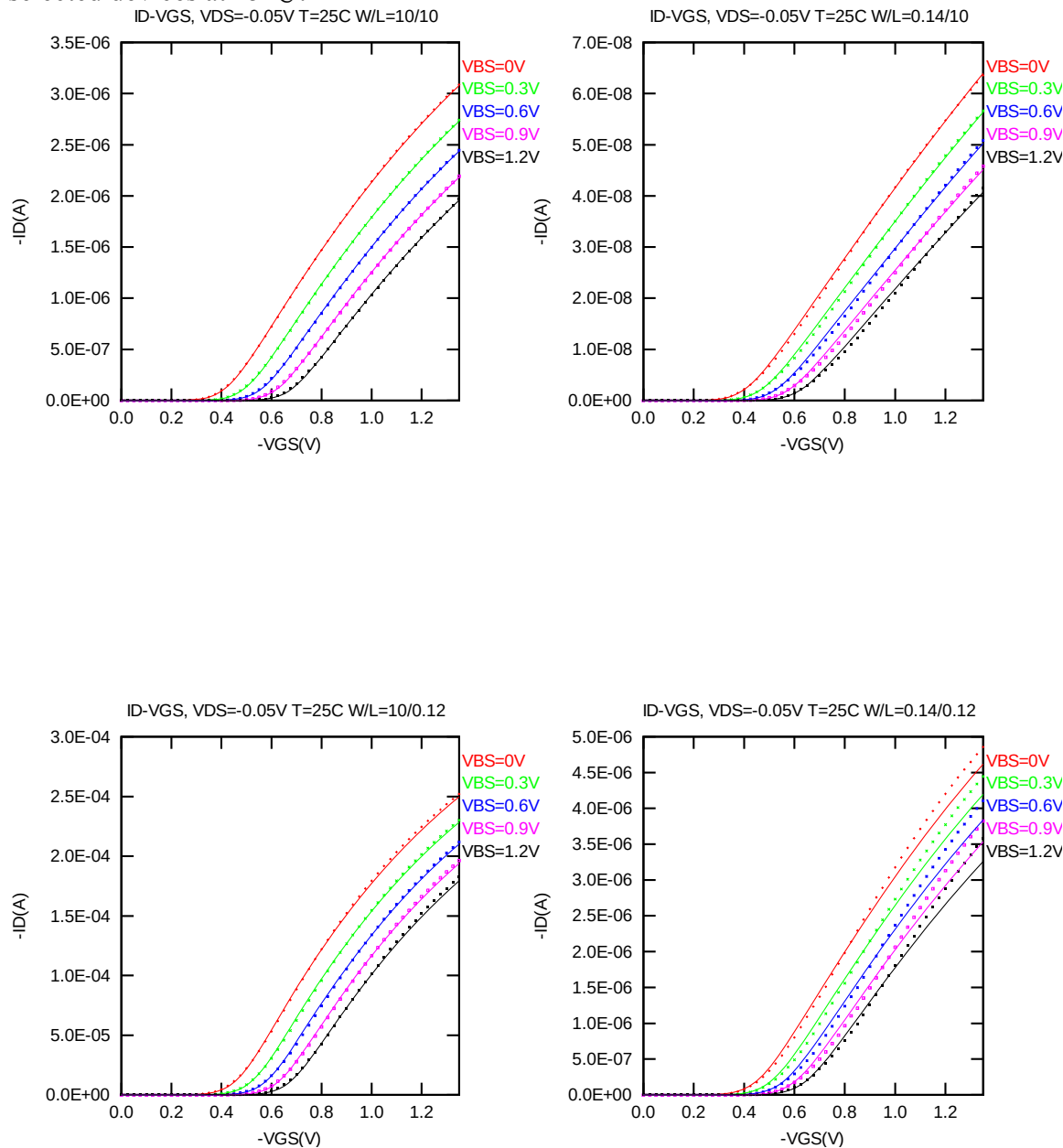


Figure 3-5 I_D vs. V_{GS} at $V_{DS} = -0.05V$ for PMOS

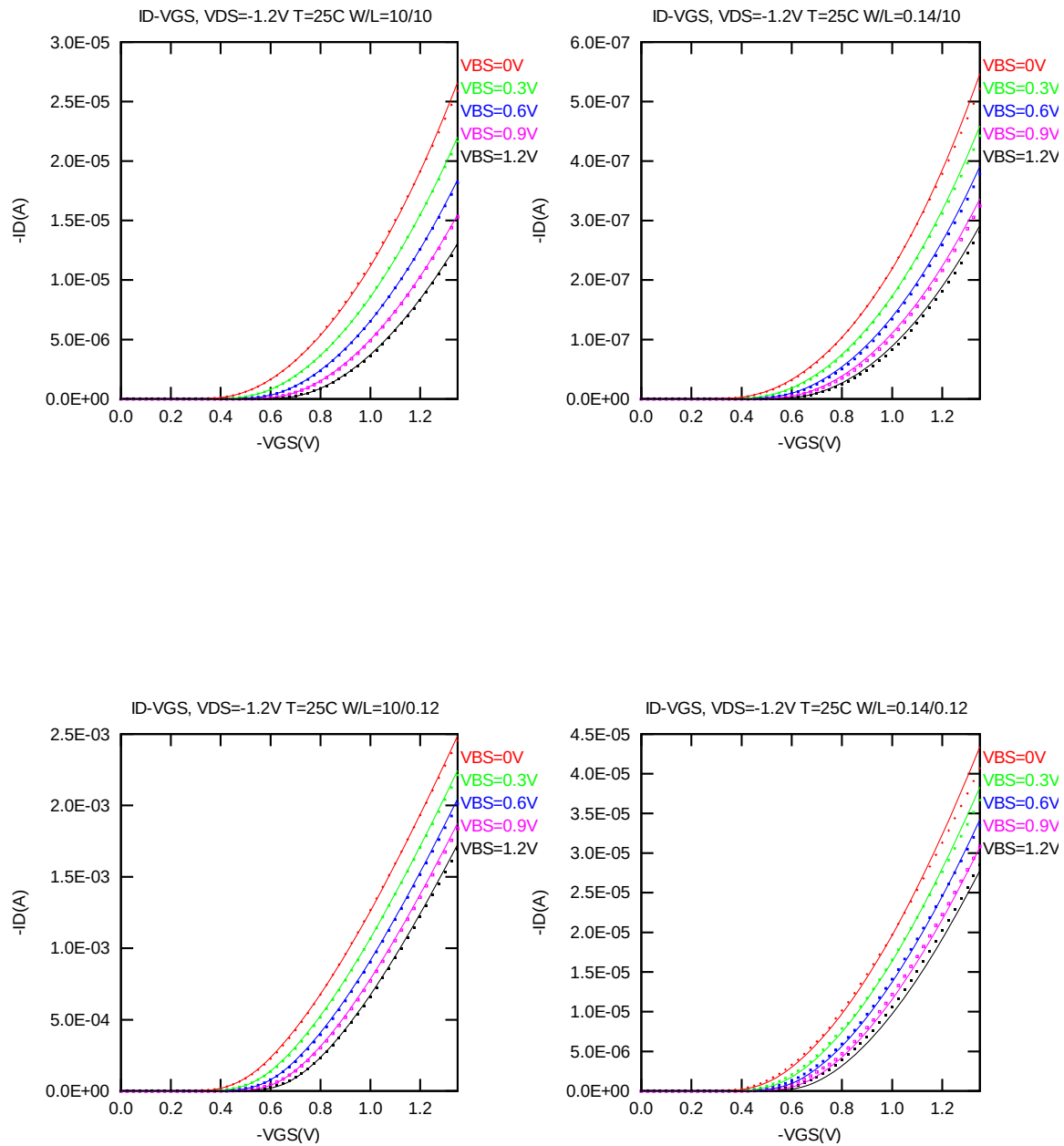


Figure 3-6 I_D vs. V_{GS} at $V_{DS} = -1.2V$ for PMOS

I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 °C.

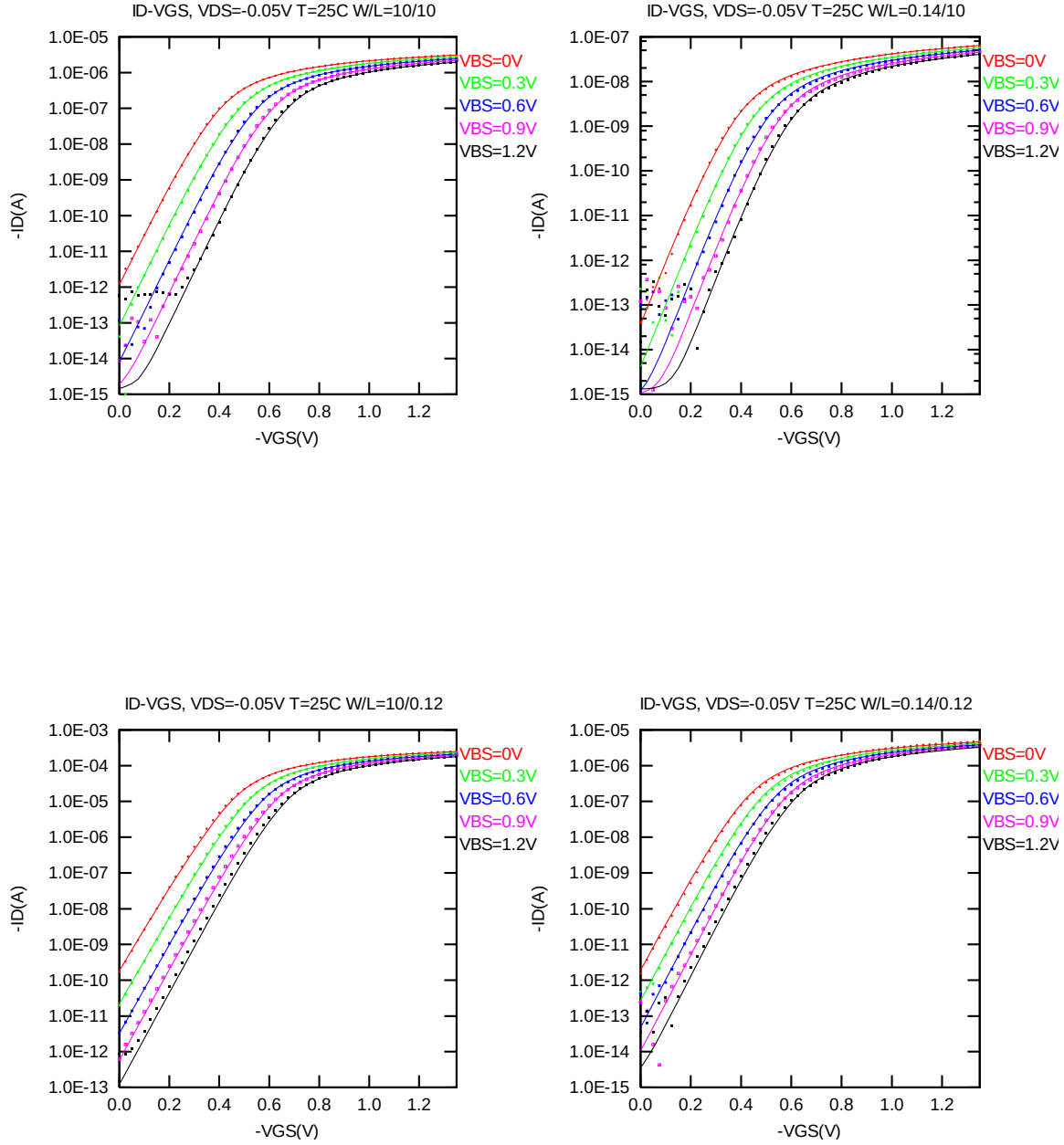


Figure 3-7 Log scaled I_D vs. V_{GS} at $V_{DS} = -0.05$ V for PMOS

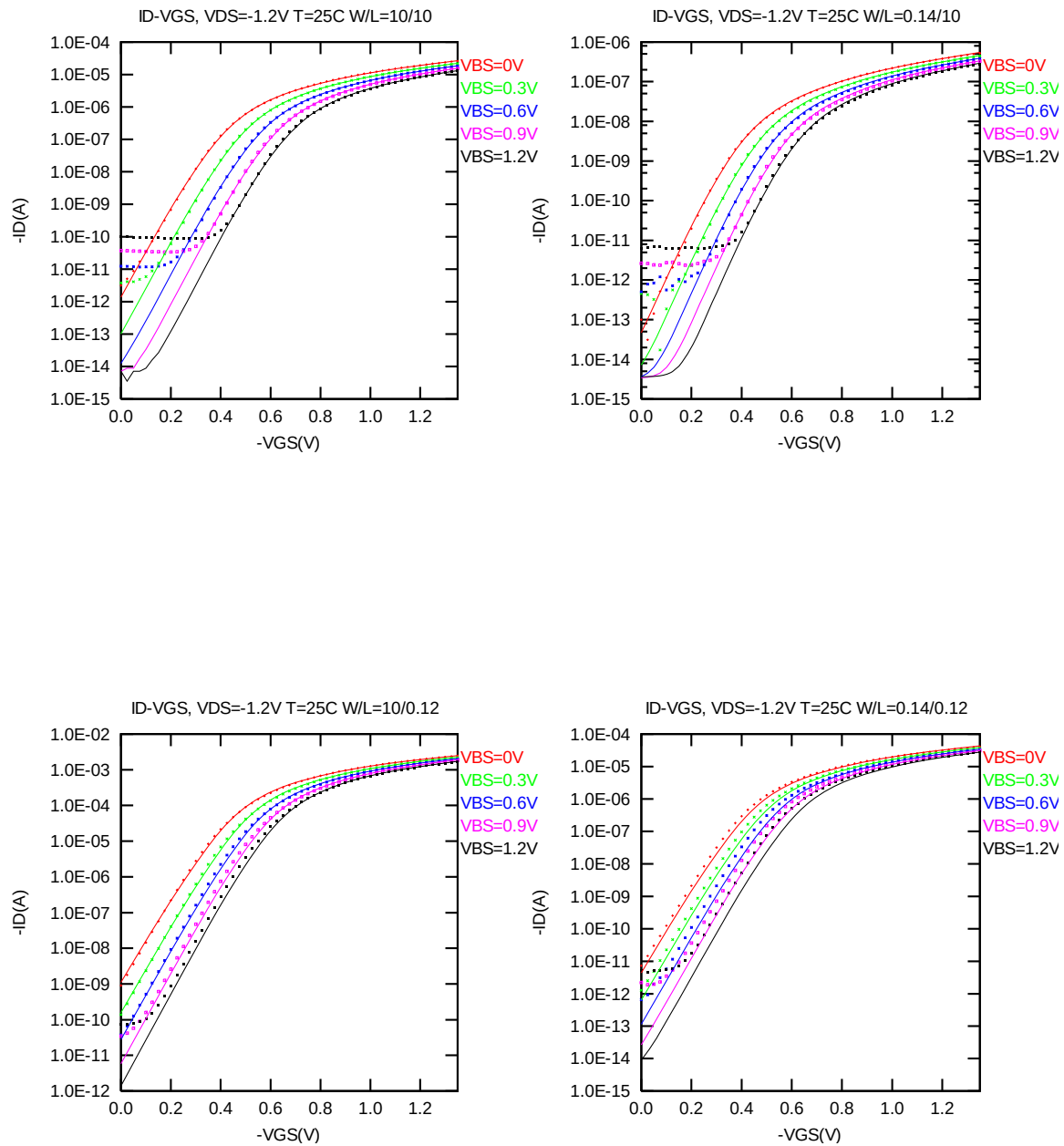


Figure 3-8 Log scaled I_D vs. V_{GS} at $V_{DS} = -1.2$ V for PMOS

G_M vs. V_{GS} Plots

The following plots show the measured and simulated transfer conductance curves in linear scale for selected devices at 25 °C .

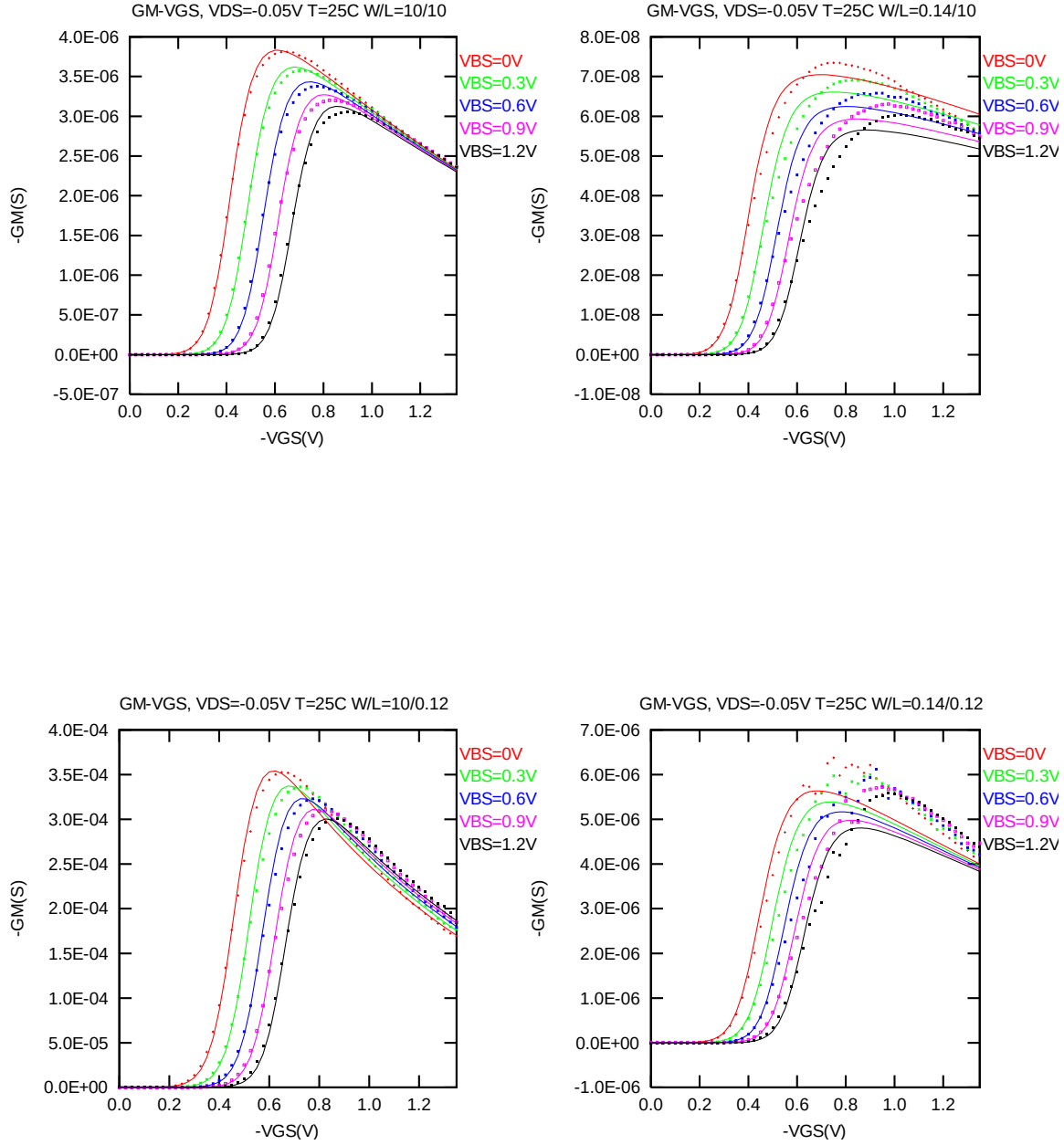


Figure 3-9 G_M vs. V_{GS} at $V_{DS} = -0.05 V$ for PMOS

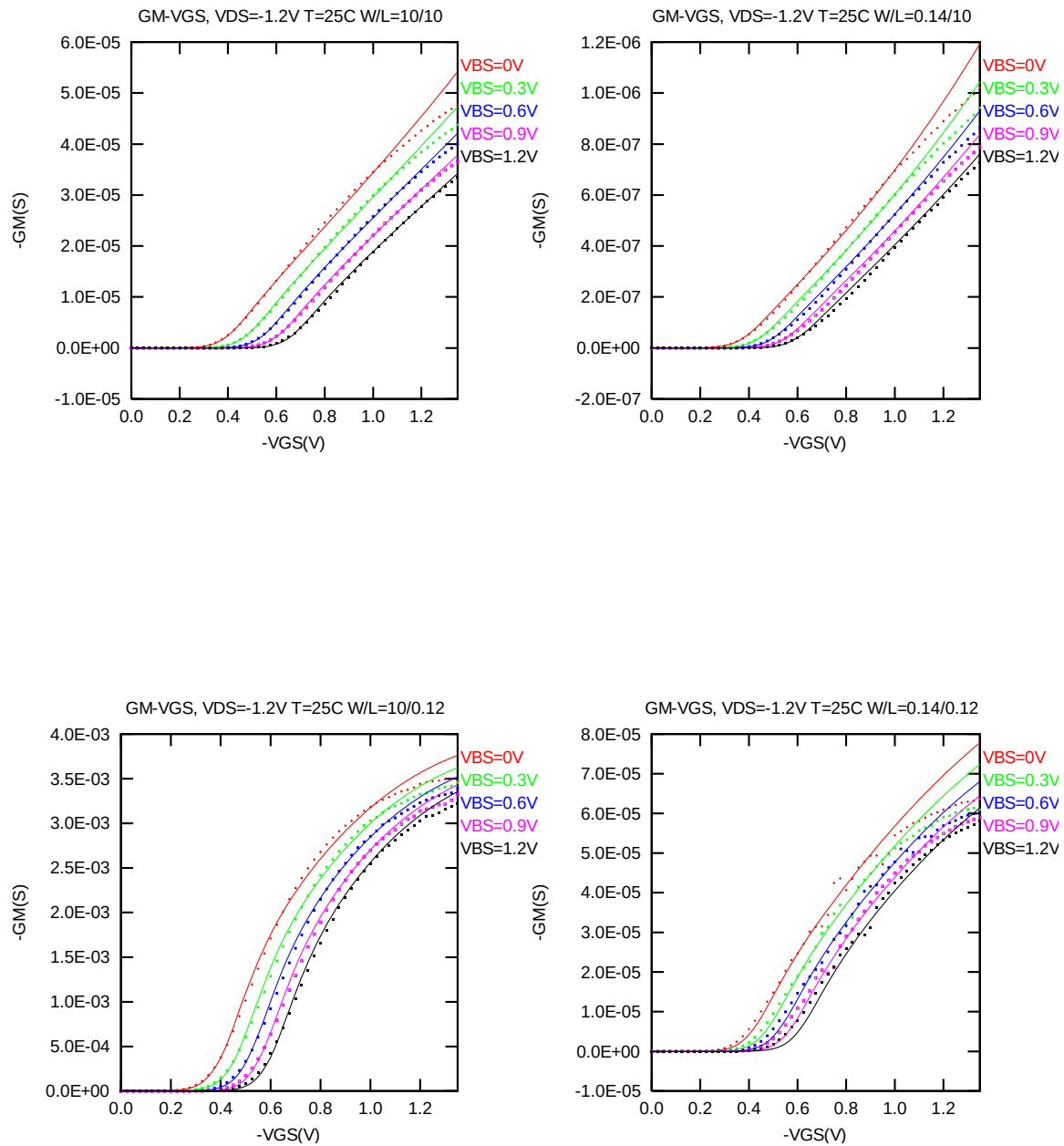


Figure 3-10 G_M vs. V_{GS} at $V_{DS} = -1.2V$ for PMOS

V_T vs. L_{DES} & V_T vs. W_{DES}

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70nA \cdot (W_{DES} \cdot 0.9) / (L_{DES} \cdot 0.9 + 0.012 \mu m)$ at $25^\circ C$.

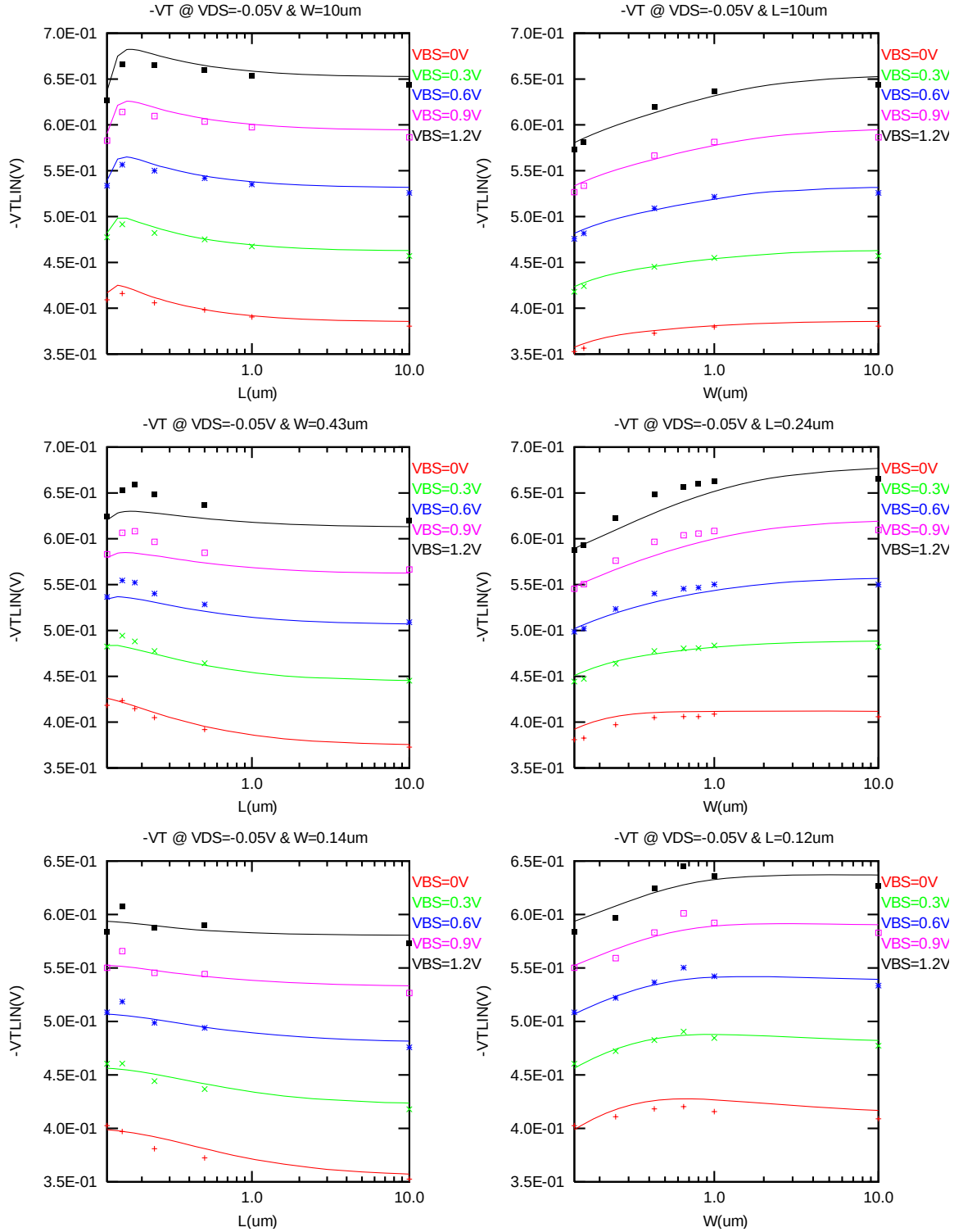


Figure 3-11 V_T vs. channel length & channel width for low drain bias ($V_{DS} = -0.05 V$) for PMOS

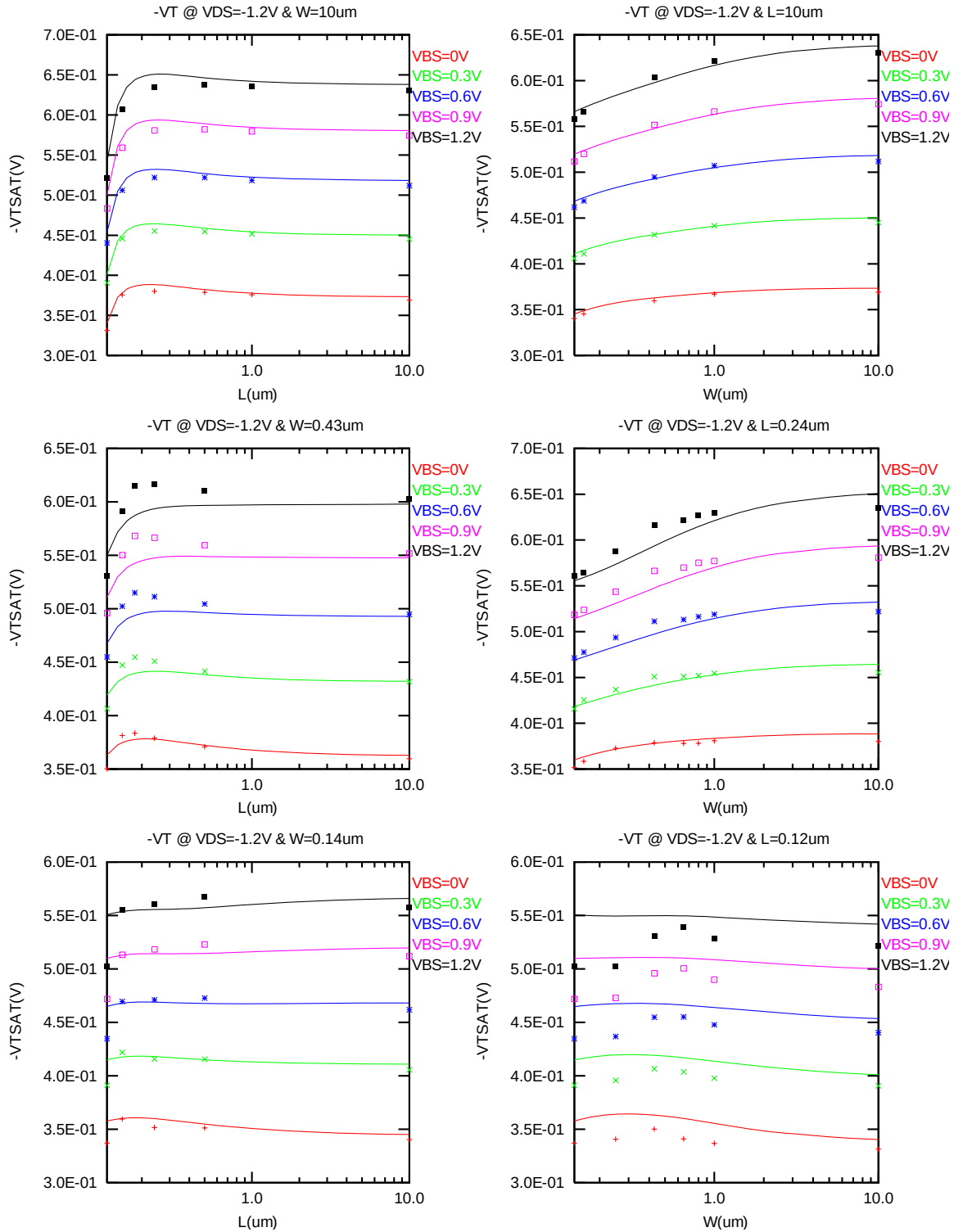


Figure 3-12 V_T vs. channel length & channel width for low drain bias ($V_{DS} = -1.2$ V) for PMOS

I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = -1.2$ for $V_{BS} = 0$ V to $V_{BS} = 1.2$ V at 25°C .

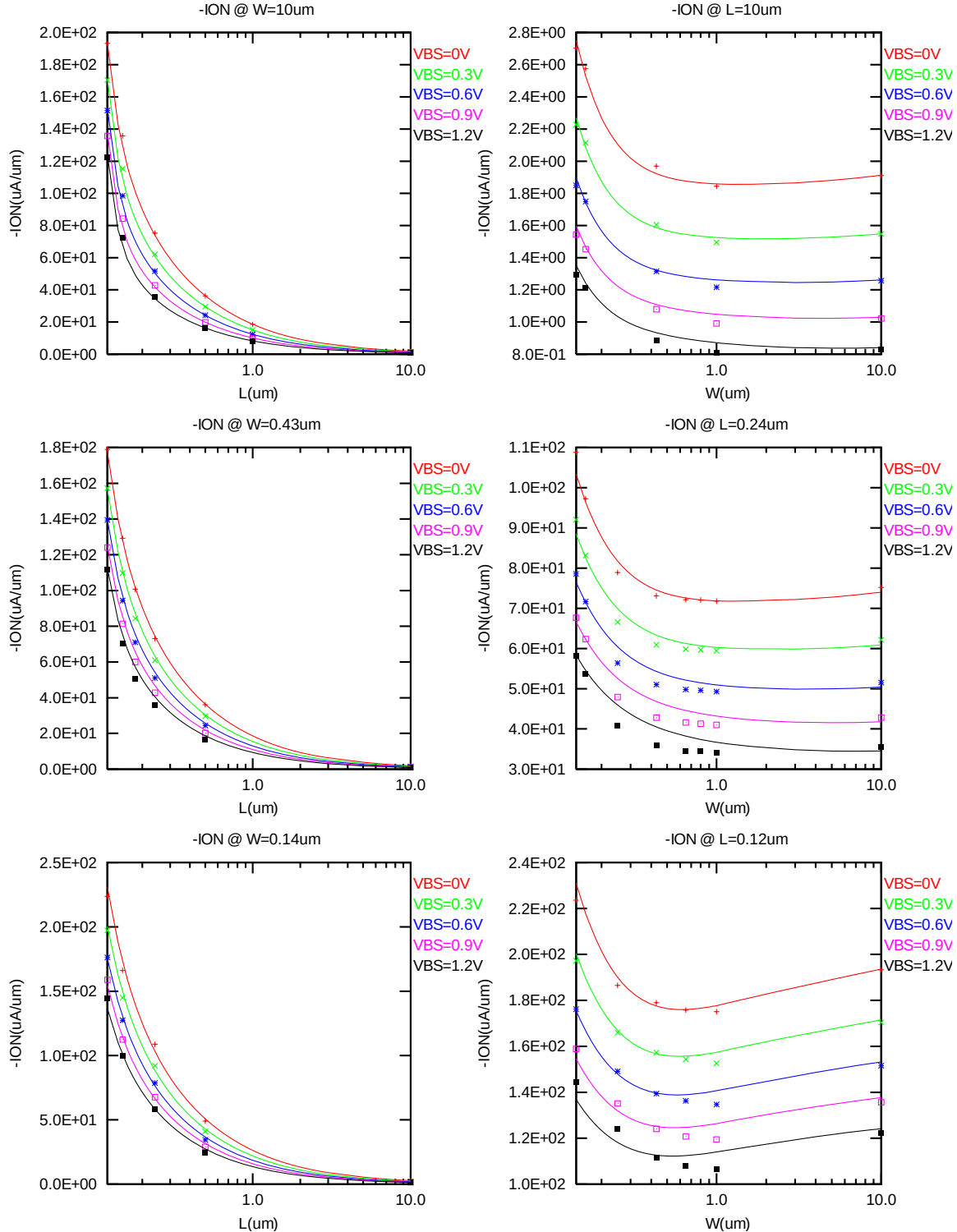


Figure 3-13 Saturation current I_{ON} vs. channel length and channel width for PMOS

V_T vs. Temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at $I_D = -70\text{nA} \cdot (W_{\text{DES}} \cdot 0.9) / (L_{\text{DES}} \cdot 0.9 + 0.012 \mu\text{m})$.

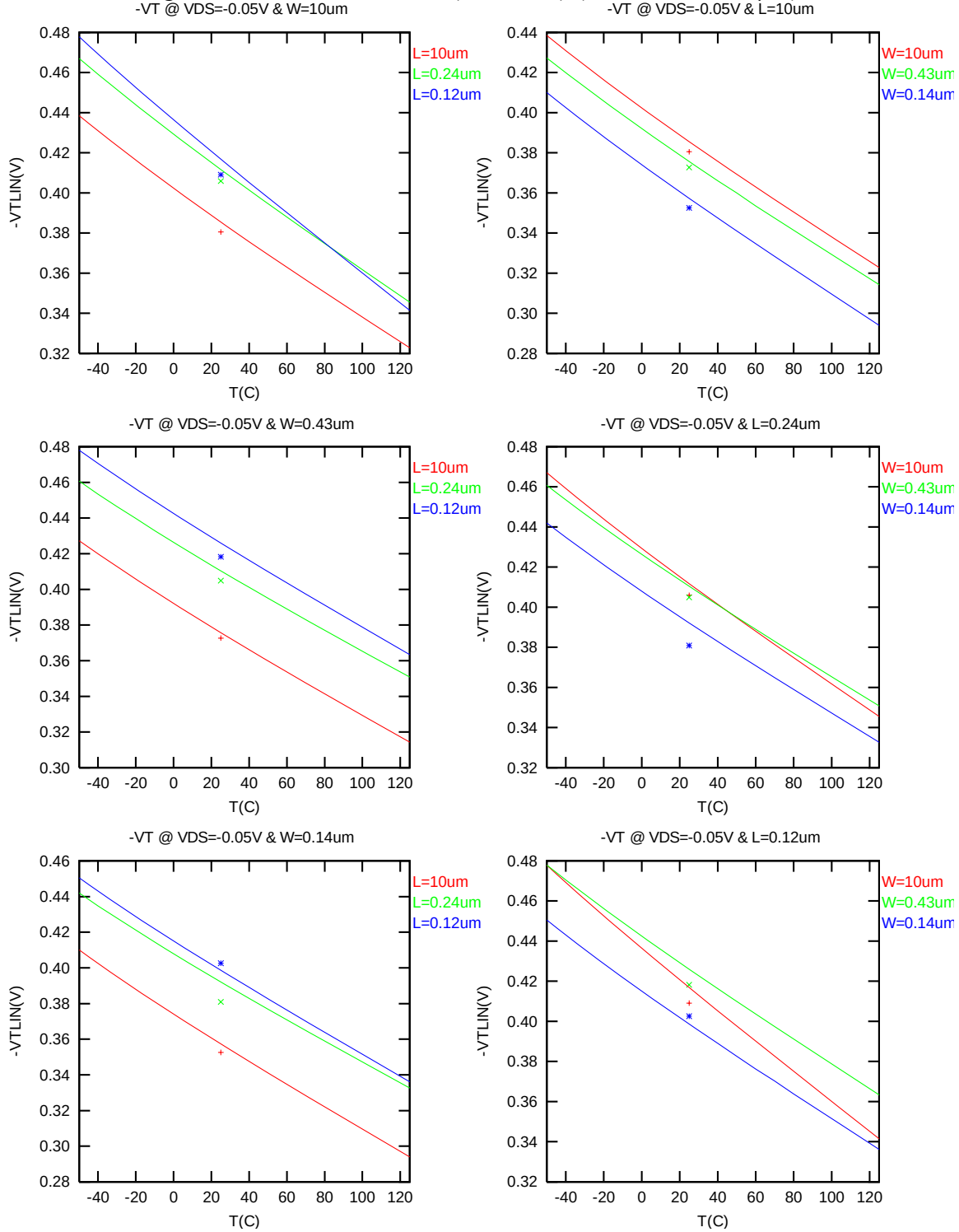


Figure 3-14 V_{TLIN} vs. temperature for PMOS

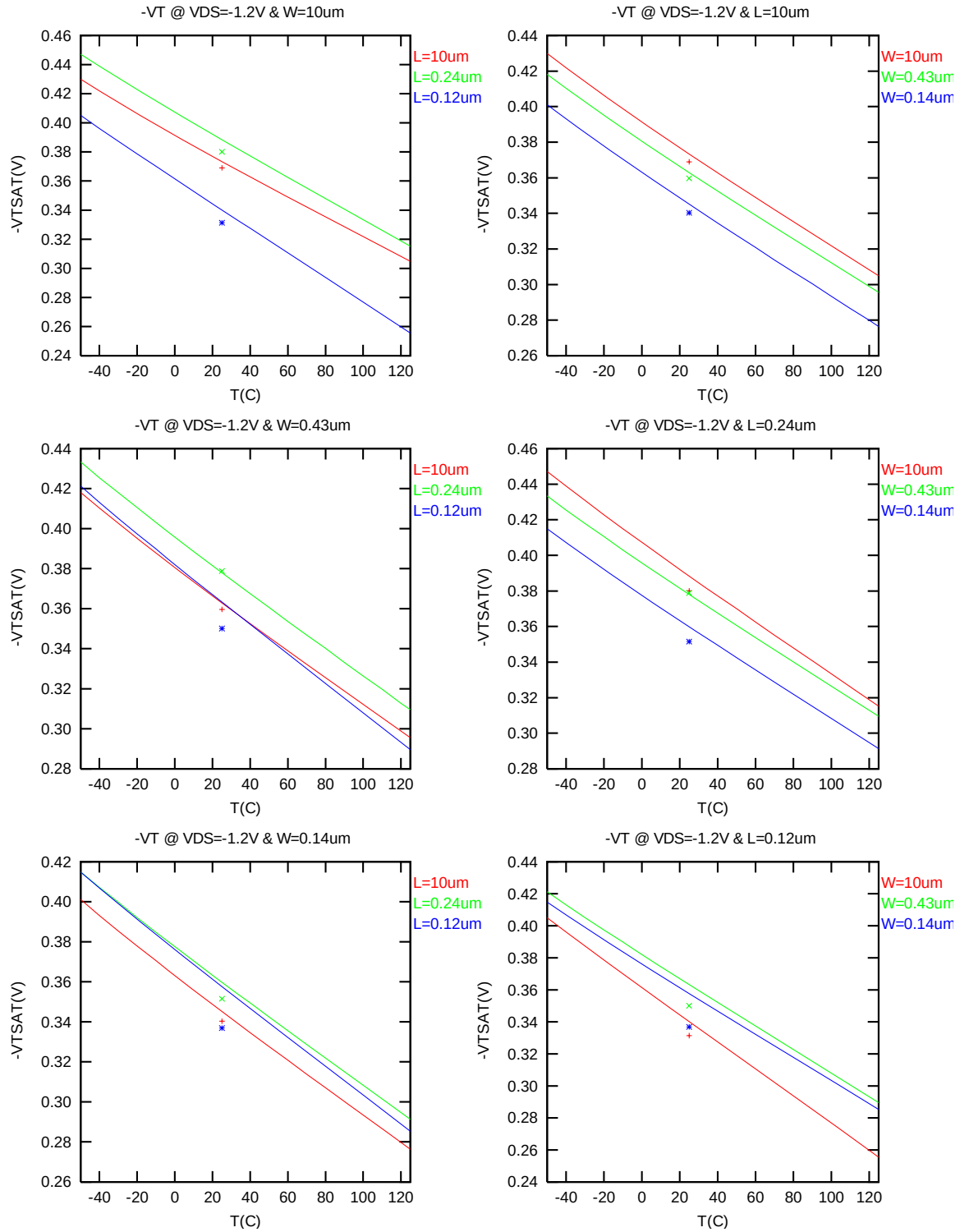


Figure 3-15 V_{TSAT} vs. temperature for PMOS

I_{ON} vs Temperature

The following plot shows the temperature dependence of I_{ON} extracted at $V_{GS}=V_{DS} = -1.2$ V, $V_{BS} = 0$ V.

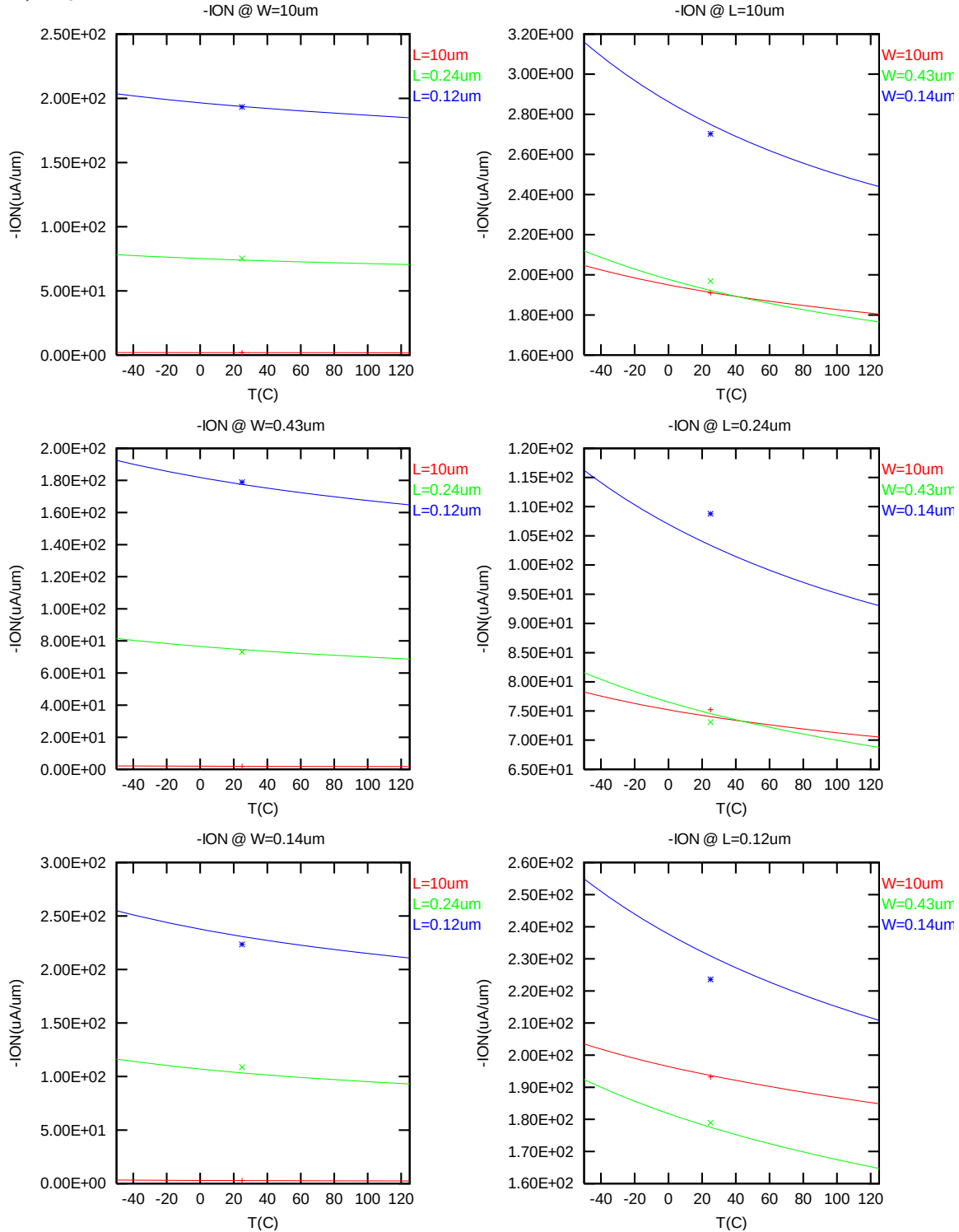


Figure 3-16 Saturation Current vs. temperature for PMOS

3.3.4 I_{GATE} Extraction plot

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ 25C

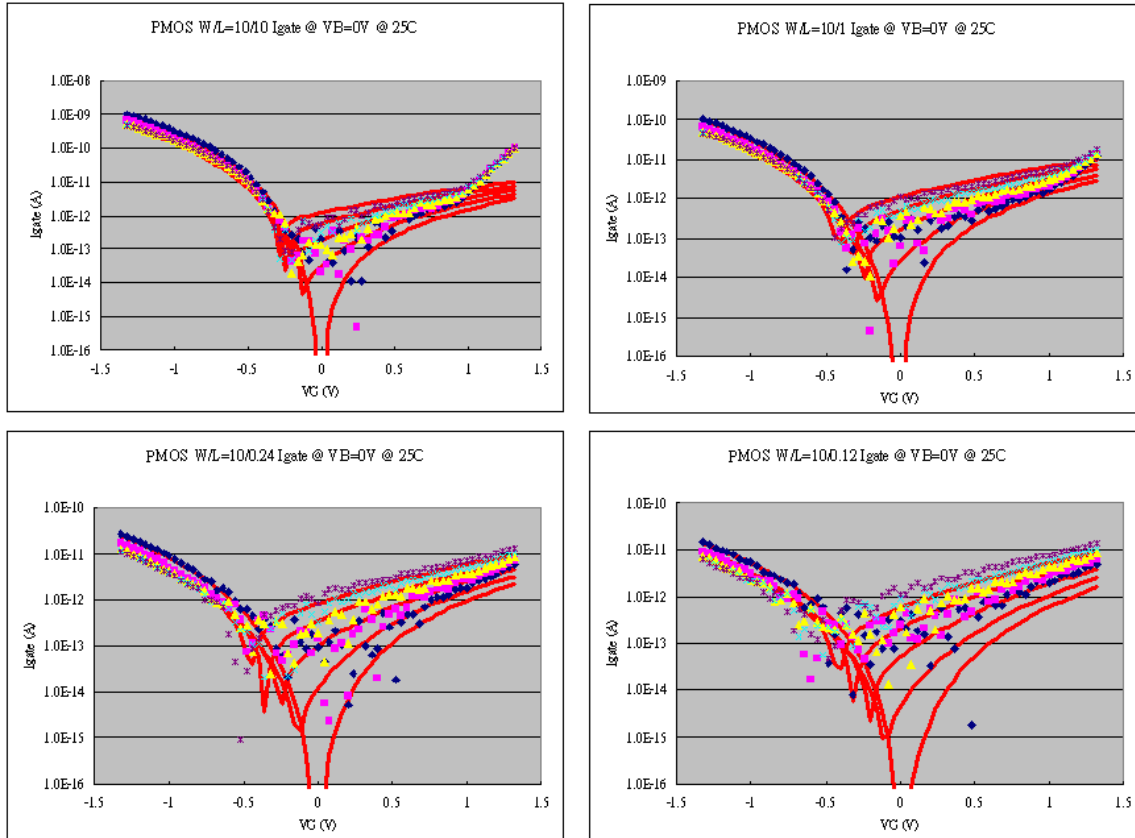


Figure 3-17 Measured and Simulated I_{GATE} for PMOS @ 25C

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ 125C

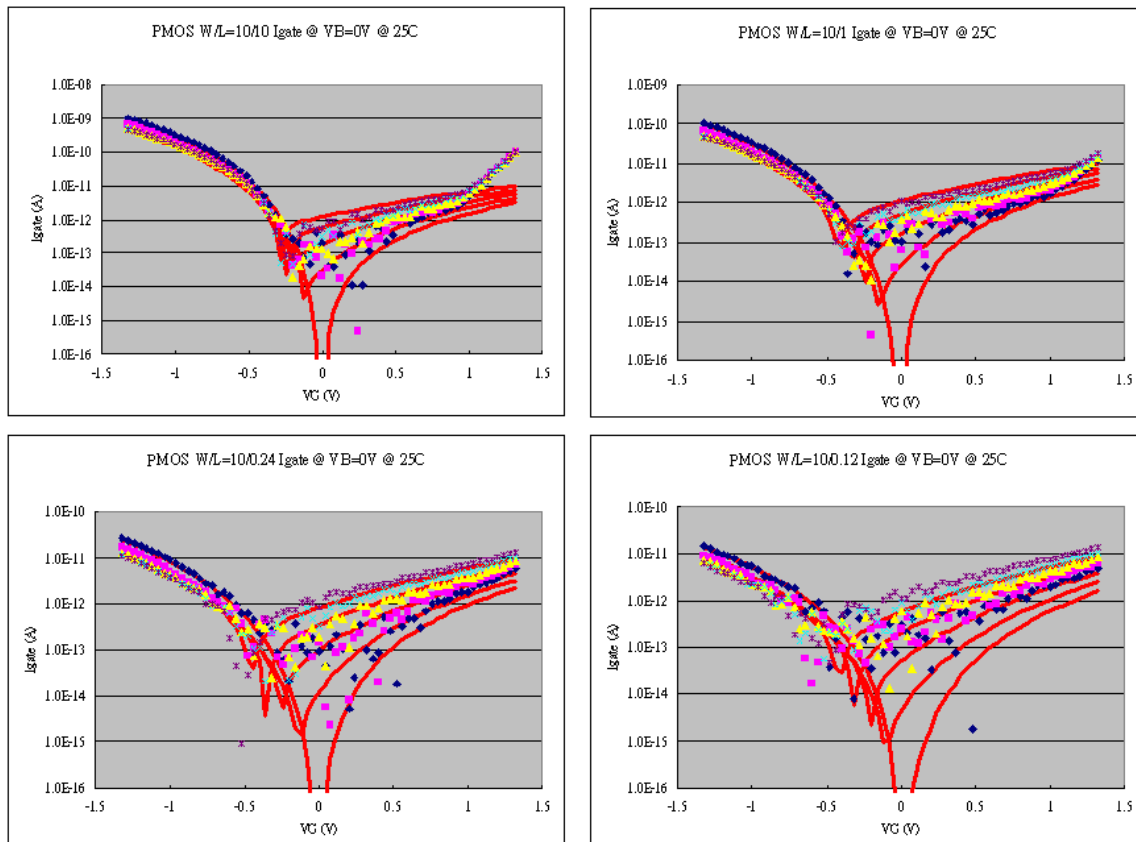


Figure 3-18 Measured and Simulated I_{GATE} for PMOS @ 125C

I_{GATE} vs V_{GS} @ $V_{BS}=0$ @ $-50C$

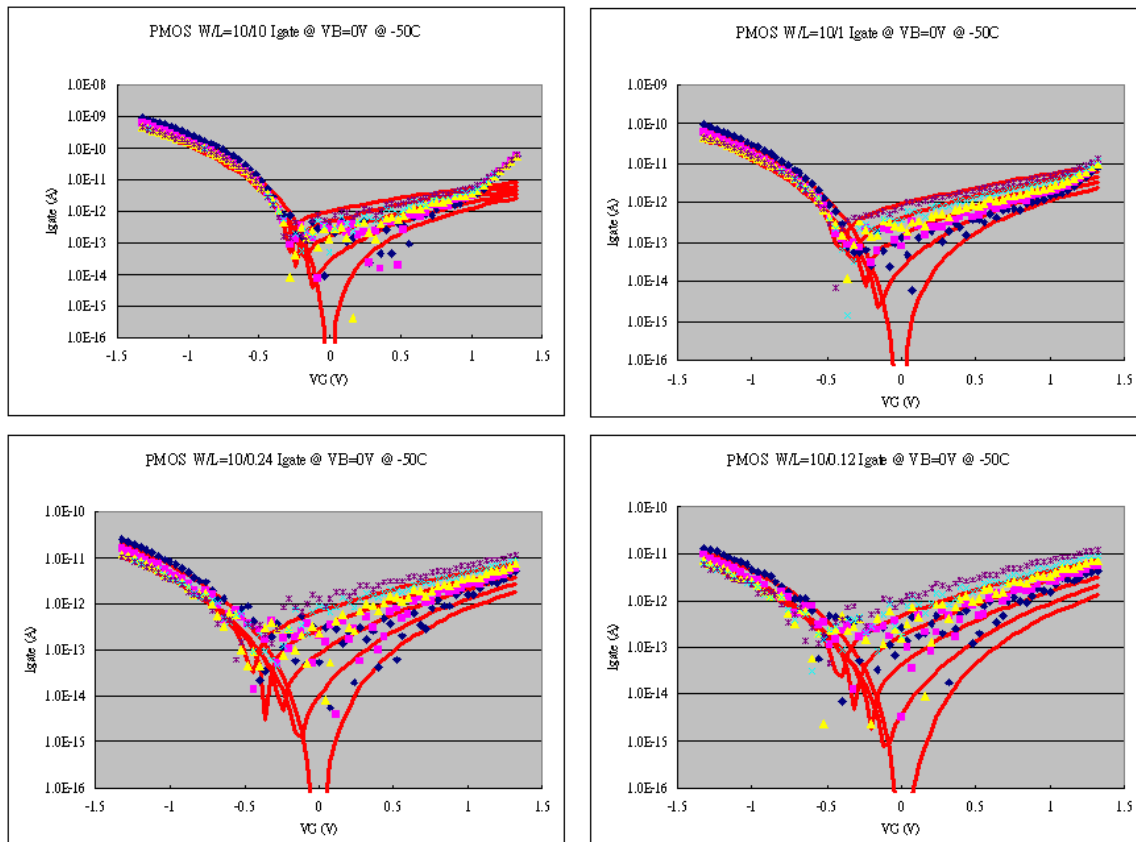


Figure 3-19 Measured and Simulated I_{GATE} for PMOS @ $-50C$

3.3.5 Model fitting accuracy

The model fitting accuracy of parameters I_D , G_M and G_{DS} is quantified using the following equation:

$$\text{error \%} = 100 * | \text{measurement} - \text{simulation} | / [(\text{measurement} + \text{simulation}) / 2]$$

Different error criteria are defined for different parameters (I_D , G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

- c. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
- d. I_D in moderate inversion region, i.e. $V_T \leq V_{GS} < V_T + V_X$
- e. I_D in strong inversion region, i.e. $V_T + V_X \leq V_{GS} \leq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$, $V_{CC} = -1.2 \text{ V}$

2) G_M and G_{DS} accuracy criteria

- f. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
- g. G_M and G_{DS} in inversion region, i.e. $V_T \leq V_{GS} \leq V_{CC}$

Table 3-5 Requirement for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T + V_X$ (moderate inversion)	$V_T + V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	$\leq 75\%$	$\leq 25\%$	$\leq 7\%$
G_M	$\leq 50\%$	$\leq 20\%$	$\leq 20\%$
G_{DS}	$\leq 100\%$	$\leq 50\%$	$\leq 50\%$

I_D Accuracy

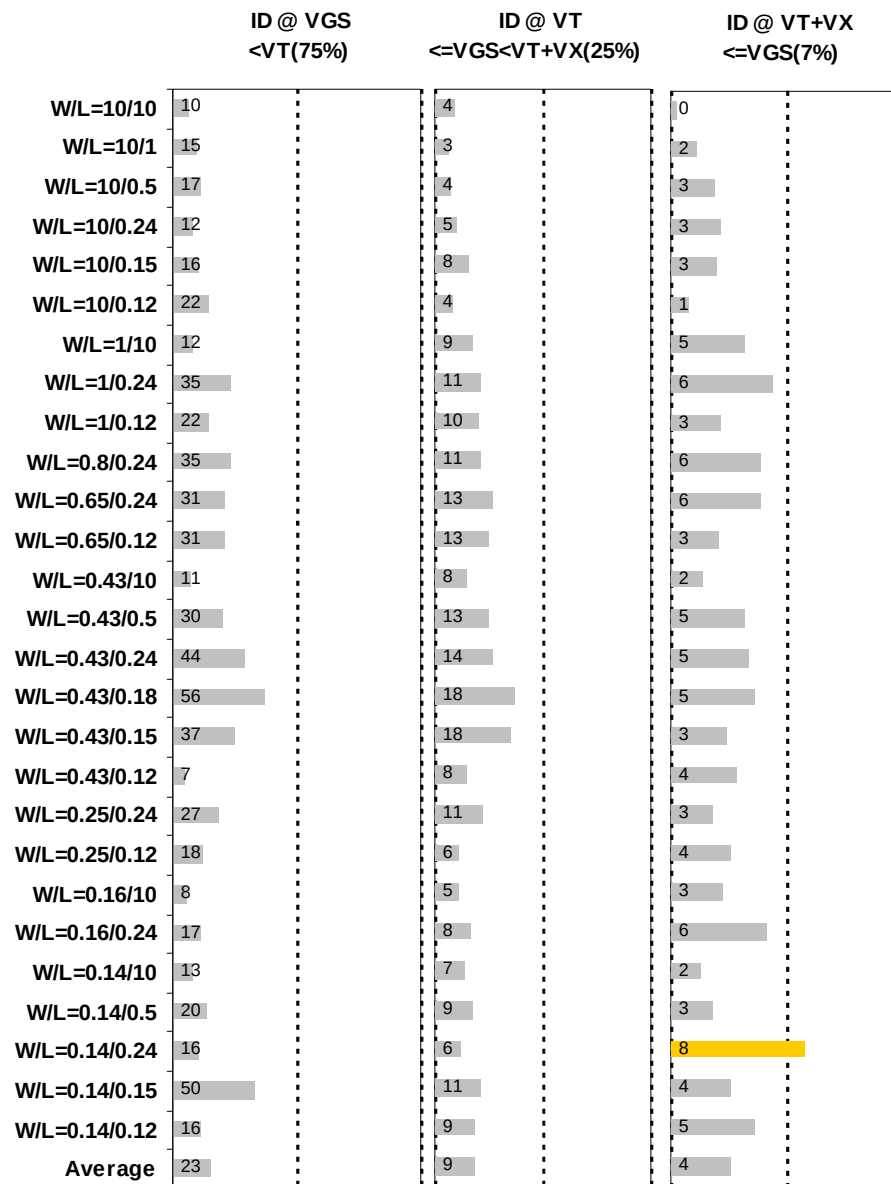


Figure 3-20 I_D accuracy at 25 °C for PMOS

G_M and G_{DS} Accuracy

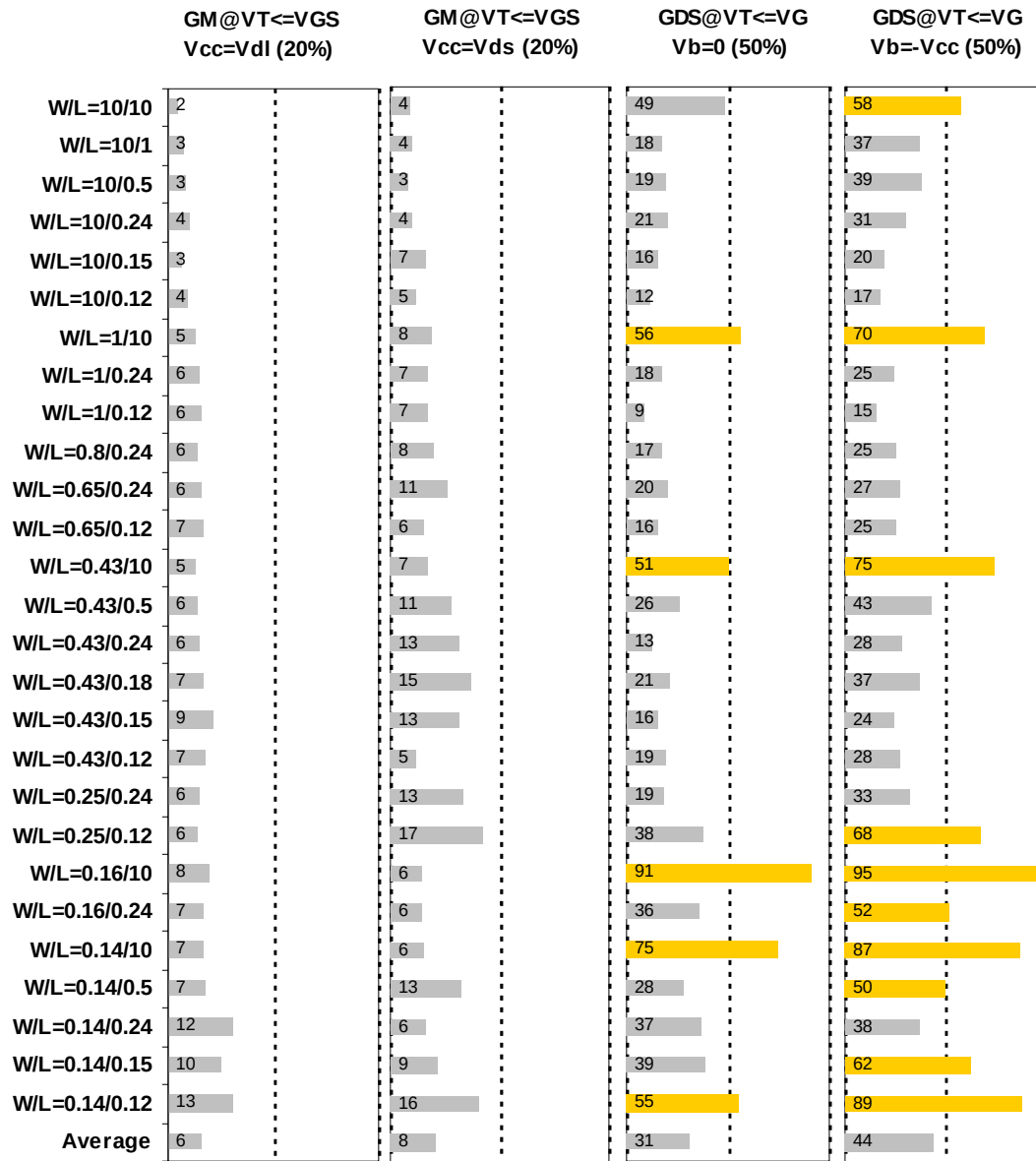


Figure 3-21 G_M and G_{DS} accuracy at 25 °C for PMOS

3.4 Parasitic source / drain junction capacitance model

3.4.1 Test devices

The following test structures were used for parameter extraction.

Table 3-6 Junction capacitor structures for PMOS

Structures	Area[m ²]	Perimeter[m]	Description
Bottom area junction capacitance	3.60E-08	7.60E-04	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.70E-08	1.83E-02	Poly finger structure

3.4.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 3-7 Junction capacitor measurement conditions for PMOS

V _{BIAS} [V]	VH: nwell & VL: p+ From 0 to 1.3 V in 0.05 V step
Frequency [kHz]	100
V _{OSC} [mV]	50
Temp [°C]	-50,25,125

3.4.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 3-8 Extracted parameter from junction capacitor measurement for PMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	Unit	Value	Parameter	Unit	Value	Parameter	Unit	Value
CJ	F/m ²	9.69E-04	CJSW	F/m	1.18E-10	CJSWG	F/m	5.43E-10
PB	V	6.87E-01	PBSW	V	4.50E-01	PBSWG	V	5.00E-01
MJ	--	3.08E-01	MJSW	--	1.00E-01	MJSWG	--	3.91E-01
TCJ	1/K	1.00E-03	TCJSW	1/K	5.24E-04	TCJSWG	1/K	2.04E-03
TPB	V/K	1.71E-03	TPBSW	V/K	2.59E-03	TPBSWG	V/K	2.03E-03

CGG vs. VBIAS Plots

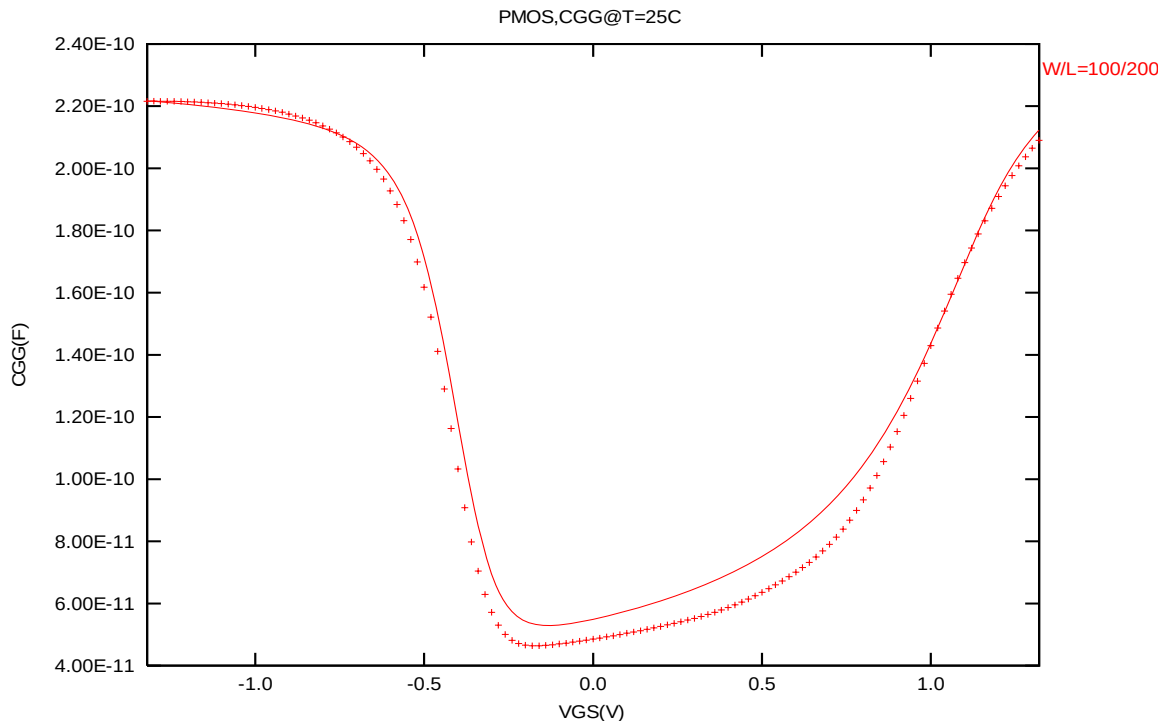


Figure 3-22 Measured and Simulated CGG curve for PMOS

CGC vs. VBIAS Plots

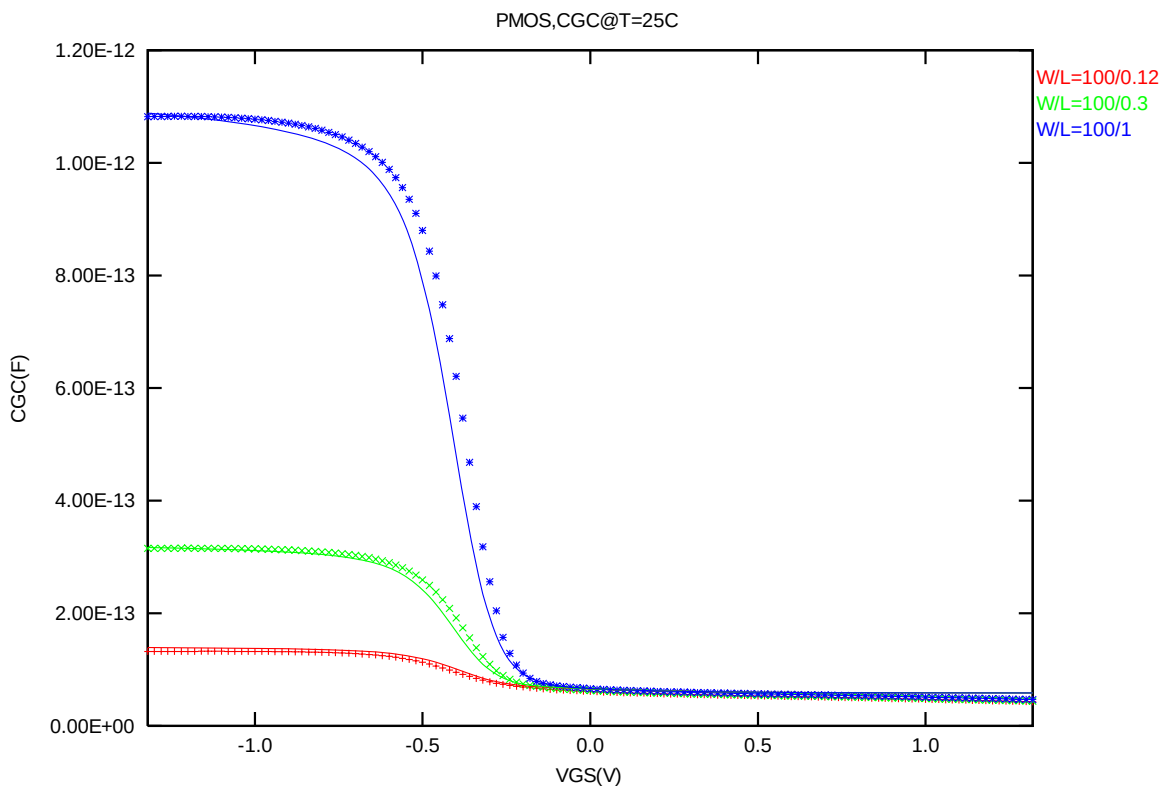


Figure 3-23 Measured and Simulated CGC curve for PMOS

CJ, CJSW vs. VBIAS Plots

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

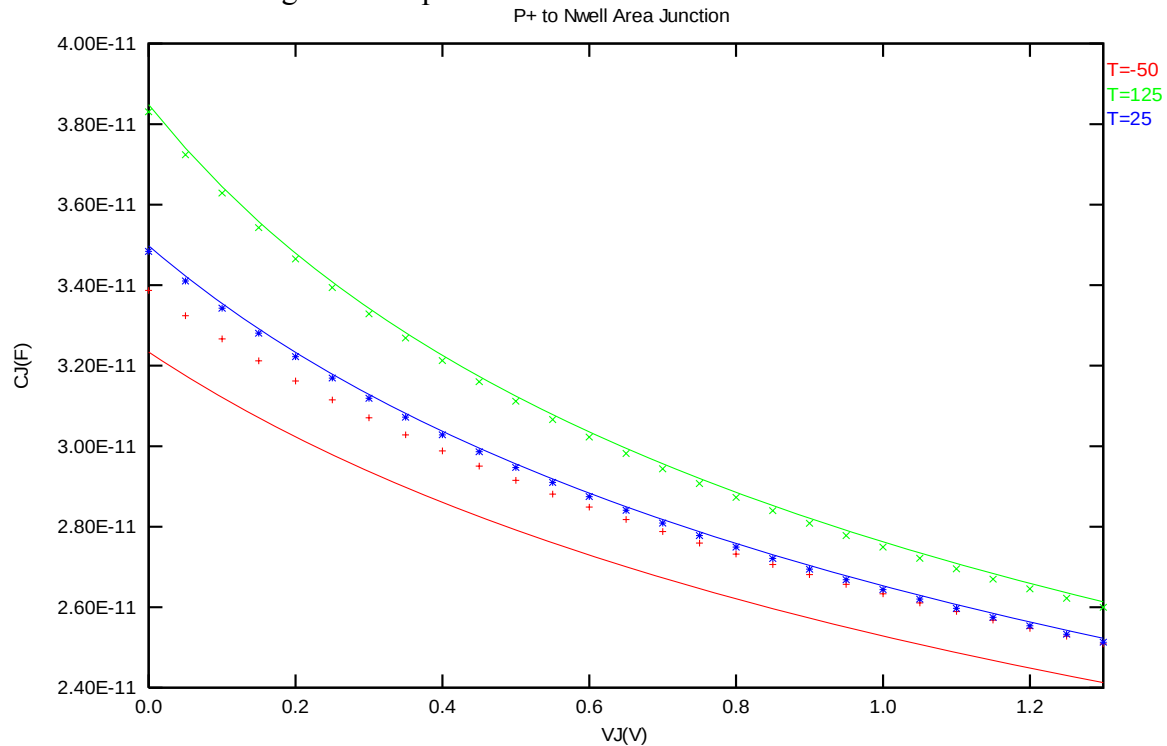


Figure 3-24 Bottom area junction capacitance at -50 °C, 25 °C and 125 °C for PMOS

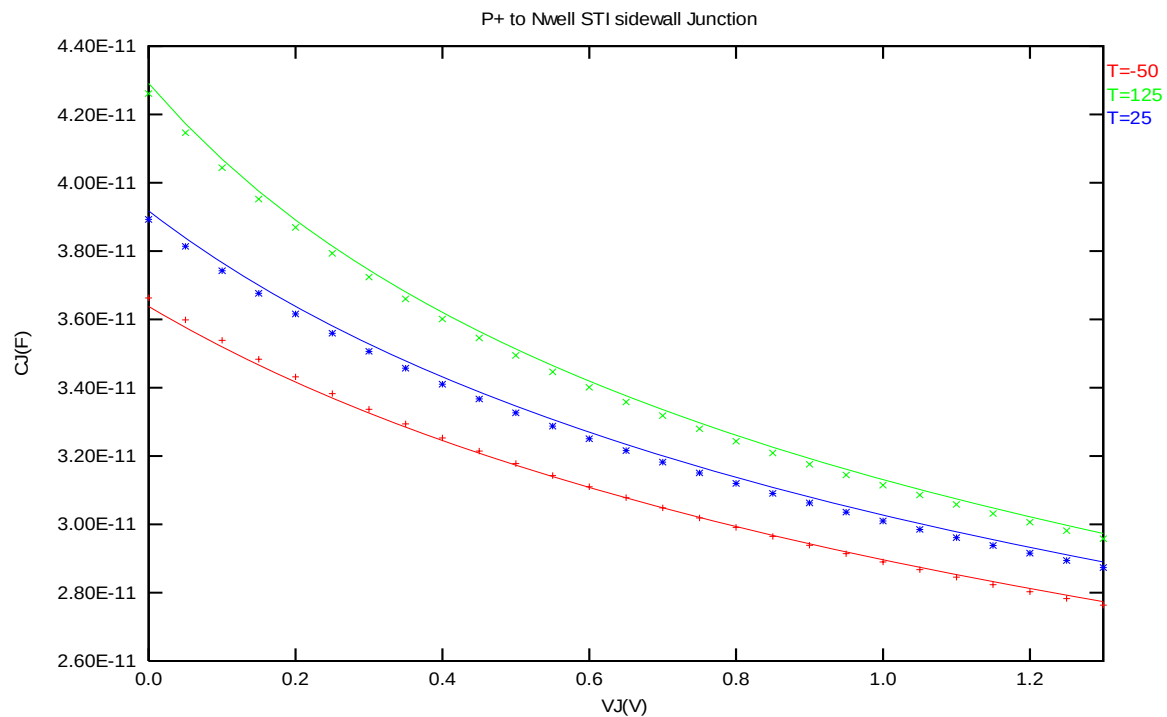


Figure 3-25 STI bounded sidewall junction capacitance at -50 °C, 25 °C and 125 °C for PMOS

4 Dynamic Performance

4.1 Test Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 4-1 Ring Oscillator (Inverter) structure description for SA=0.34 μ m

R.O. (Inverter) Structure description		
Contact size	0.16 μ m X 0.16 μ m	
Polysilicon gate to contact spacing	0.12 μ m	
Contact to diffusion edge spacing	0.06 μ m	
Without RC loading	R = 0 ohm	C = 0 fF
Inverter		
Logic gate type	Inverter	
Channel width for n-ch MOSFET	2.5 μ m X 2	
Channel length for n-ch MOSFET	0.12 μ m	
Channel width for p-ch MOSFET	3.75 μ m X 2	
Channel Length for p-ch MOSFET	0.12 μ m	

The simulation results at different temperatures and using different worst-case models are listed below.

Table 4-2 Ring Oscillator simulation results (ps/gate)

Inverter, R=0, C=0f, Fanout=1, Vcc=1.2V				Inverter, R=0, C=0f, Fanout=1, T=25C			
Temp(C)	TT	FF	SS	Vcc(V)	TT	FF	SS
-50	16.192	13.34	20.568	0.90	30.469	23	43.175
0	17.089	14.14	21.618	1.00	24.262	19.068	32.697
25	17.501	14.518	22.083	1.10	20.265	16.412	26.313
85	18.399	15.358	23.071	1.20	17.501	14.518	22.083
125	18.916	15.858	23.612	1.32	15.158	12.868	18.614

5 Worst case model

5.1 Parameter Variations

Table 5-1 Corner parameters

	n_12_spl110e	TT	SS	FF	SNFP	FNSP
1	xl	-6.00E-09	+3.45E-09	-7.40E-10	+1.30E-09	-6.20E-10
2	xw	1.40E-08	-3.00E-08	+1.27E-08	-1.00E-08	+7.47E-09
3	tox	2.64E-09	+5.28E-11	-5.28E-11	+0.00E+00	+0.00E+00
4	wwl	8.25E-25	-1.27E-22	+9.83E-23	-4.87E-23	+4.97E-23
5	cgso	2.65E-10	-10.0%	+10.0%	+0.0%	+0.0%
6	cgdo	2.65E-10	-10.0%	+10.0%	+0.0%	+0.0%
7	vth0	3.15E-01	+2.96E-02	-3.00E-02	+1.89E-02	-1.89E-02
8	pvth0	0.00E+00	+2.11E-15	-1.67E-15	+9.40E-16	-8.70E-16
9	k3	-2.66E+00	+1.74E+00	-2.10E+00	+9.10E-01	-9.70E-01
10	nlx	1.47E-07	+1.21E-08	-1.42E-08	+7.05E-09	-7.48E-09
11	u0	4.40E-02	-2.13E-03	+1.77E-03	-1.09E-03	+1.01E-03
12	rdsw	1.15E+01	+2.96E+00	-2.96E+00	+1.48E+00	-1.48E+00
13	ags	7.73E-01	+1.10E-01	-1.05E-01	+5.26E-02	-5.04E-02
14	wags	8.80E-09	+2.03E-09	-5.05E-09	+1.65E-09	-2.58E-09
15	pags	-5.55E-14	+3.00E-14	+0.00E+00	+1.00E-14	+0.00E+00
16	vsat	1.03E+05	-1.81E+03	+1.47E+03	-1.73E+03	+1.70E+03
17	pvsat	-7.00E-10	+5.80E-10	-3.14E-10	+2.27E-10	-2.09E-10
18	wu0	0.00E+00	-4.83E-11	+0.00E+00	-1.50E-11	+0.00E+00
19	cgdl	5.80E-11	-10.0%	+10.0%	+0.0%	+0.0%
20	cgs1	5.80E-11	-10.0%	+10.0%	+0.0%	+0.0%
21	cj	7.20E-04	+10.0%	-10.0%	+10.0%	-10.0%

22	cjsw	9.94E-11	+10.0%	-10.0%	+10.0%	-10.0%
23	cjswg	3.76E-10	+10.0%	-10.0%	+10.0%	-10.0%

	p_12_spl110e	TT	SS	FF	SNFP	FNSP
1	xl	-6.00E-09	+4.50E-09	-1.82E-09	-1.21E-09	+1.85E-09
2	xw	1.70E-08	-2.72E-08	+1.67E-08	+9.92E-09	-4.40E-09
3	tox	2.84E-09	+5.68E-11	-5.68E-11	+0.00E+00	+0.00E+00
4	wwl	-3.79E-22	+2.99E-23	+3.23E-23	+1.85E-23	+6.12E-23
5	cgso	1.75E-10	-10.0%	+10.0%	+0.0%	+0.0%
6	cgdo	1.75E-10	-10.0%	+10.0%	+0.0%	+0.0%
7	vth0	-3.74E-01	-2.60E-02	+2.65E-02	+1.86E-02	-1.84E-02
8	pvth0	-4.45E-16	-3.18E-15	+2.75E-15	+1.39E-15	-1.35E-15
9	k3	-4.65E+00	+3.54E+00	-4.15E+00	-1.91E+00	+1.88E+00
10	nlx	2.31E-08	+4.81E-09	-6.00E-09	-2.68E-09	+2.55E-09
11	u0	9.36E-03	-4.57E-04	+3.71E-04	+2.36E-04	-2.60E-04
12	rdsw	2.76E+02	+5.12E+01	-5.12E+01	-2.56E+01	+2.56E+01
13	ags	7.97E-01	+1.46E-01	-1.44E-01	-6.68E-02	+6.74E-02
14	lags	1.87E-07	+2.19E-07	-1.58E-07	-8.22E-08	+1.03E-07
15	vsat	2.55E+05	+1.00E+04	-8.00E+03	-8.00E+03	+5.00E+03
16	pvsat	-1.50E-08	+1.15E-10	+1.13E-09	+1.13E-09	+1.05E-11
17	wags	0.00E+00	+1.16E-08	-9.92E-09	-4.25E-09	+6.48E-09
18	pags	0.00E+00	+1.30E-14	-2.66E-14	-1.87E-14	+2.50E-15
19	wu0	0.00E+00	-3.05E-11	+0.00E+00	+0.00E+00	-3.05E-11
20	cgdl	6.10E-11	-10.0%	+10.0%	+0.0%	+0.0%
21	cgs1	6.10E-11	-10.0%	+10.0%	+0.0%	+0.0%
22	cj	9.69E-04	+10.0%	-10.0%	-10.0%	+10.0%
23	cjsw	1.18E-10	+10.0%	-10.0%	-10.0%	+10.0%

24	cjswg	5.43E-10	+10.0%	-10.0%	-10.0%	+10.0%
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	n_12bpw_spl110e	TT	SS	FF	SNFP	FNSP
1	xl	-6.00E-09	+3.45E-09	-7.40E-10	+1.30E-09	-6.20E-10
2	xw	1.40E-08	-3.00E-08	+1.27E-08	-1.00E-08	+7.47E-09
3	tox	2.64E-09	+5.28E-11	-5.28E-11	+0.00E+00	+0.00E+00
4	wwl	8.25E-25	-1.27E-22	+9.83E-23	-4.87E-23	+4.97E-23
5	cgso	2.65E-10	-10.0%	+10.0%	+0.0%	+0.0%
6	cgdo	2.65E-10	-10.0%	+10.0%	+0.0%	+0.0%
7	vth0	3.15E-01	+2.96E-02	-3.00E-02	+1.89E-02	-1.89E-02
8	pvth0	0.00E+00	+2.11E-15	-1.67E-15	+9.40E-16	-8.70E-16
9	k3	-2.66E+00	+1.74E+00	-2.10E+00	+9.10E-01	-9.70E-01
10	nlx	1.47E-07	+1.21E-08	-1.42E-08	+7.05E-09	-7.48E-09
11	u0	4.40E-02	-2.13E-03	+1.77E-03	-1.09E-03	+1.01E-03
12	rdsw	1.15E+01	+2.96E+00	-2.96E+00	+1.48E+00	-1.48E+00
13	ags	7.73E-01	+1.10E-01	-1.05E-01	+5.26E-02	-5.04E-02
14	wags	8.80E-09	+2.03E-09	-5.05E-09	+1.65E-09	-2.58E-09
15	pags	-5.55E-14	+3.00E-14	+0.00E+00	+1.00E-14	+0.00E+00
16	vsat	1.03E+05	-1.81E+03	+1.47E+03	-1.73E+03	+1.70E+03
17	pvsat	-7.00E-10	+5.80E-10	-3.14E-10	+2.27E-10	-2.09E-10
18	wu0	0.00E+00	-4.83E-11	+0.00E+00	-1.50E-11	+0.00E+00
19	cgdl	5.80E-11	-10.0%	+10.0%	+0.0%	+0.0%
20	cgs1	5.80E-11	-10.0%	+10.0%	+0.0%	+0.0%
21	cj	7.20E-04	+10.0%	-10.0%	+10.0%	-10.0%
22	cjsw	9.94E-11	+10.0%	-10.0%	+10.0%	-10.0%
23	cjswg	3.76E-10	+10.0%	-10.0%	+10.0%	-10.0%

Threshold voltage in linear region V_{TLIN}

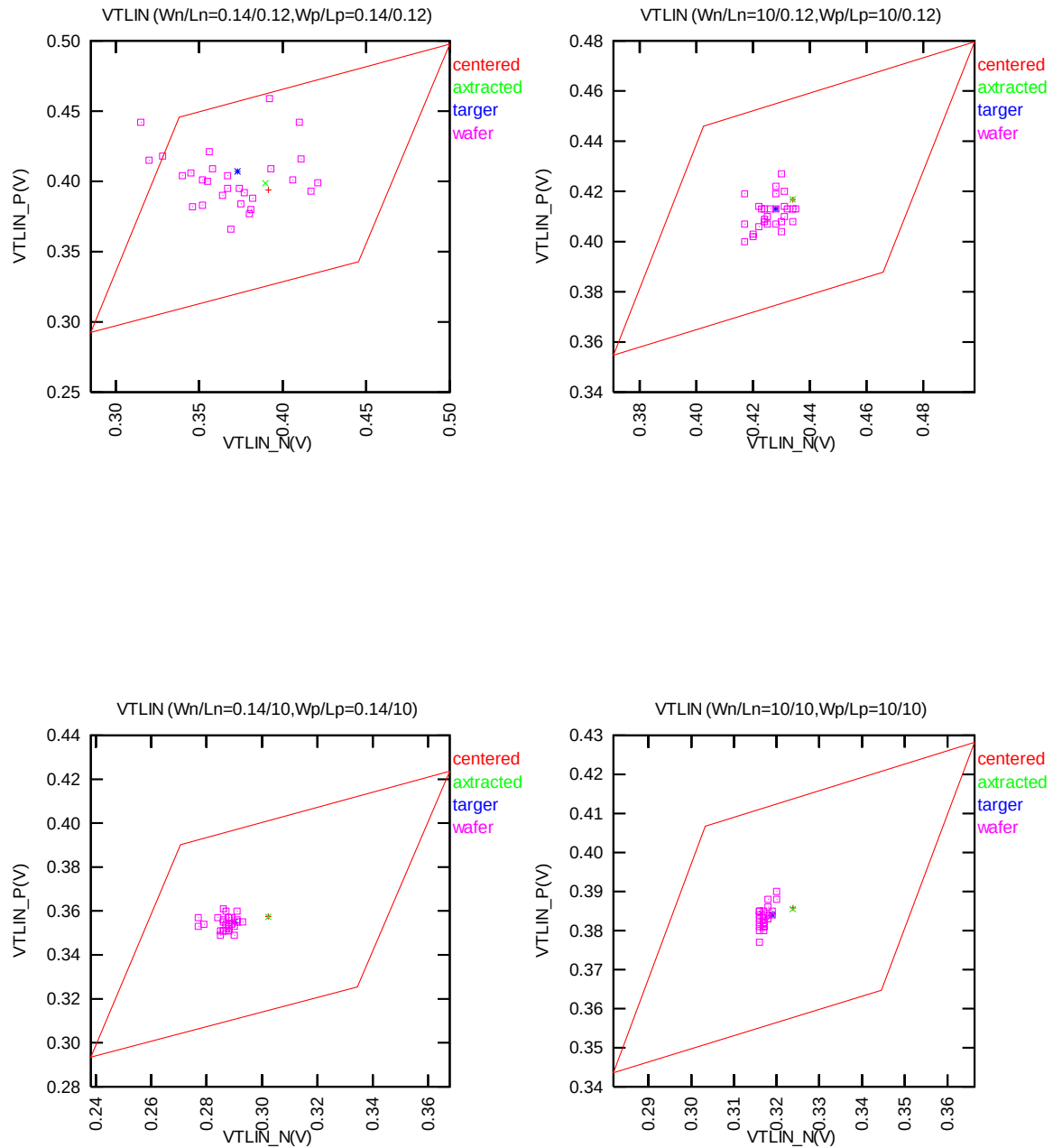


Figure 5-1 Corner graph of threshold voltage at low drain bias

Saturation Current I_{ON}

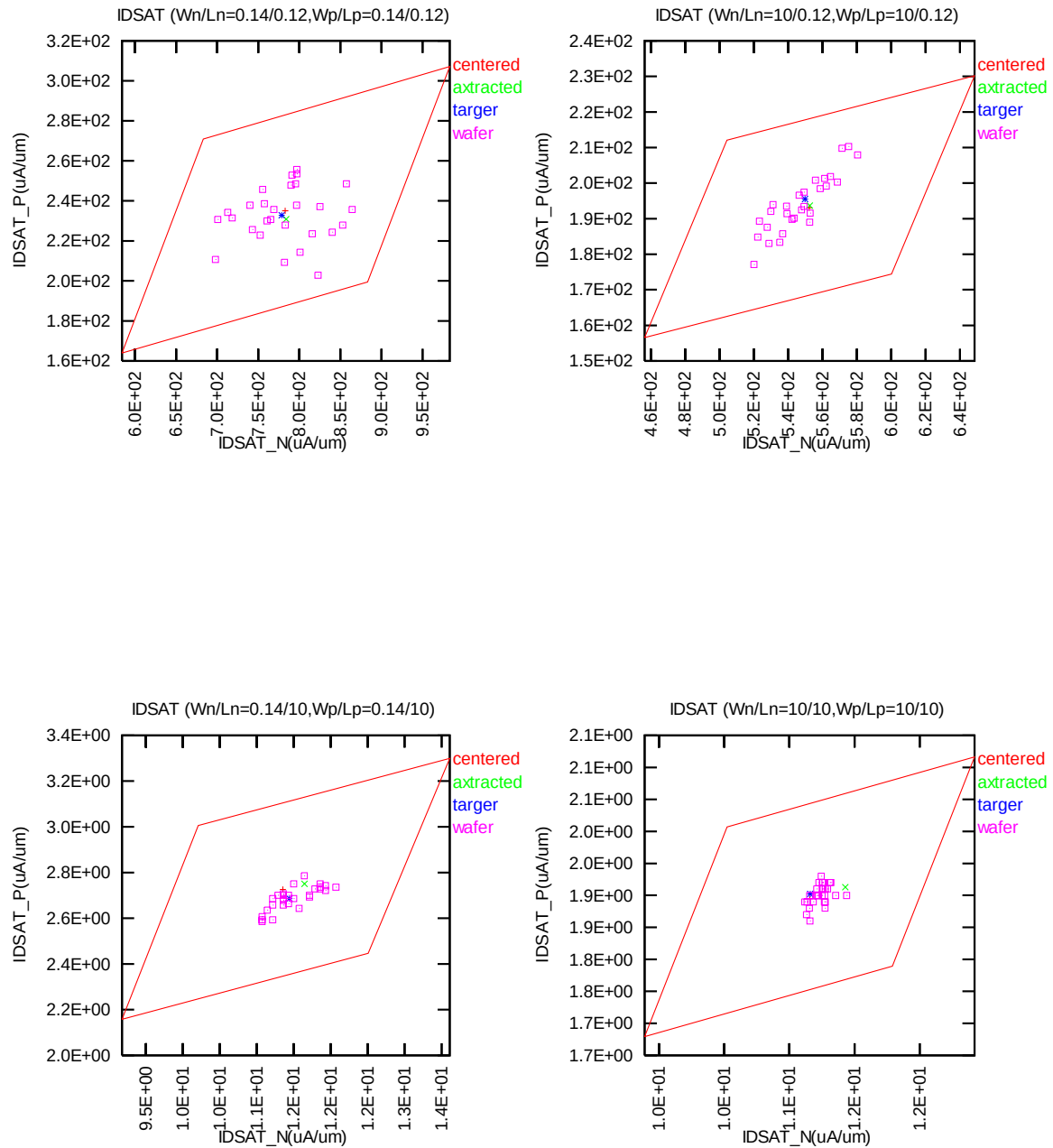


Figure 5-2 Corner graph of on-current

5.2 Wafer Data vs. Corner graph for V101

Threshold voltage in linear region V_{TLIN}

The threshold voltage extracted at $ID = 300nA * W_{DES} * 0.9 / (L_{DES} * 0.9 + 0.012 \mu m)$ for NMOS and $ID = -70nA * W_{DES} * 0.9 / (L_{DES} * 0.9 + 0.012 \mu m)$ for PMOS at 25 °C . (The dimensions are after 90% shrink and 12nm sizing channel length back.)

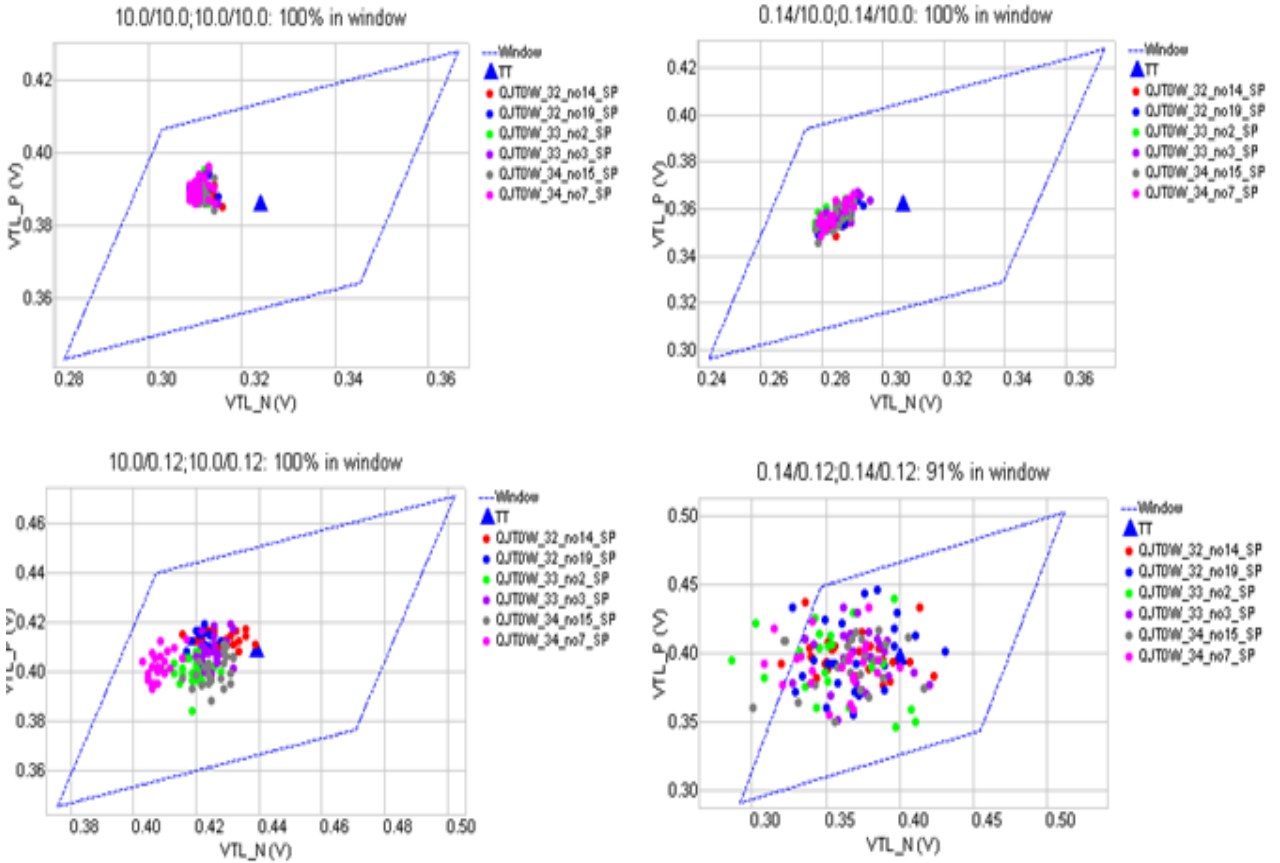


Figure 5-3 Corner graph of threshold voltage at low drain bias

Saturation Current I_{ON}

(The dimensions are after 90% shrink and 12nm sizing channel length back.)

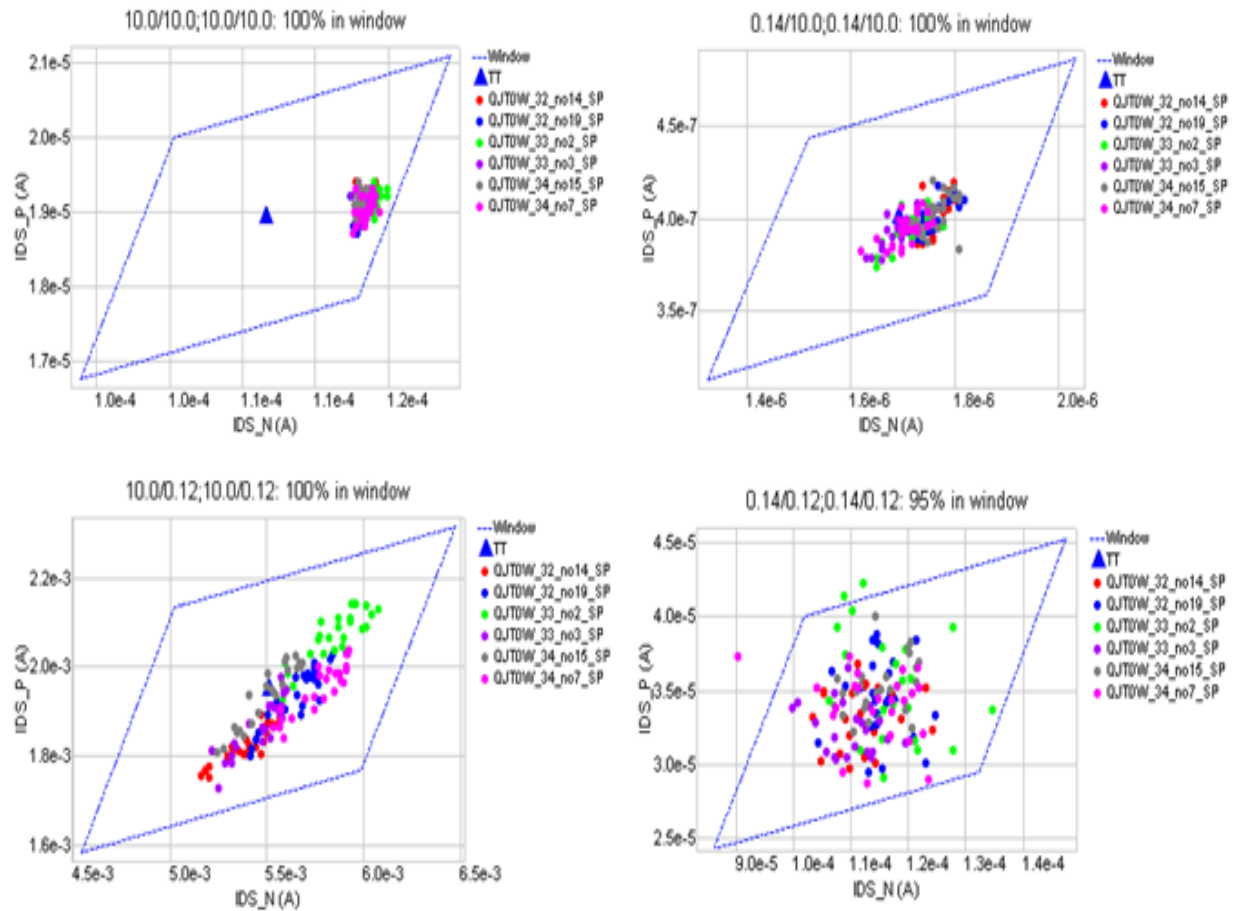


Figure 5-4 Corner graph of on-current

**DEVICE PERFORMANCE LOOK UP TABLE (SIMULATION RESULTS FROM
 HSPICE VER.05.3):**

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
VTH (V)	VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	0.384	0.367	0.324	0.302	0.257	0.228
SS	9/9	0.425	0.409	0.367	0.344	0.301	0.270
FF	9/9	0.342	0.325	0.282	0.260	0.214	0.185
SNFP	9/9	0.405	0.388	0.345	0.323	0.279	0.249
FNSP	9/9	0.363	0.346	0.303	0.281	0.236	0.206
TT	9/0.108	0.496	0.412	0.438	0.350	0.376	0.279
SS	9/0.108	0.559	0.476	0.502	0.414	0.441	0.344
FF	9/0.108	0.433	0.350	0.374	0.287	0.312	0.215
SNFP	9/0.108	0.527	0.444	0.470	0.382	0.408	0.311
FNSP	9/0.108	0.464	0.380	0.406	0.318	0.344	0.247
TT	0.144/9	0.362	0.344	0.303	0.281	0.238	0.208
SS	0.144/9	0.425	0.407	0.369	0.345	0.305	0.273
FF	0.144/9	0.298	0.282	0.239	0.217	0.172	0.143
SNFP	0.144/9	0.394	0.376	0.336	0.313	0.272	0.240
FNSP	0.144/9	0.330	0.313	0.271	0.249	0.205	0.175
TT	0.144/0.108	0.448	0.397	0.395	0.340	0.338	0.272
SS	0.144/0.108	0.557	0.510	0.503	0.452	0.446	0.385
FF	0.144/0.108	0.340	0.288	0.287	0.230	0.229	0.162
SNFP	0.144/0.108	0.502	0.453	0.449	0.395	0.392	0.328
FNSP	0.144/0.108	0.394	0.342	0.341	0.284	0.284	0.217

Table 5-2 Corner window of NMOS Vth

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-VTH (V)	-VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	0.439	0.430	0.386	0.374	0.323	0.305
SS	9/9	0.481	0.471	0.428	0.415	0.367	0.348
FF	9/9	0.397	0.388	0.343	0.332	0.280	0.262
SNFP	9/9	0.418	0.409	0.365	0.353	0.301	0.284
FNSP	9/9	0.460	0.451	0.407	0.395	0.345	0.327
TT	9/0.108	0.469	0.395	0.408	0.330	0.332	0.245
SS	9/0.108	0.530	0.462	0.471	0.399	0.399	0.318
FF	9/0.108	0.406	0.332	0.344	0.266	0.267	0.178
SNFP	9/0.108	0.438	0.362	0.376	0.297	0.299	0.211
FNSP	9/0.108	0.500	0.429	0.439	0.364	0.365	0.282
TT	0.144/9	0.411	0.402	0.359	0.346	0.295	0.278
SS	0.144/9	0.475	0.466	0.425	0.411	0.364	0.345
FF	0.144/9	0.347	0.338	0.293	0.281	0.227	0.211
SNFP	0.144/9	0.379	0.370	0.326	0.314	0.261	0.244
FNSP	0.144/9	0.443	0.434	0.392	0.379	0.330	0.311
TT	0.144/0.108	0.443	0.399	0.391	0.341	0.328	0.268
SS	0.144/0.108	0.544	0.506	0.494	0.450	0.435	0.381
FF	0.144/0.108	0.343	0.296	0.289	0.236	0.222	0.158
SNFP	0.144/0.108	0.393	0.347	0.340	0.289	0.275	0.213
FNSP	0.144/0.108	0.494	0.451	0.443	0.394	0.381	0.322

Table 5-3 Corner window of PMOS Vth

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IDSAT (μA/um)	VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	1.37	16.20	1.42	12.36	1.48	9.90
SS	9/9	0.89	14.16	1.00	10.96	1.11	8.94
FF	9/9	2.00	18.27	1.93	13.77	1.90	10.85
SNFP	9/9	1.12	15.15	1.20	11.66	1.30	9.44
FNSP	9/9	1.66	17.25	1.65	13.07	1.67	10.36
TT	9/0.108	42.20	566.91	59.92	550.38	76.90	528.79
SS	9/0.108	16.50	471.50	30.36	453.76	45.32	433.88
FF	9/0.108	82.14	662.34	100.27	647.04	116.89	623.72
SNFP	9/0.108	27.43	519.00	43.78	502.09	60.25	481.94
FNSP	9/0.108	60.54	614.80	78.68	598.64	95.51	575.57
TT	0.144/9	1.55	16.26	1.53	12.36	1.54	9.85
SS	0.144/9	0.81	12.81	0.90	9.95	0.99	8.16
FF	0.144/9	2.63	19.82	2.38	14.78	2.22	11.50
SNFP	0.144/9	1.15	14.50	1.19	11.15	1.26	9.03
FNSP	0.144/9	2.04	18.05	1.92	13.57	1.85	10.67
TT	0.144/0.108	71.59	866.18	88.11	767.08	99.53	669.44
SS	0.144/0.108	13.49	653.52	26.10	570.97	39.13	501.16
FF	0.144/0.108	184.06	1069.31	188.97	963.75	187.77	845.42
SNFP	0.144/0.108	35.21	760.00	51.97	669.06	65.86	585.60
FNSP	0.144/0.108	122.03	968.40	133.99	865.28	140.03	756.11

Table 5-4 Corner window of NMOS Ion

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-IDSAT (μA/um)	-VDS	0.6V	1.2V	0.6V	1.2V	0.6V	1.2V
TT	9/9	0.14	2.24	0.18	2.09	0.23	1.97
SS	9/9	0.08	1.96	0.12	1.85	0.16	1.78
FF	9/9	0.22	2.53	0.26	2.33	0.31	2.17
SNFP	9/9	0.17	2.39	0.22	2.21	0.27	2.07
FNSP	9/9	0.11	2.09	0.15	1.97	0.20	1.88
TT	9/0.108	12.97	204.88	18.44	194.33	24.66	184.71
SS	9/0.108	5.26	162.66	9.22	157.42	14.18	153.40
FF	9/0.108	25.10	247.72	31.12	231.07	37.79	215.30
SNFP	9/0.108	18.52	226.78	24.35	212.72	30.86	199.66
FNSP	9/0.108	8.55	183.20	13.38	175.91	19.10	169.58
TT	0.144/9	0.21	3.41	0.25	2.96	0.31	2.63
SS	0.144/9	0.09	2.65	0.13	2.34	0.18	2.12
FF	0.144/9	0.39	4.18	0.43	3.59	0.48	3.12
SNFP	0.144/9	0.29	3.80	0.33	3.28	0.39	2.88
FNSP	0.144/9	0.15	3.01	0.19	2.65	0.24	2.38
TT	0.144/0.108	17.53	253.68	22.16	230.60	27.26	210.11
SS	0.144/0.108	3.79	171.81	6.74	160.51	10.45	151.74
FF	0.144/0.108	45.21	338.06	48.35	300.42	52.31	266.08
SNFP	0.144/0.108	29.56	296.29	33.86	265.56	38.73	237.95
FNSP	0.144/0.108	9.13	212.23	13.25	195.61	17.95	181.30

Table 5-5 Corner window of PMOS Ion

NMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
IOFF (A)	VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	1.42E-13	2.10E-13	2.72E-11	3.97E-11	1.27E-09	1.91E-09
SS	9/9	2.28E-14	3.55E-14	6.80E-12	9.92E-12	4.57E-10	6.84E-10
FF	9/9	9.05E-13	1.31E-12	1.08E-10	1.58E-10	3.48E-09	5.24E-09
SNFP	9/9	5.48E-14	8.17E-14	1.33E-11	1.94E-11	7.55E-10	1.13E-09
FNSP	9/9	3.68E-13	5.35E-13	5.53E-11	8.07E-11	2.13E-09	3.20E-09
TT	9/0.108	4.05E-13	7.00E-12	1.44E-10	1.43E-09	1.13E-08	7.35E-08
SS	9/0.108	3.45E-14	5.60E-13	2.24E-11	2.11E-10	2.77E-09	1.75E-08
FF	9/0.108	4.39E-12	7.55E-11	8.81E-10	8.62E-09	4.41E-08	2.82E-07
SNFP	9/0.108	1.11E-13	1.87E-12	5.49E-11	5.30E-10	5.50E-09	3.53E-08
FNSP	9/0.108	1.43E-12	2.50E-11	3.73E-10	3.71E-09	2.29E-08	1.49E-07
TT	0.144/9	7.72E-15	1.34E-14	1.01E-12	1.48E-12	3.55E-11	5.32E-11
SS	0.144/9	5.92E-16	3.11E-15	1.28E-13	1.89E-13	7.67E-12	1.15E-11
FF	0.144/9	1.23E-13	1.81E-13	7.91E-12	1.16E-11	1.56E-10	2.36E-10
SNFP	0.144/9	1.96E-15	5.05E-15	3.51E-13	5.15E-13	1.64E-11	2.45E-11
FNSP	0.144/9	3.14E-14	4.76E-14	2.88E-12	4.21E-12	7.55E-11	1.13E-10
TT	0.144/0.108	1.05E-13	4.46E-13	1.59E-11	5.51E-11	6.52E-10	1.95E-09
SS	0.144/0.108	1.63E-15	7.94E-15	6.77E-13	2.06E-12	6.31E-11	1.69E-10
FF	0.144/0.108	5.55E-12	2.47E-11	3.07E-10	1.11E-09	5.73E-09	1.77E-08
SNFP	0.144/0.108	1.34E-14	5.57E-14	3.40E-12	1.13E-11	2.09E-10	6.01E-10
FNSP	0.144/0.108	8.04E-13	3.54E-12	7.25E-11	2.60E-10	1.99E-09	6.09E-09

Table 5-6 Corner window of NMOS Ioff

PMOS	Temp. (°C)	-50 °C		25 °C		125 °C	
-IOFF (A)	-VDS	0.05V	1.2V	0.05V	1.2V	0.05V	1.2V
TT	9/9	4.55E-15	7.11E-15	1.15E-12	1.37E-12	7.78E-11	9.56E-11
SS	9/9	1.05E-15	3.55E-15	3.28E-13	3.93E-13	3.00E-11	3.69E-11
FF	9/9	2.31E-14	3.02E-14	4.13E-12	4.90E-12	2.04E-10	2.51E-10
SNFP	9/9	1.05E-14	1.42E-14	2.23E-12	2.64E-12	1.28E-10	1.57E-10
FNSP	9/9	2.05E-15	3.55E-15	5.99E-13	7.11E-13	4.75E-11	5.84E-11
TT	9/0.108	9.01E-13	9.02E-12	2.11E-10	1.31E-09	1.26E-08	5.60E-08
SS	9/0.108	1.02E-13	8.31E-13	3.94E-11	2.09E-10	3.45E-09	1.36E-08
FF	9/0.108	7.66E-12	8.20E-11	1.08E-09	7.04E-09	4.38E-08	2.01E-07
SNFP	9/0.108	2.76E-12	2.95E-11	4.94E-10	3.23E-09	2.41E-08	1.11E-07
FNSP	9/0.108	2.90E-13	2.62E-12	8.88E-11	5.12E-10	6.52E-09	2.73E-08
TT	0.144/9	3.02E-16	2.58E-15	4.13E-14	5.21E-14	2.27E-12	2.84E-12
SS	0.144/9	1.17E-16	2.41E-15	5.85E-15	9.33E-15	5.12E-13	6.43E-13
FF	0.144/9	2.84E-15	5.63E-15	3.00E-13	3.64E-13	1.02E-11	1.27E-11
SNFP	0.144/9	8.57E-16	3.28E-15	1.13E-13	1.38E-13	4.86E-12	6.07E-12
FNSP	0.144/9	1.54E-16	2.44E-15	1.51E-14	2.05E-14	1.06E-12	1.32E-12
TT	0.144/0.108	1.17E-14	3.94E-14	2.40E-12	6.92E-12	1.32E-10	3.61E-10
SS	0.144/0.108	3.43E-16	3.03E-15	1.25E-13	2.96E-13	1.36E-11	3.10E-11
FF	0.144/0.108	5.56E-13	1.99E-12	4.47E-11	1.43E-10	1.19E-09	3.59E-09
SNFP	0.144/0.108	8.35E-14	2.86E-13	1.07E-11	3.25E-11	4.06E-10	1.17E-09
FNSP	0.144/0.108	1.76E-15	7.41E-15	5.43E-13	1.50E-12	4.22E-11	1.11E-10

Table 5-7 Corner window of PMOS Ioff

6 Shallow Trench Isolation (STI) or LOD Stress Effect

6.1 Simulation results for NMOS at SA=0.38 μ m (SAREF=2.2 μ m)

Note: The LOD stress effect model was based on very limit and early silicon data. SB is equal to SA, SD=0 in all following plots and tables. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = 300\text{nA} * (W_{DES} * 0.9) / (L_{DES} * 0.9 + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

Table 6-1 Simulation results for NMOS

All parameters are given for T=25 °C , $V_{BS}=0$ V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
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Long channel device ($W_{DES}/L_{DES}= 10 \mu\text{m} / 10 \mu\text{m}$)

V_{TLIN}	$V_{DS} = 0.05 \text{ V}$	V	0.324
I_{ON}	$V_{DS} = 1.2 \text{ V}$	$\mu\text{A} / \mu\text{m}$	11.09

Wide and short channel device ($W_{DES}/L_{DES}= 10 \mu\text{m} / 0.12 \mu\text{m}$)

V_{TLIN}	$V_{DS} = 0.05 \text{ V}$	V	0.452
V_{TSAT}	$V_{DS} = 1.2 \text{ V}$	V	0.363
Body Effect V_T shift	$V_{DS} = 1.2 \text{ V}$ $V_{BS} = 0 \sim -1.2 \text{ V}$	V	0.110
DIBL V_T shift	$V_{DS} = 0.05 \sim 1.2 \text{ V}$	V	0.089
I_{ON}	$V_{DS}=V_{GS}= 1.2 \text{ V}$	$\mu\text{A}/\mu\text{m}$	522.18
I_{OFF}	$V_{DS}= 1.2 \text{ V}, V_{GS}=0 \text{ V}$	A/ μm	1.1E-10

Narrow and long channel device ($W_{DES}/L_{DES}= 0.14 \mu\text{m} / 10 \mu\text{m}$)

V_{TLIN}	$V_{DS} = 0.05 \text{ V}$	V	0.303
V_{TSAT}	$V_{DS} = 1.2 \text{ V}$	V	0.280
I_{ON}	$V_{DS}=V_{GS}= 1.2 \text{ V}$	$\mu\text{A}/\mu\text{m}$	11.28

Narrow and short channel device ($W_{DES}/L_{DES}= 0.14 \mu\text{m} / 0.12 \mu\text{m}$)

V_{TLIN}	$V_{DS} = 0.05 \text{ V}$	V	0.408
V_{TSAT}	$V_{DS} = 1.2 \text{ V}$	V	0.353

I_{ON}	$V_{DS}=V_{GS}= 1.2 \text{ V}$	$\mu\text{A}/\mu\text{m}$	736.43
I_{OFF}	$V_{DS}= 1.2 \text{ V}, V_{GS}=0 \text{ V}$	$\text{A}/\mu\text{m}$	2.6E-10

6.1.1 Model verification plots for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D = 300\text{nA} * (W_{DES} * 0.9) / (L_{DES} * 0.9 + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

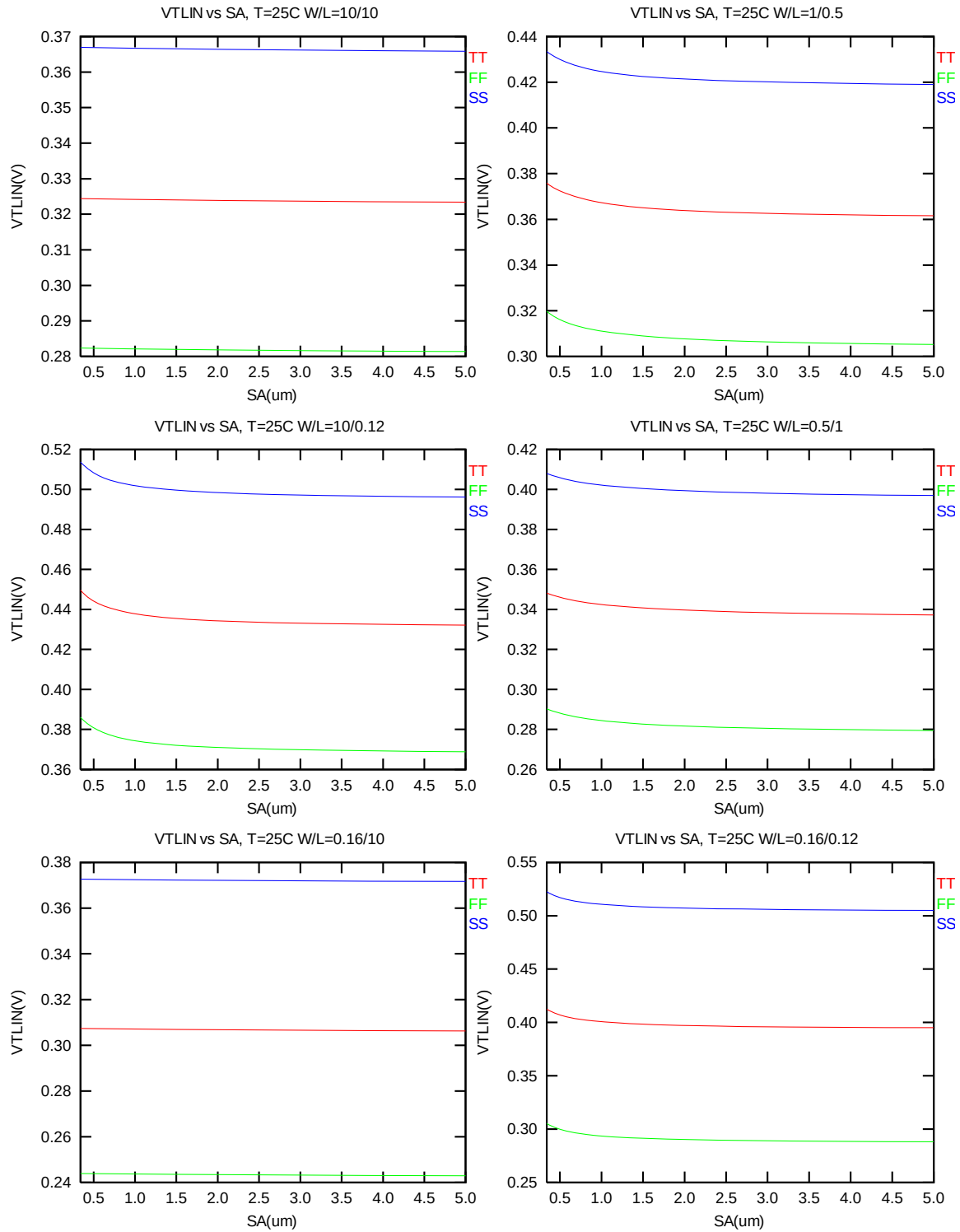


Figure 6-1 V_T vs. SA for low drain bias ($V_{DS} = 0.05$ V) for NMOS

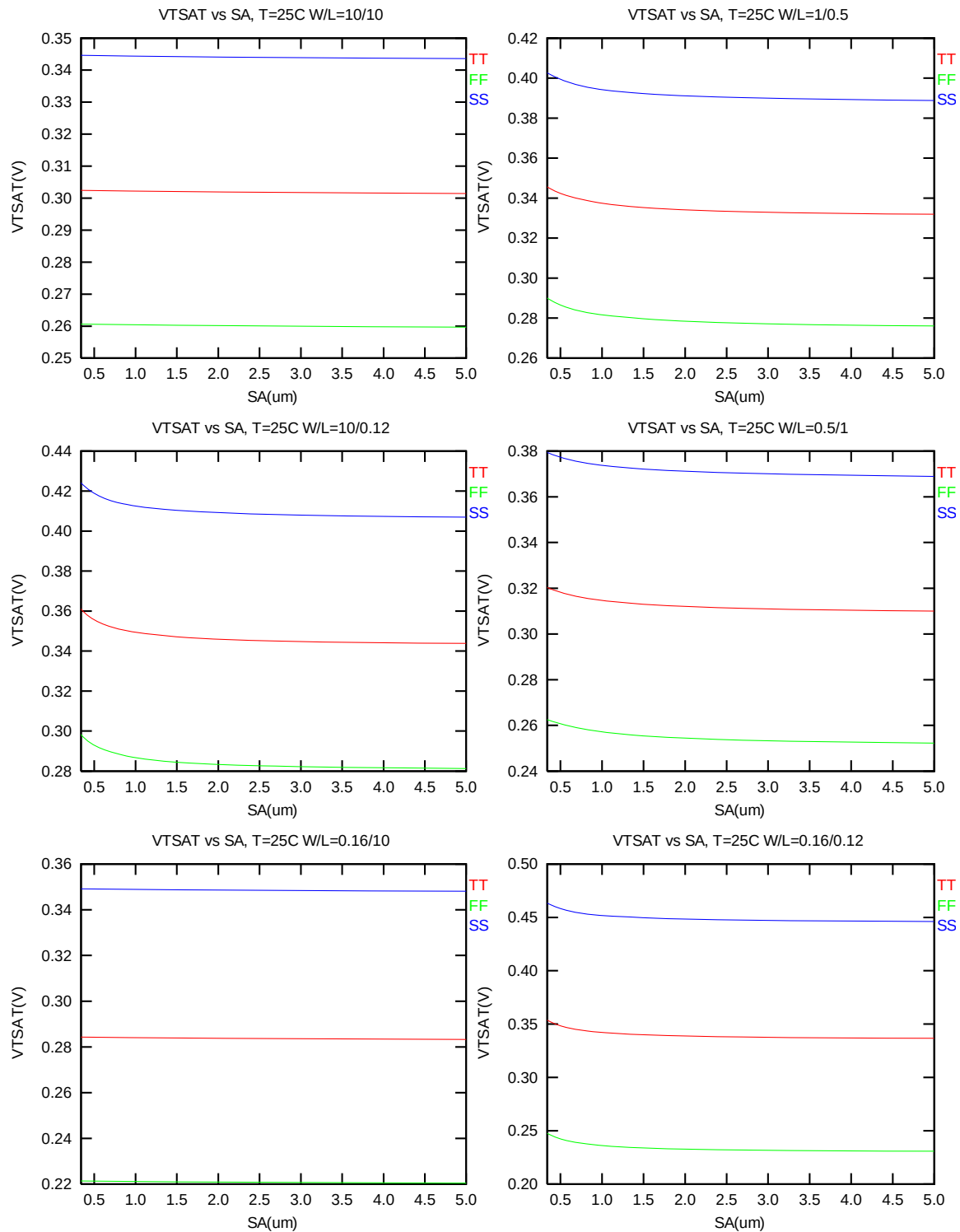


Figure 6-2 V_T vs. SA for high drain bias ($V_{DS} = 1.2$ V) for NMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}=1.2$ V for $V_{BS}=0$ at 25°C .

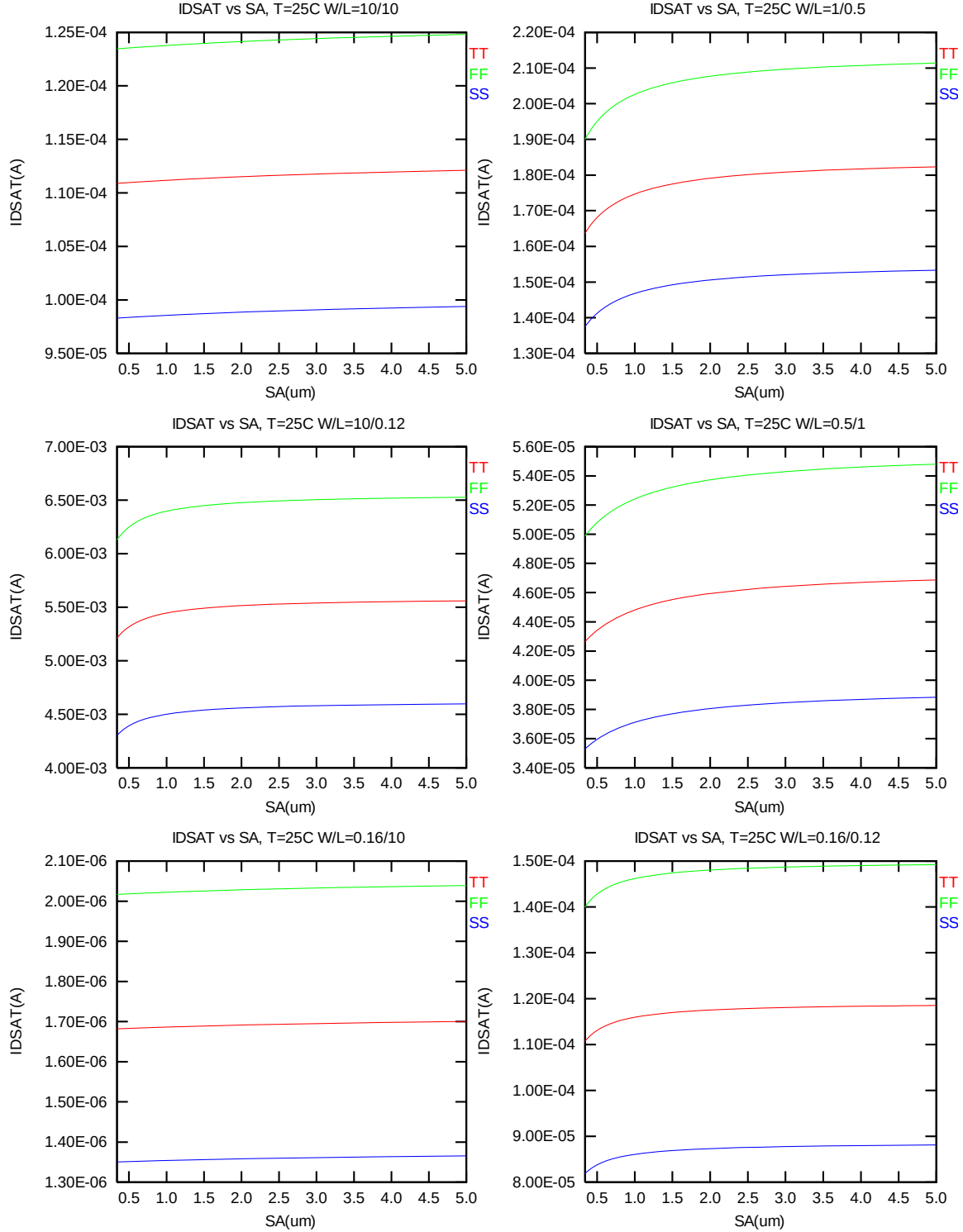


Figure 6-3 Saturation current I_{ON} vs. SA for NMOS

6.2 Simulation results for PMOS at SA=0.38 μ m (SAREF=2.2 μ m)

The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at $I_D = -70\text{nA} * (W_{DES} * 0.9) / (L_{DES} * 0.9 + 0.012\mu\text{m})$. The dimension is after 90% shrink and 12nm sizing channel length back.

All parameters are given for $T=25\text{ }^{\circ}\text{C}$, $V_{BS}=0\text{ V}$ unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
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Long channel device ($W_{DES}/L_{DES}= 10\text{ }\mu\text{ m} / 10\text{ }\mu\text{ m}$)

V_{TLIN}	$V_{DS} = -0.05\text{ V}$	V	-0.385
I_{ON}	$V_{DS} = -1.2\text{ V}$	$\mu\text{A}/\mu\text{m}$	-1.91

Wide and short channel device ($W_{DES}/L_{DES}= 10\text{ }\mu\text{ m} / 0.12\text{ }\mu\text{ m}$)

V_{TLIN}	$V_{DS} = -0.05\text{ V}$	V	-0.403
V_{TSAT}	$V_{DS} = -1.2\text{ V}$	V	-0.326
Body Effect V_T shift	$V_{DS} = -1.2\text{ V}$ $V_{BS} = 0 \sim -1.2\text{ V}$	V	-0.202
DIBL V_T shift	$V_{DS} = -0.05 \sim -1.2\text{ V}$	V	-0.077
I_{ON}	$V_{DS}=V_{GS} = -1.2\text{ V}$	$\mu\text{A}/\mu\text{m}$	-210.49
I_{OFF}	$V_{DS} = -1.2\text{ V}$, $V_{GS}=0\text{ V}$	$\text{A}/\mu\text{m}$	-1.6E-10

Narrow and long channel device ($W_{DES}/L_{DES}= 0.14\text{ }\mu\text{ m} / 10\text{ }\mu\text{ m}$)

V_{TLIN}	$V_{DS} = -0.05\text{ V}$	V	-0.357
V_{TSAT}	$V_{DS} = -1.2\text{ V}$	V	-0.345
I_{ON}	$V_{DS}=V_{GS} = -1.2\text{ V}$	$\mu\text{A}/\mu\text{m}$	-2.74

Narrow and short channel device ($W_{DES}/L_{DES}= 0.14\text{ }\mu\text{ m} / 0.12\text{ }\mu\text{ m}$)

V_{TLIN}	$V_{DS} = -0.05\text{ V}$	V	-0.386
V_{TSAT}	$V_{DS} = -1.2\text{ V}$	V	-0.345
I_{ON}	$V_{DS}=V_{GS} = -1.2\text{ V}$	$\mu\text{A}/\mu\text{m}$	-253.64
I_{OFF}	$V_{DS} = -1.2\text{ V}$, $V_{GS}=0\text{ V}$	$\text{A}/\mu\text{m}$	-5.2E-11

Table 6-2 Simulation results for PMOS

6.2.1 Model verification plots for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at $I_D = -70nA * (W_{DES} * 0.9) / (L_{DES} * 0.9 + 0.012\mu m)$. The dimension is after 90% shrink and 12nm sizing channel length back.

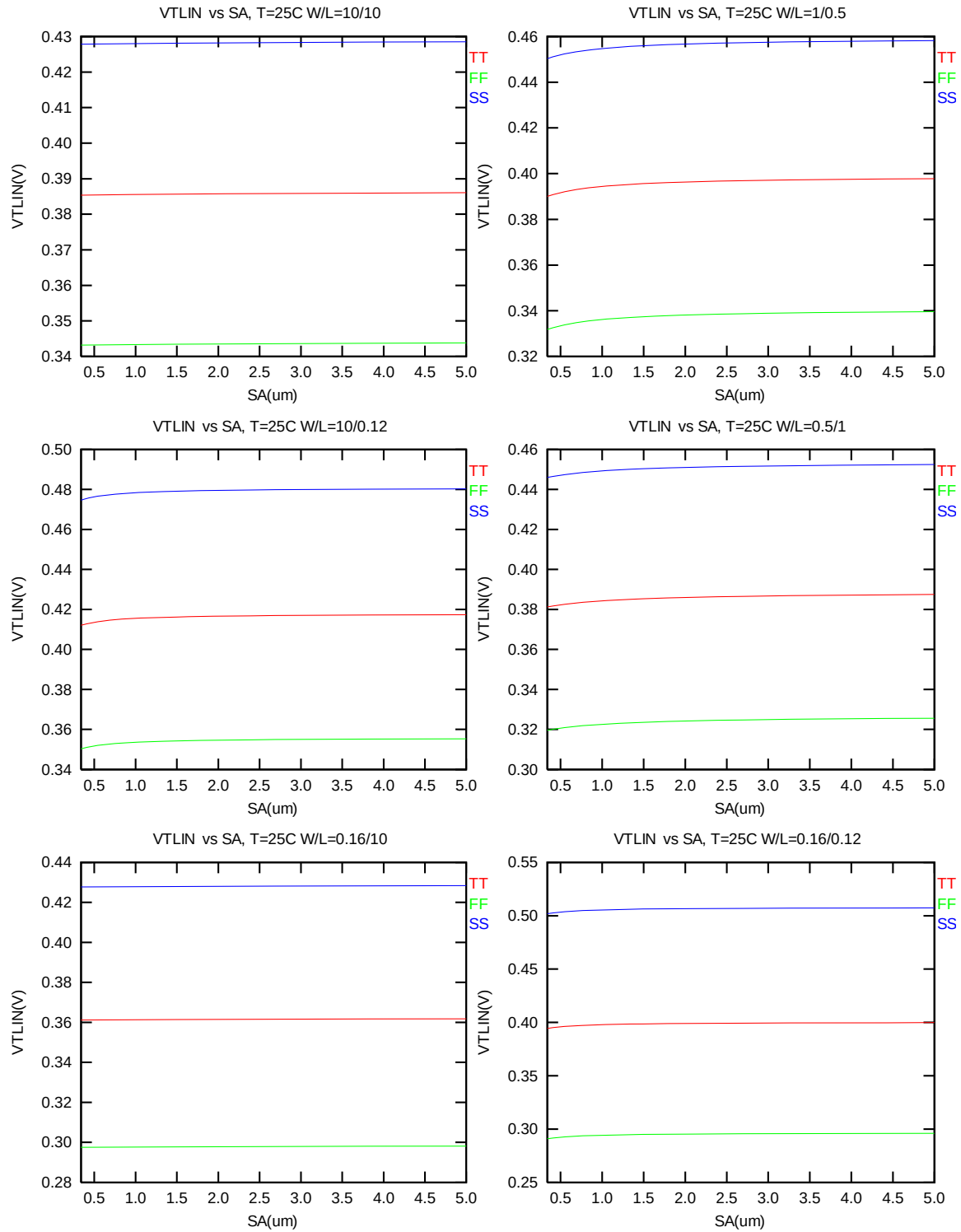


Figure 6-4 V_T vs. SA for low drain bias ($V_{DS} = -0.05$ V) for PMOS

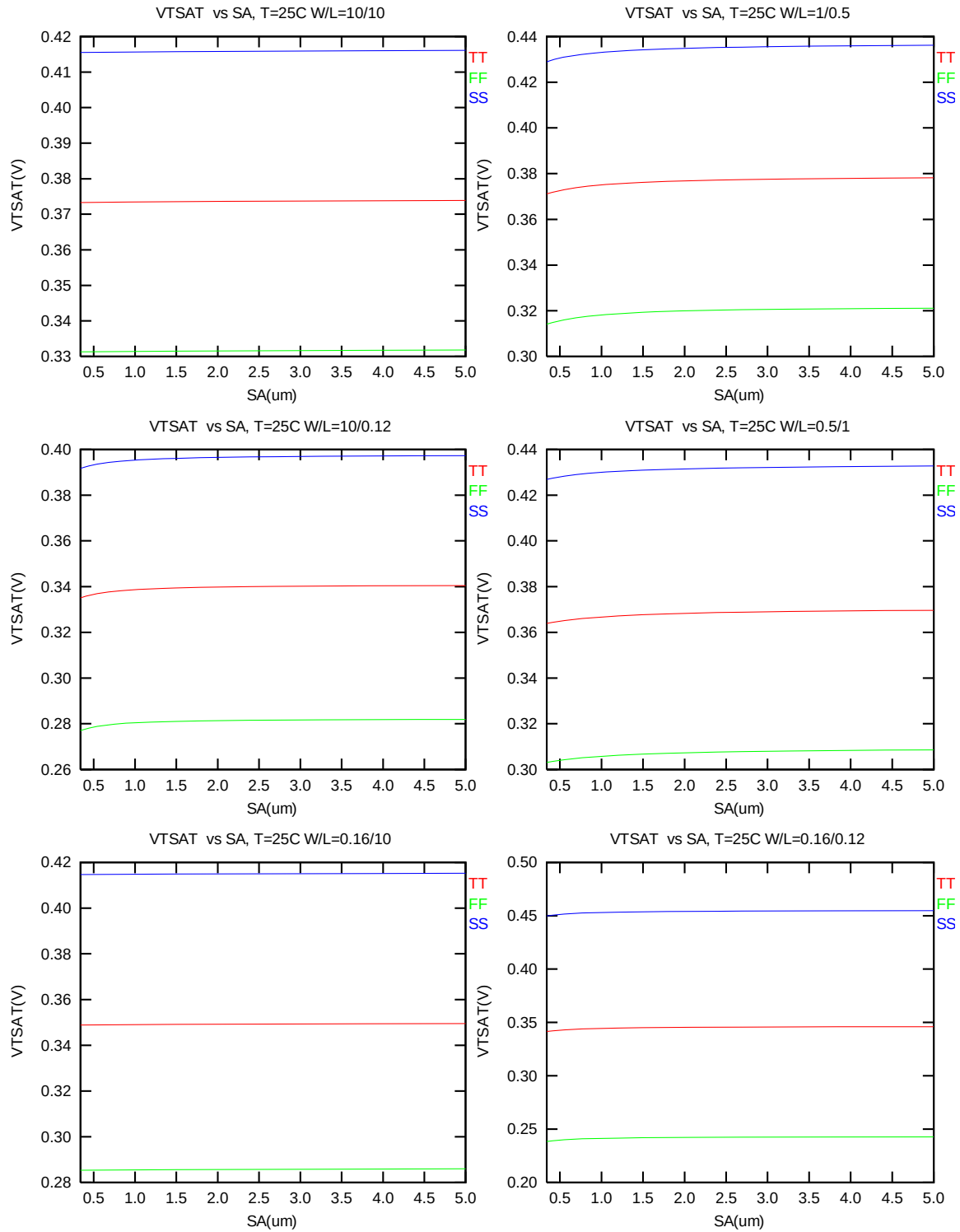


Figure 6-5 V_T vs. SA for high drain bias ($V_{DS} = -1.2$ V) for PMOS

The following plots show the SA dependence of I_{ON} extracted at $V_{DS}=V_{GS}= -1.2$ V for $V_{BS}=0$ at 25°C .

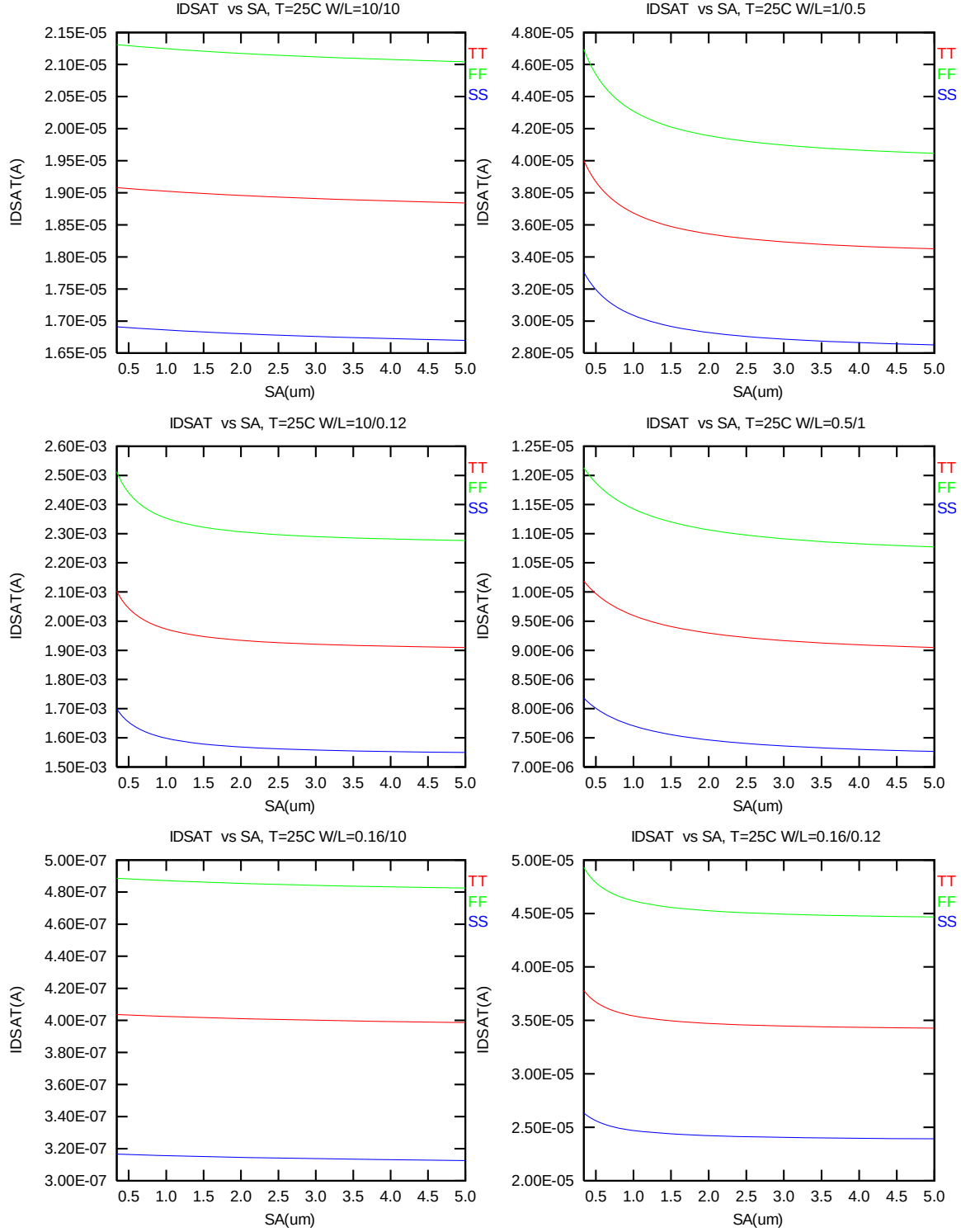


Figure 6-6 Saturation current I_{ON} vs. SA for PMOS

6.3 V_T and I_{ON} variation v. SA for both N/PMOS

Delta V_T is defined as $V_T(SA) - V_T(2.2\mu m)$ for NMOS and $-[V_T(SA) - V_T(2.2\mu m)]$ for PMOS. Delta I_{ON} is defined as $[I_{ON}(SA) - I_{ON}(2.2\mu m)] / I_{ON}(2.2\mu m) * 100\%$.

Delta V_T and I_{ON} vs. SA

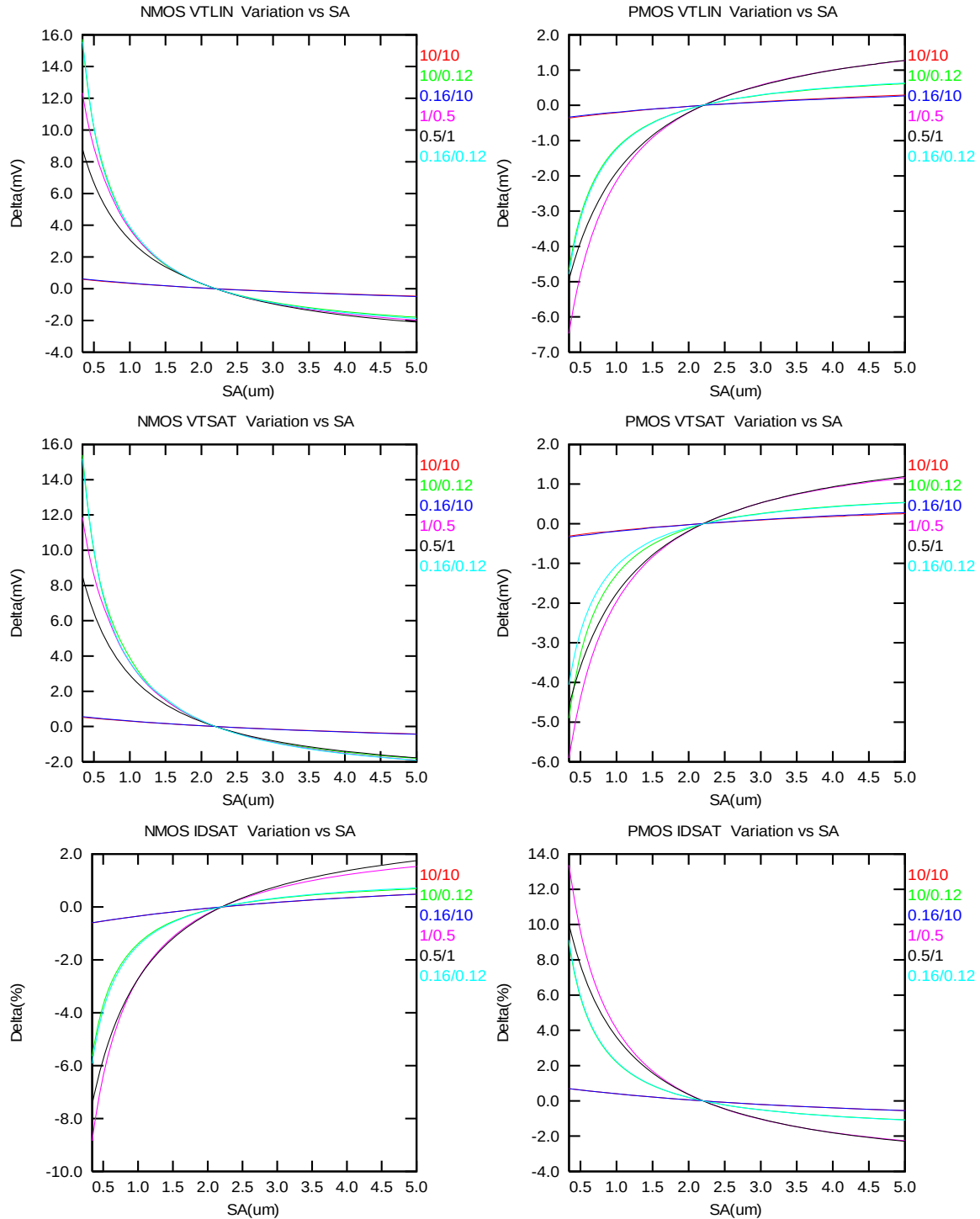


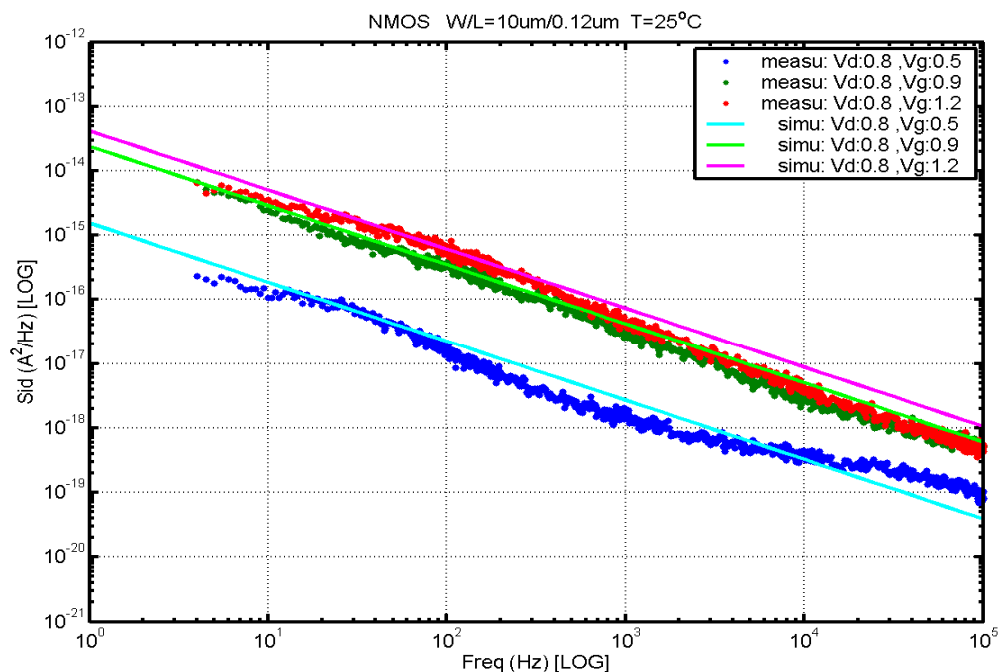
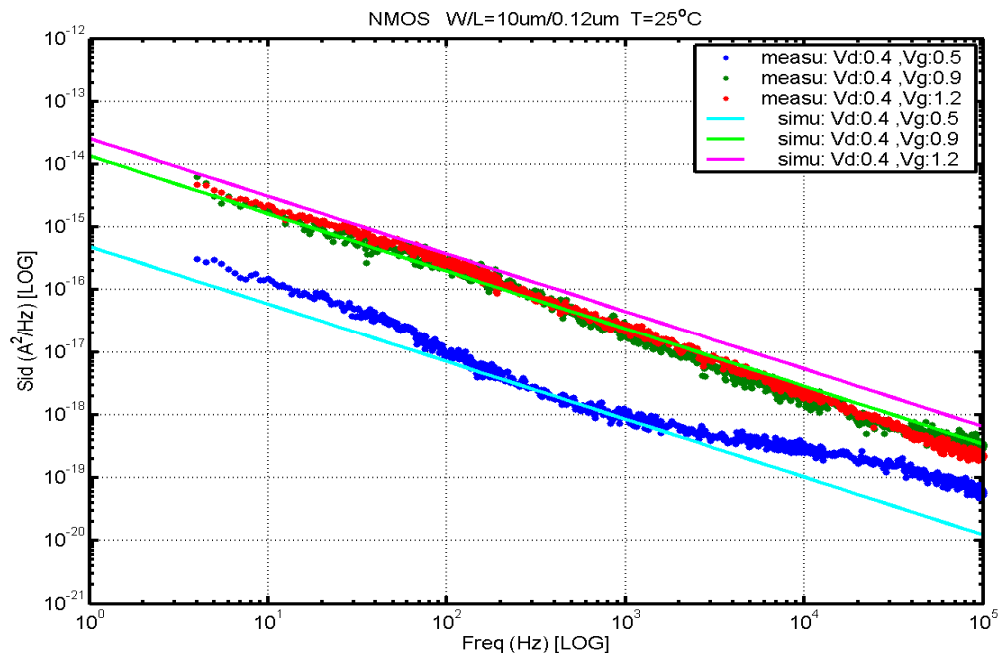
Figure 6-7 Delta V_T and I_{ON} vs. SA for both N/PMOS

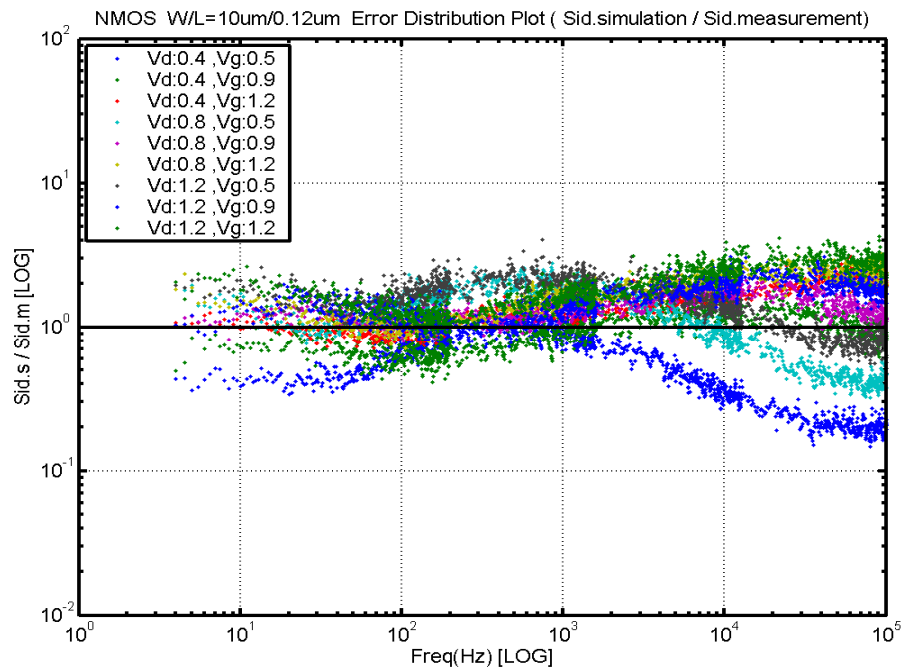
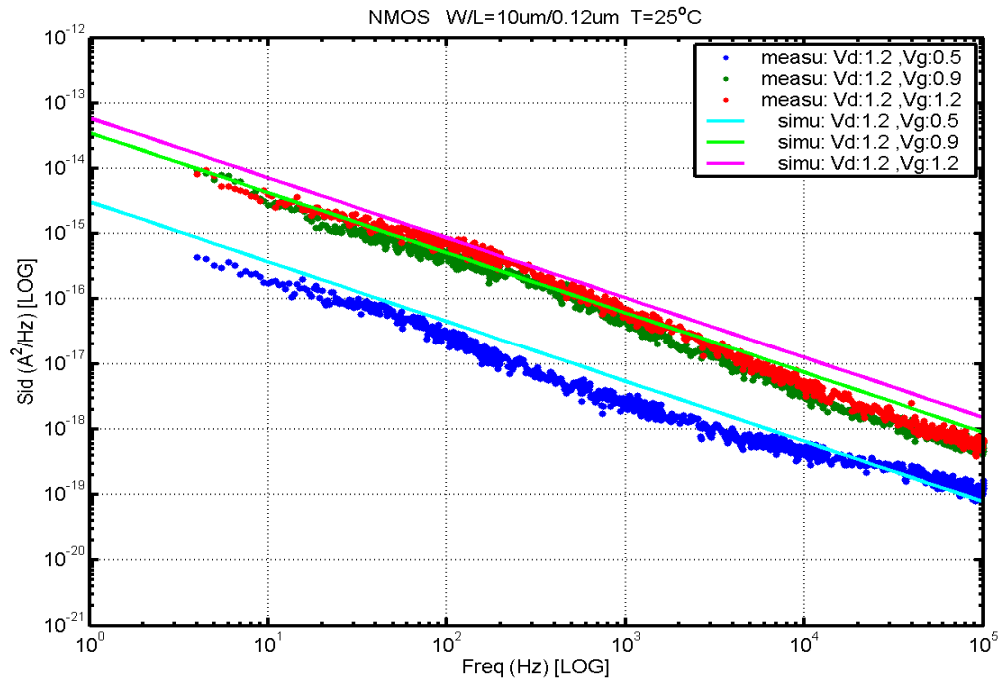
7 Noise Characteristic

7.1 Verification plots

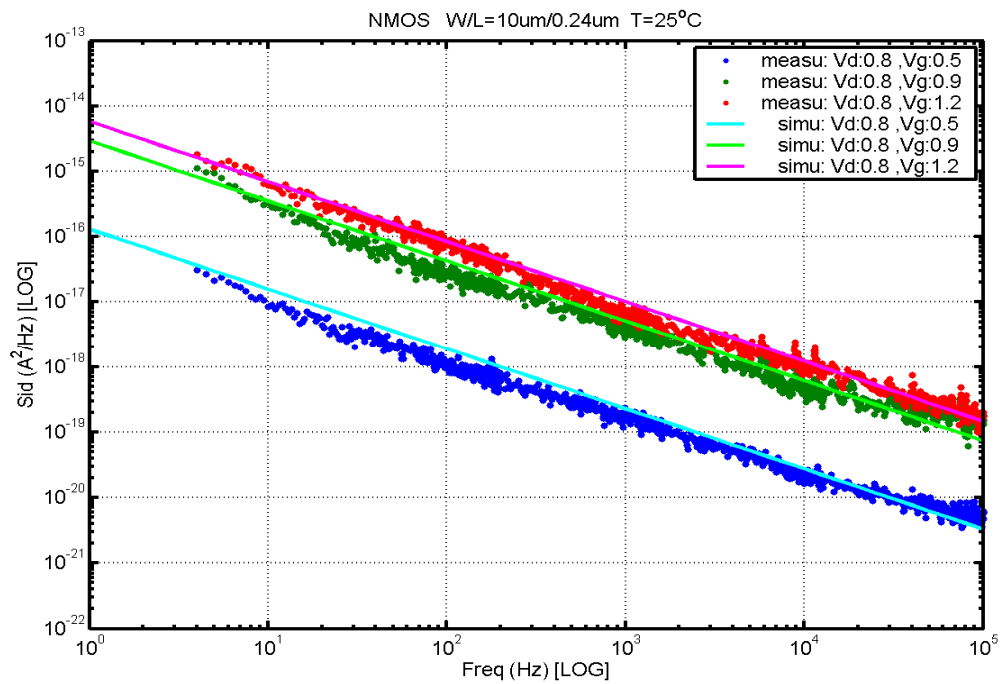
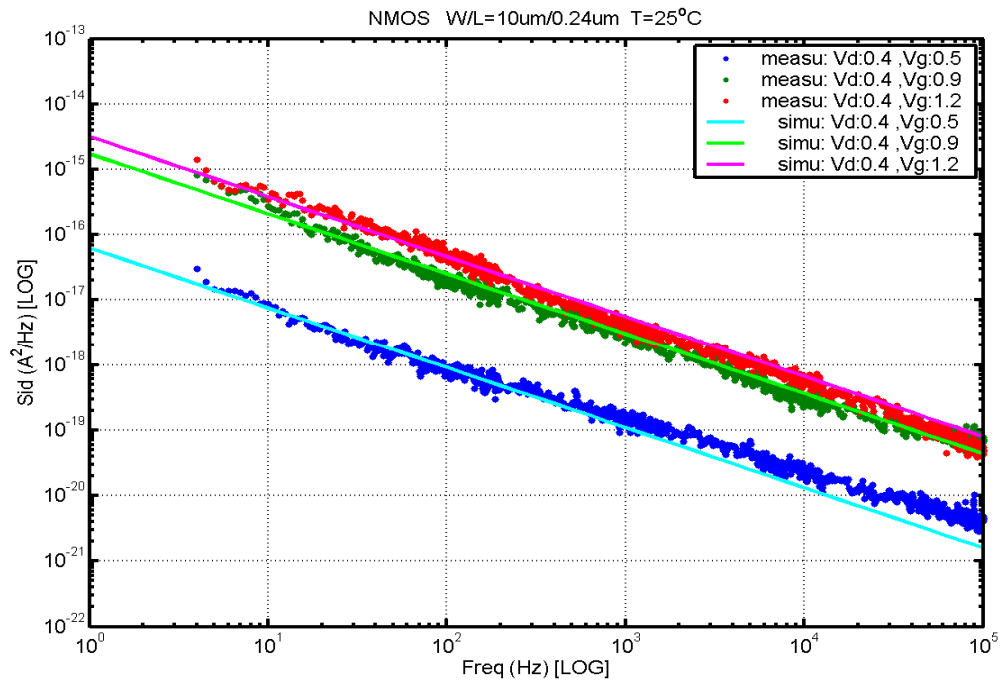
* The dimensions in noise plots are including 12nm sizing.

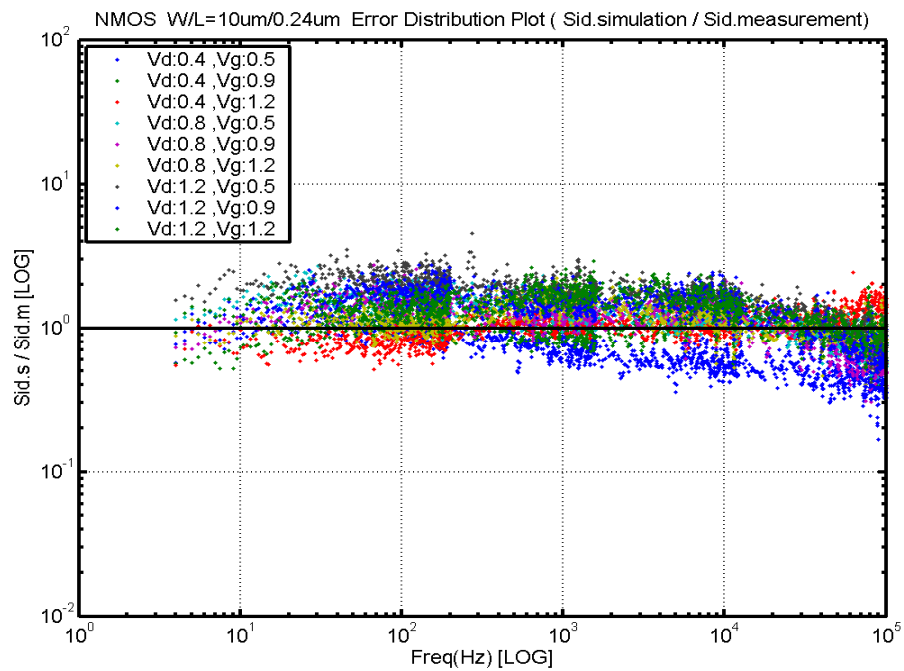
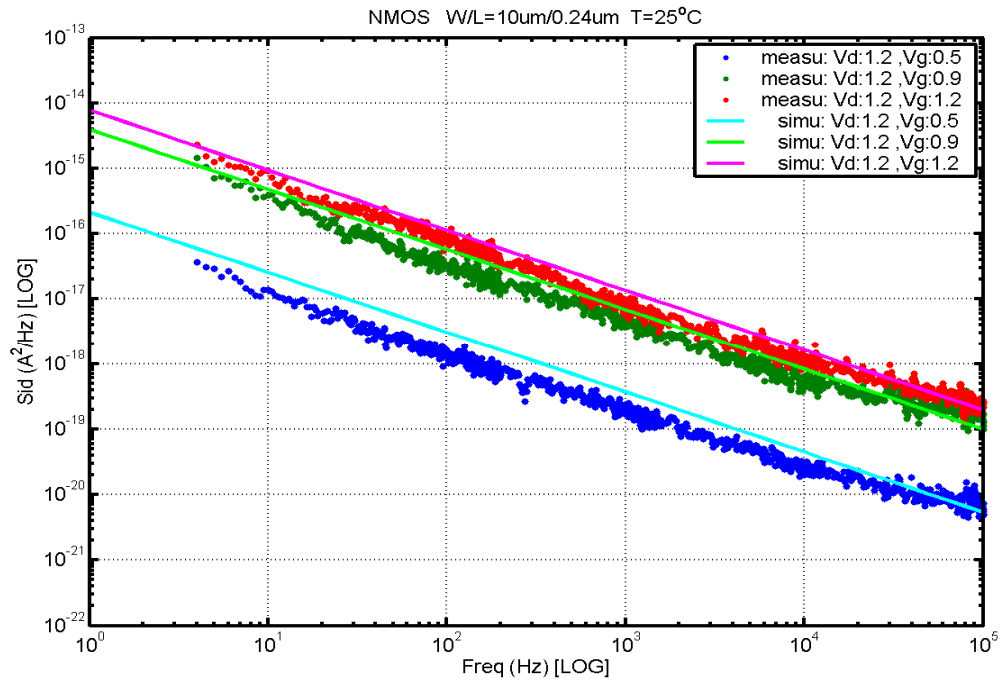
NMOS, $W=10\mu\text{m}$ $L=0.12\mu\text{m}$



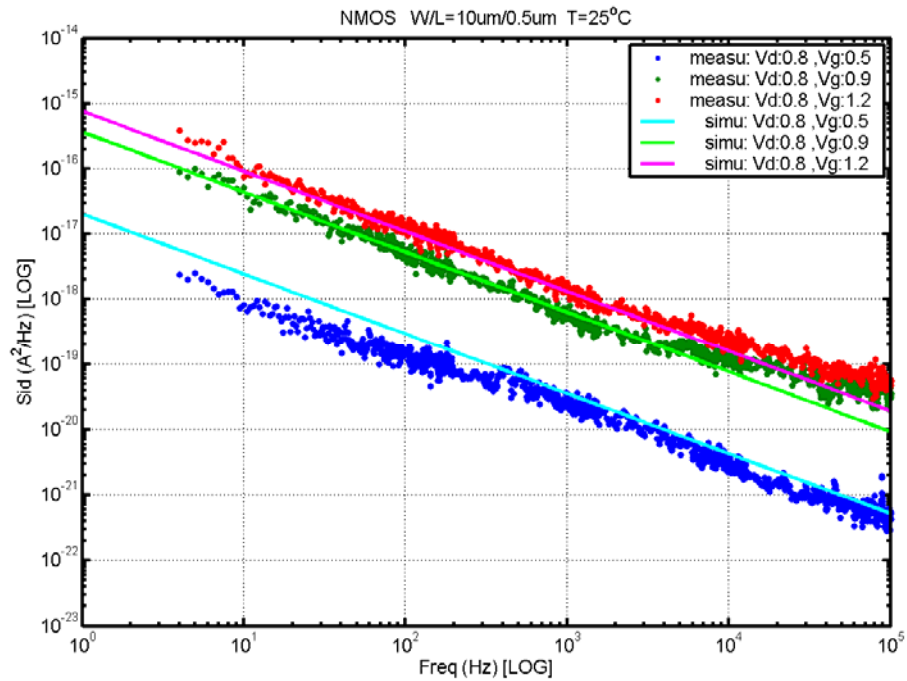
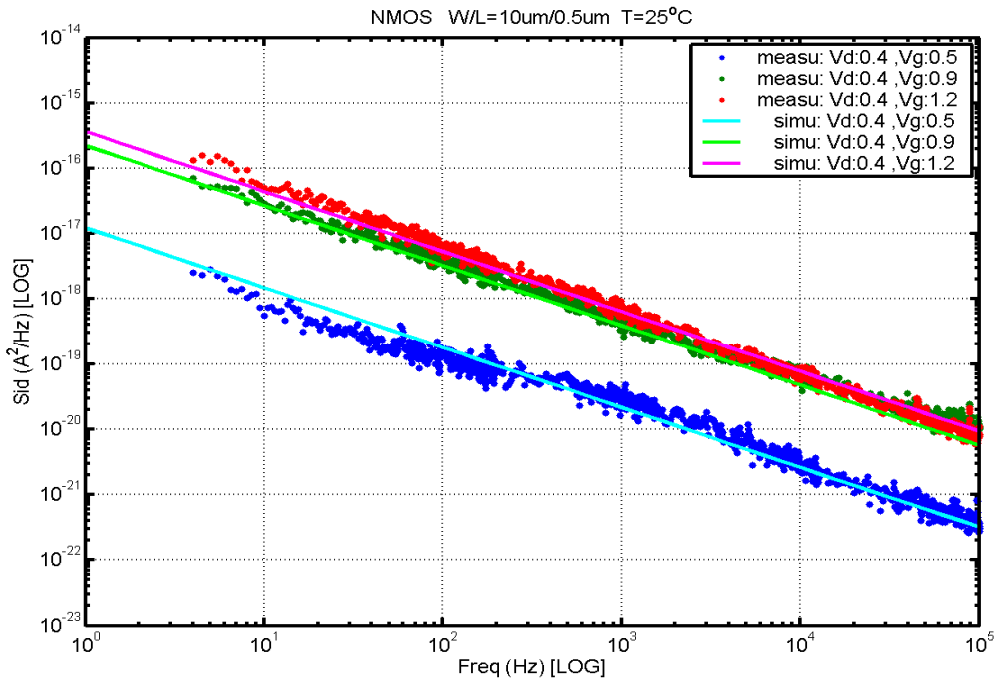


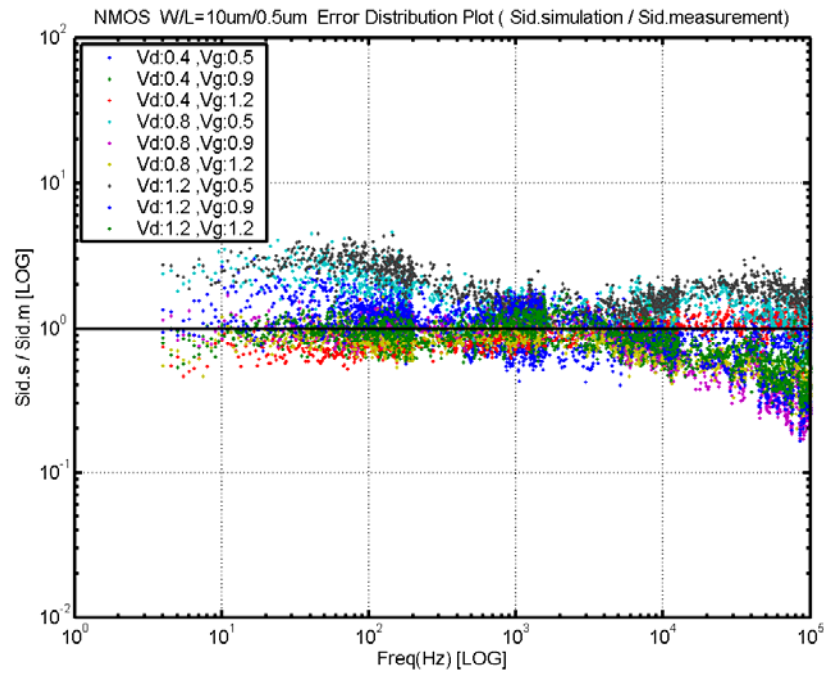
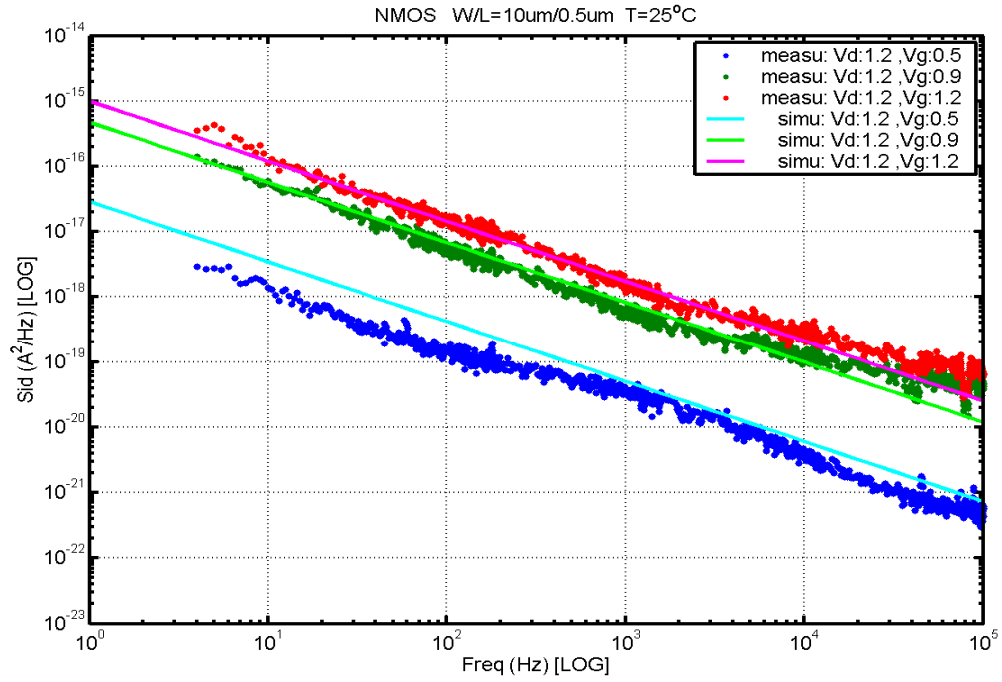
NMOS, $W=10\mu\text{m}$ $L=0.24\mu\text{m}$



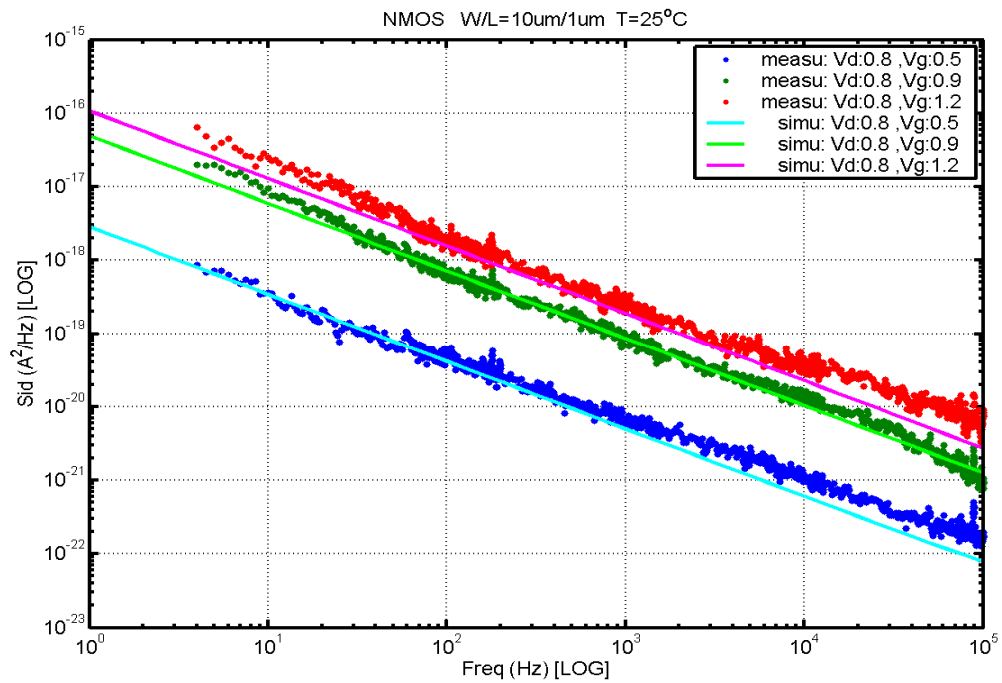
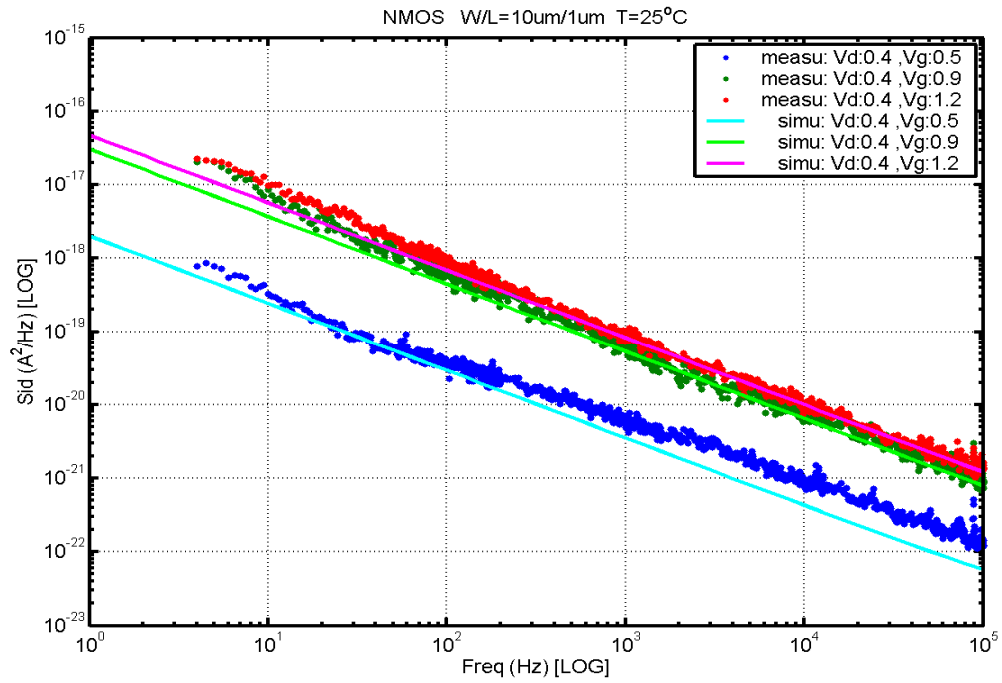


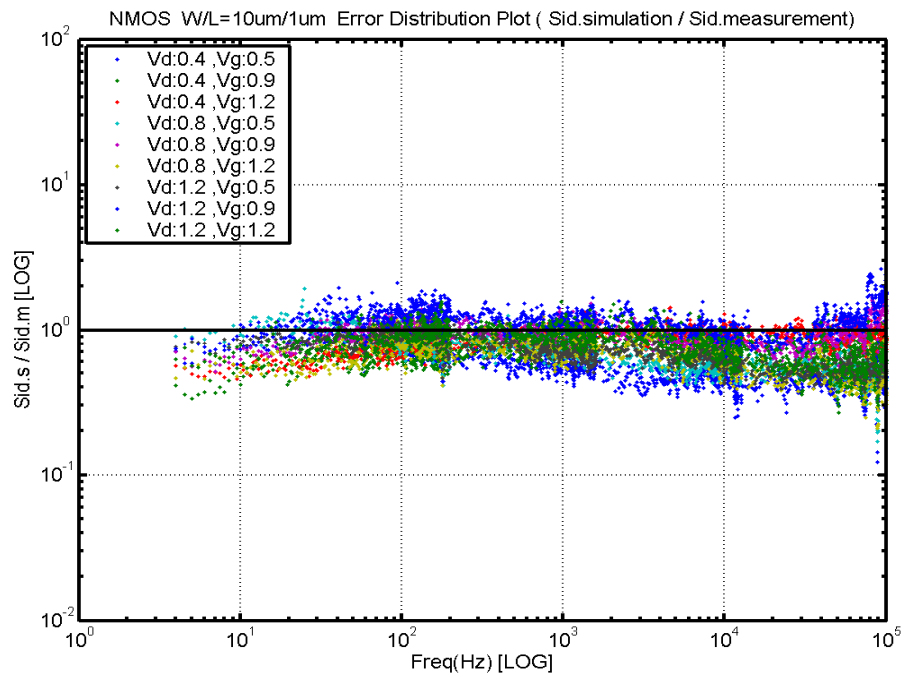
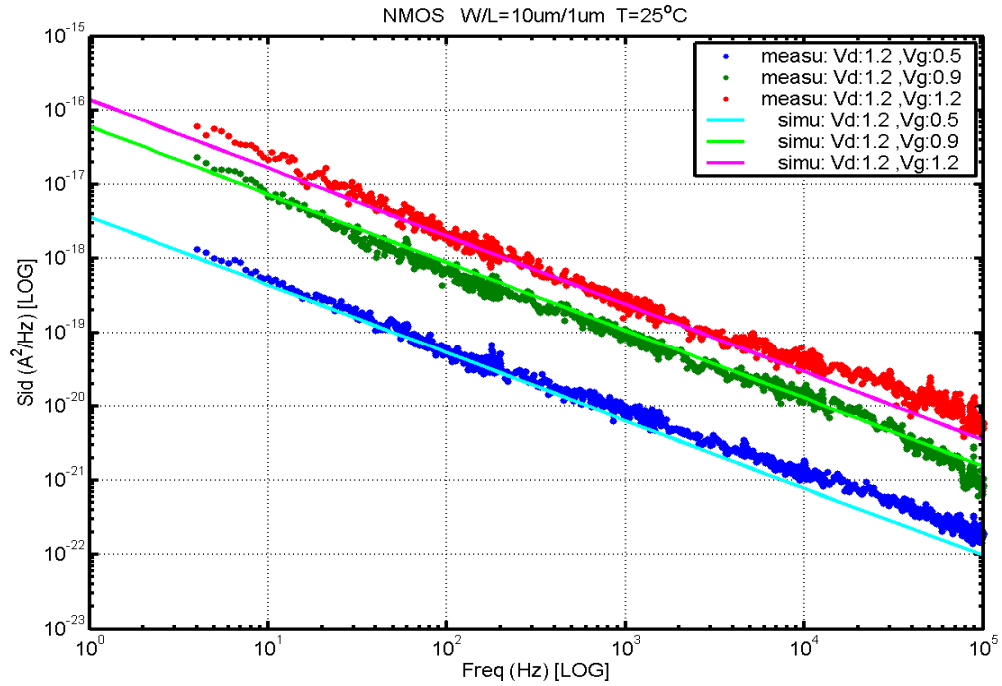
NMOS, $W=10\mu\text{m}$ $L=0.5\mu\text{m}$



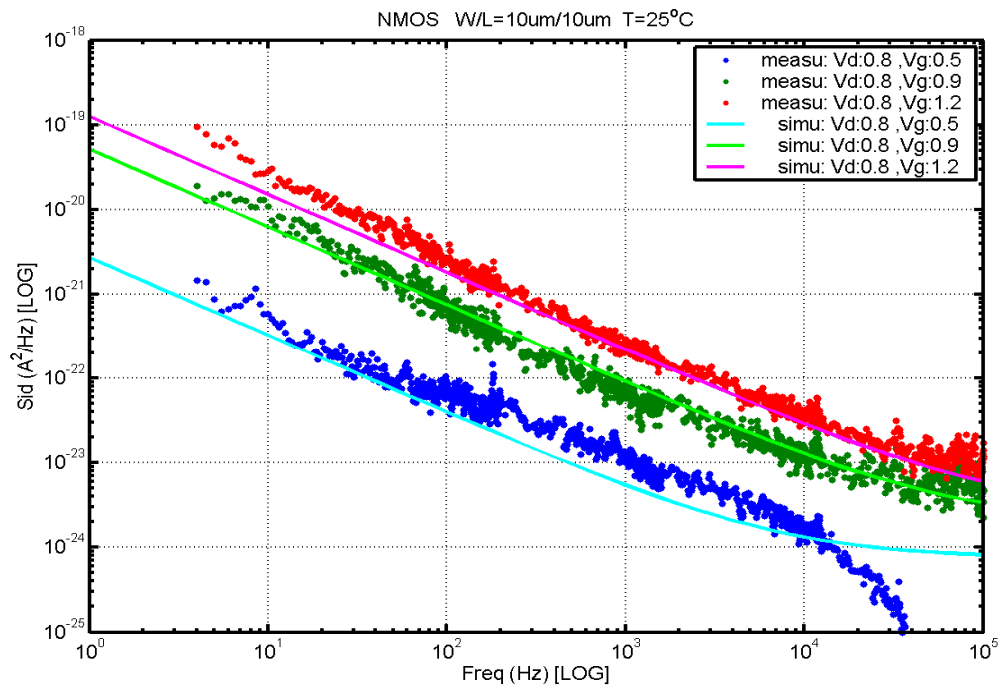
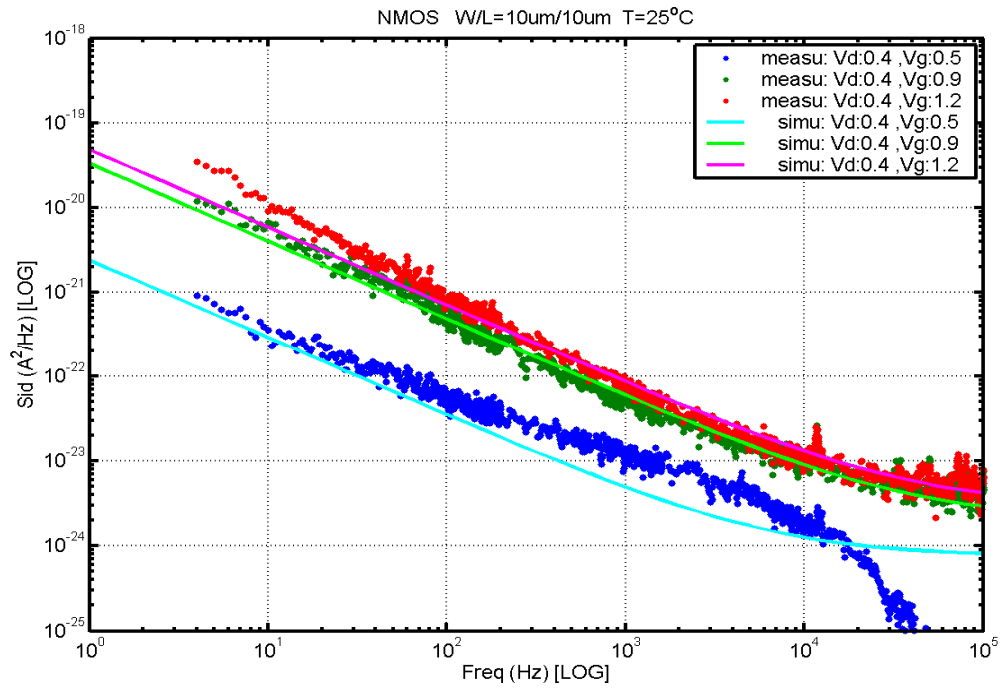


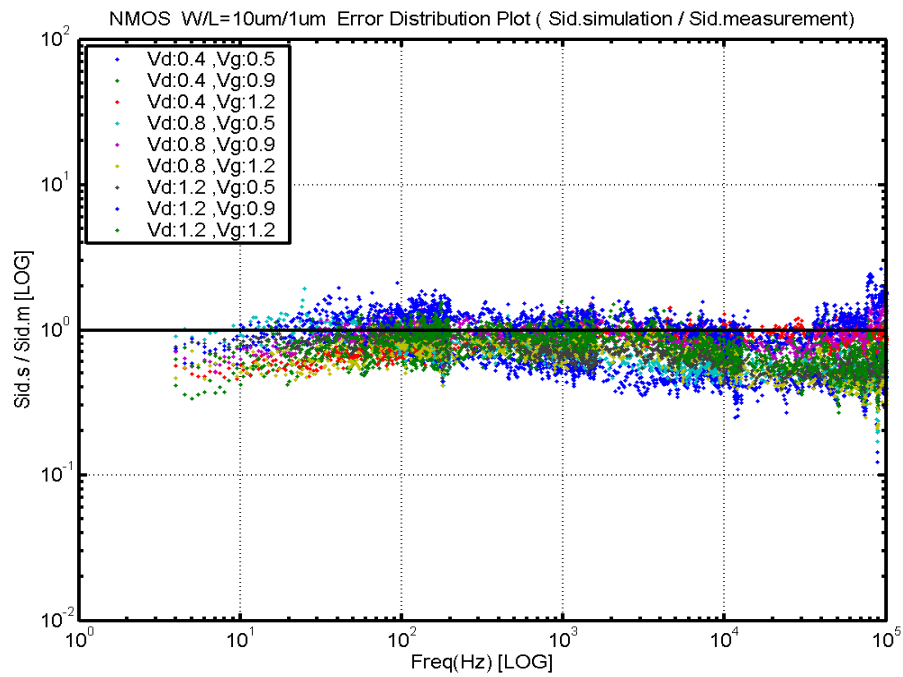
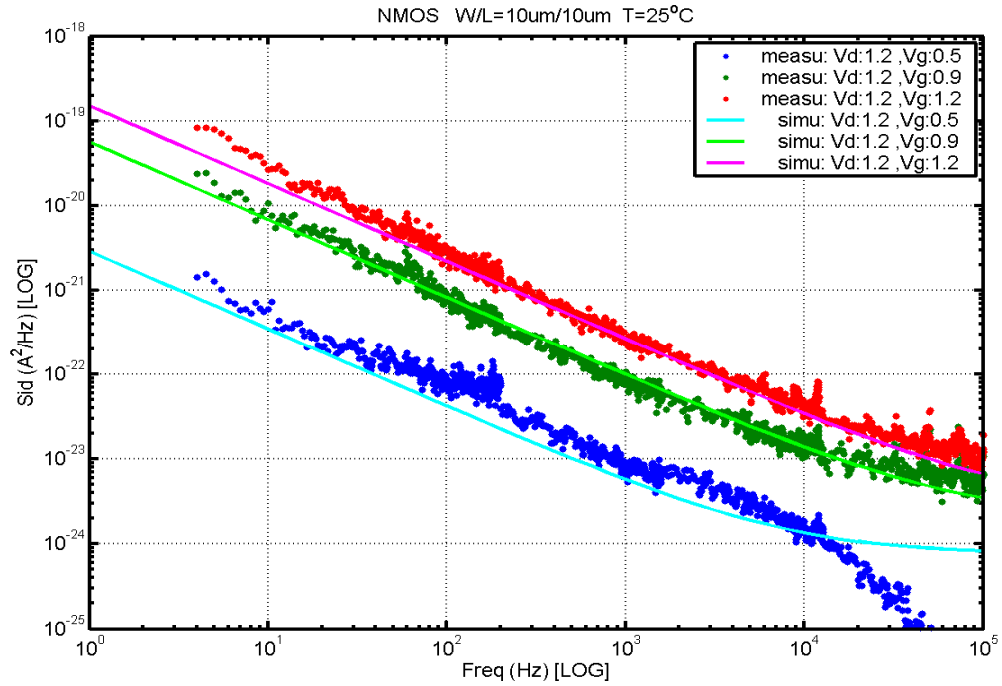
NMOS, $W=10\mu\text{m}$ $L=1\mu\text{m}$



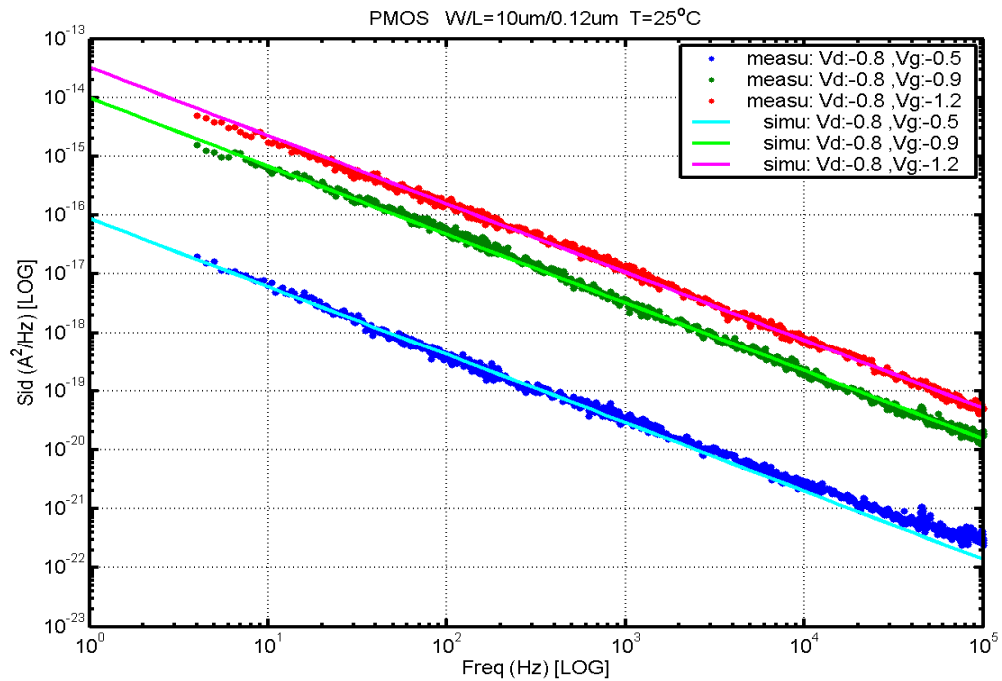
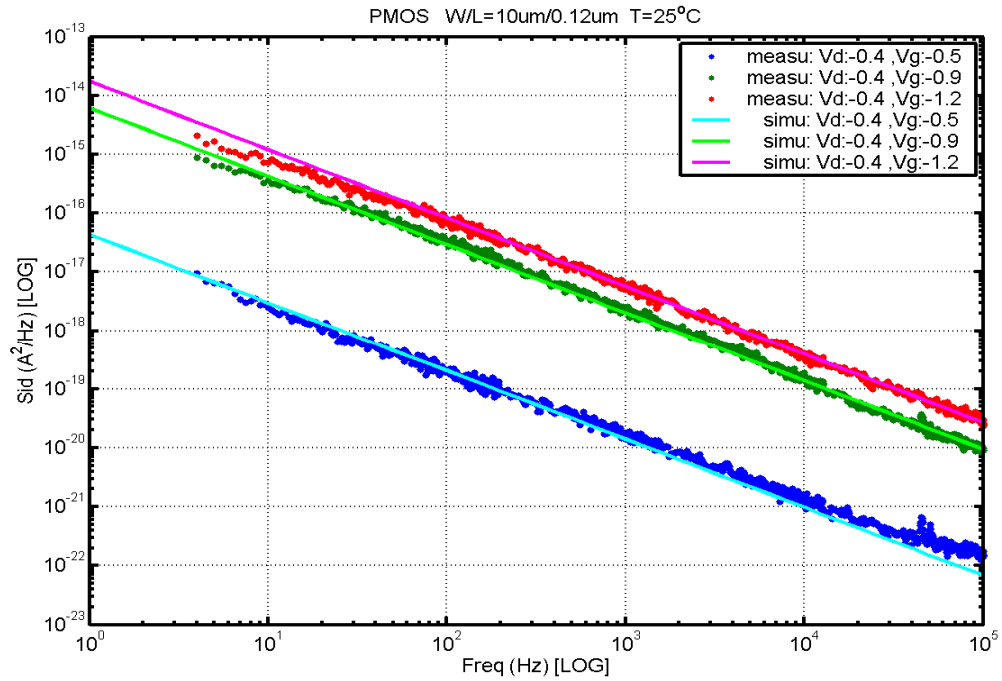


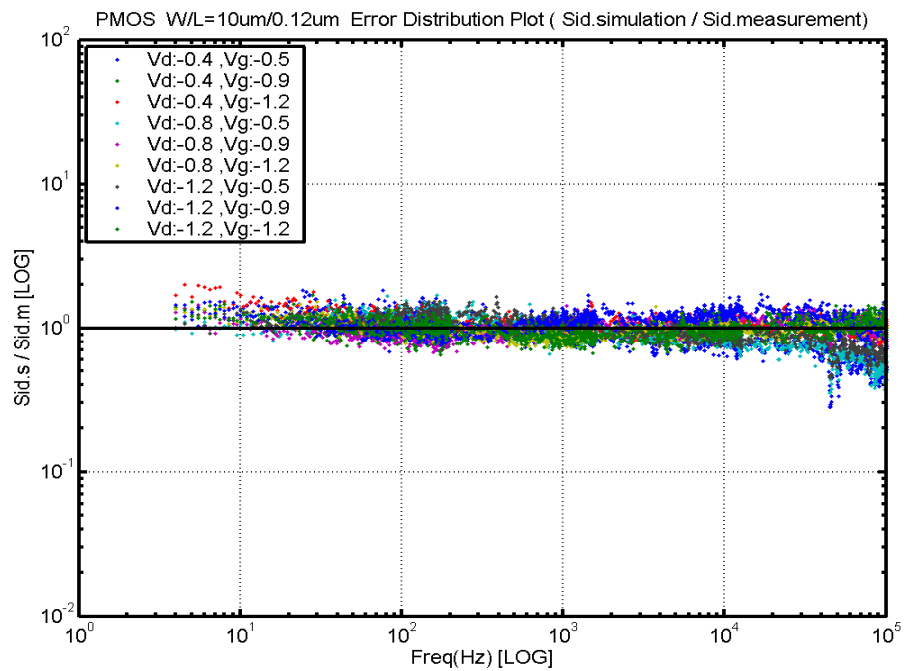
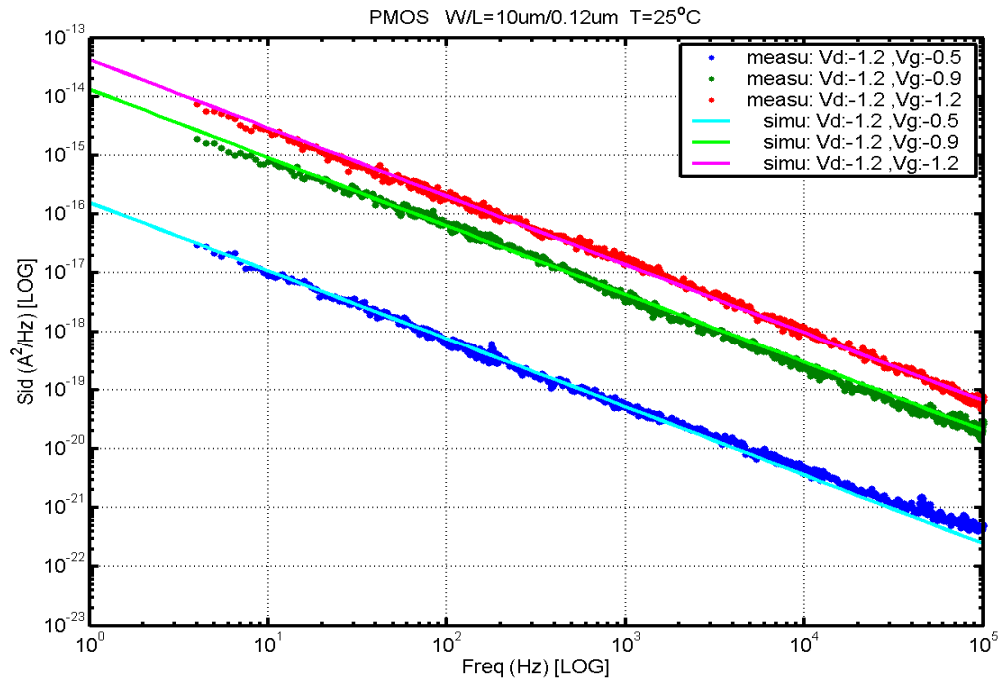
NMOS, $W=10\mu\text{m}$ $L=10\mu\text{m}$



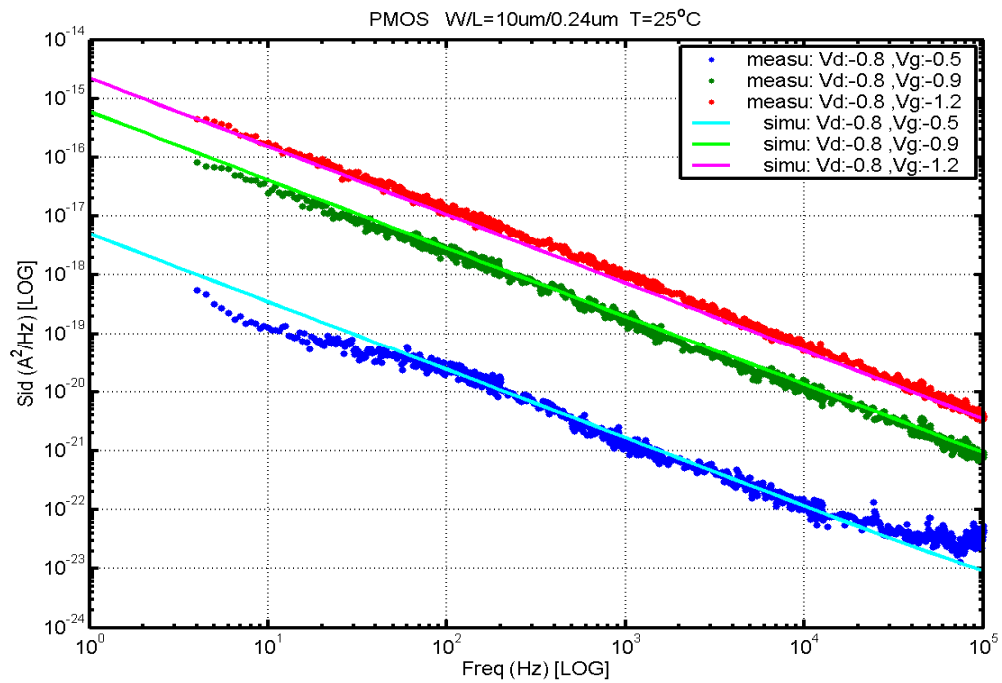
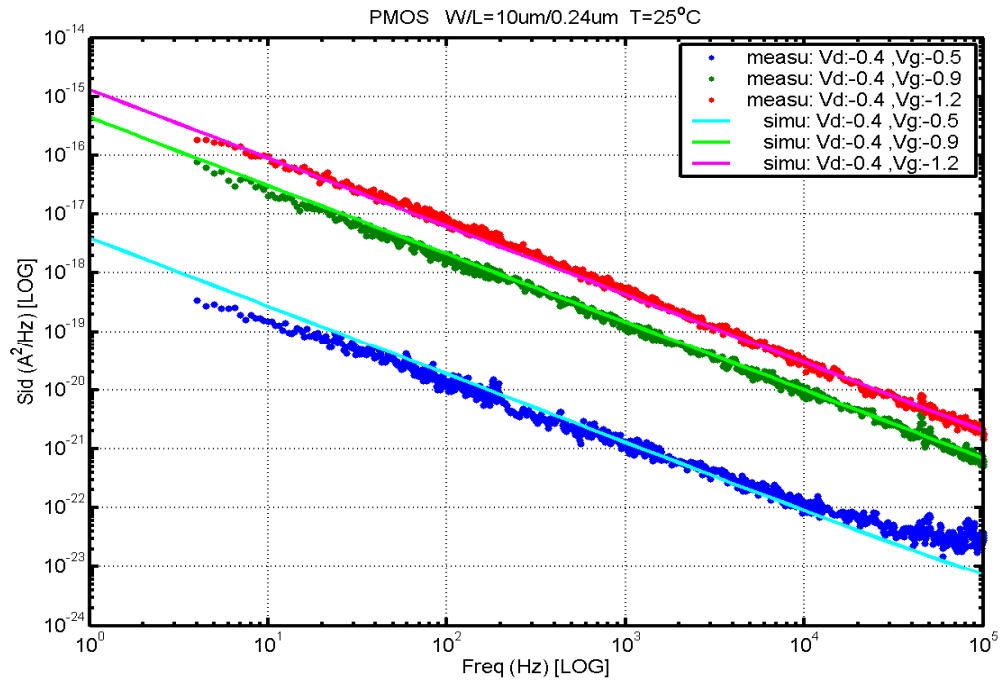


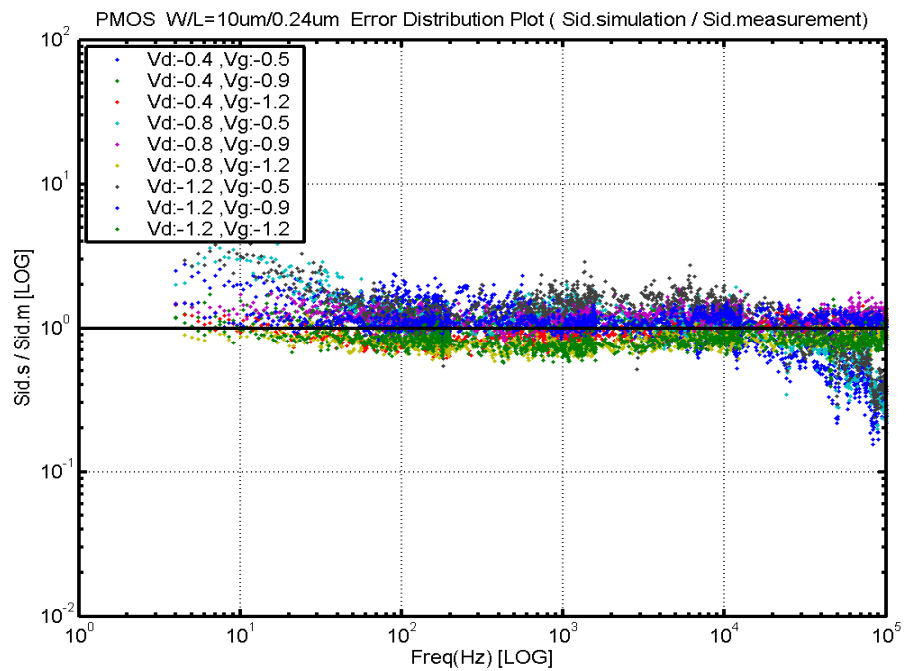
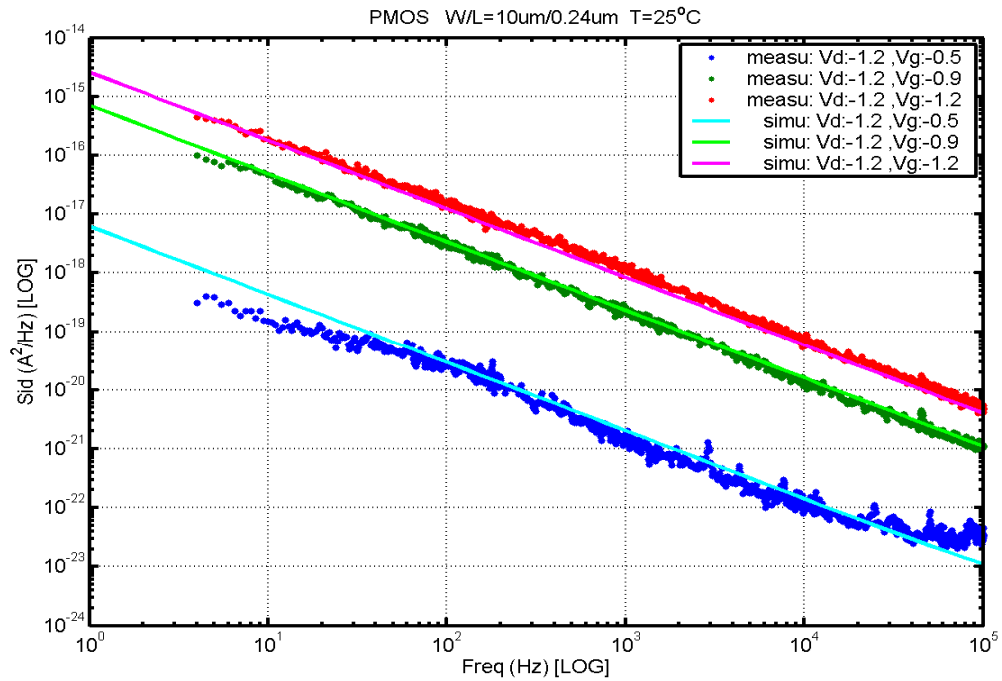
PMOS, $W=10\mu\text{m}$ $L=0.12\mu\text{m}$



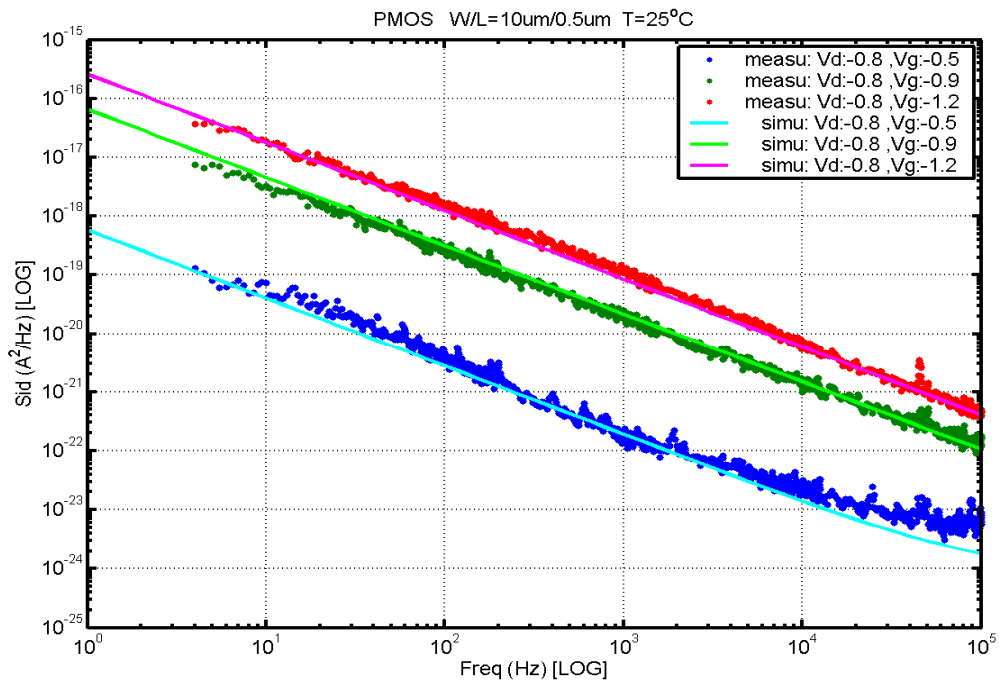
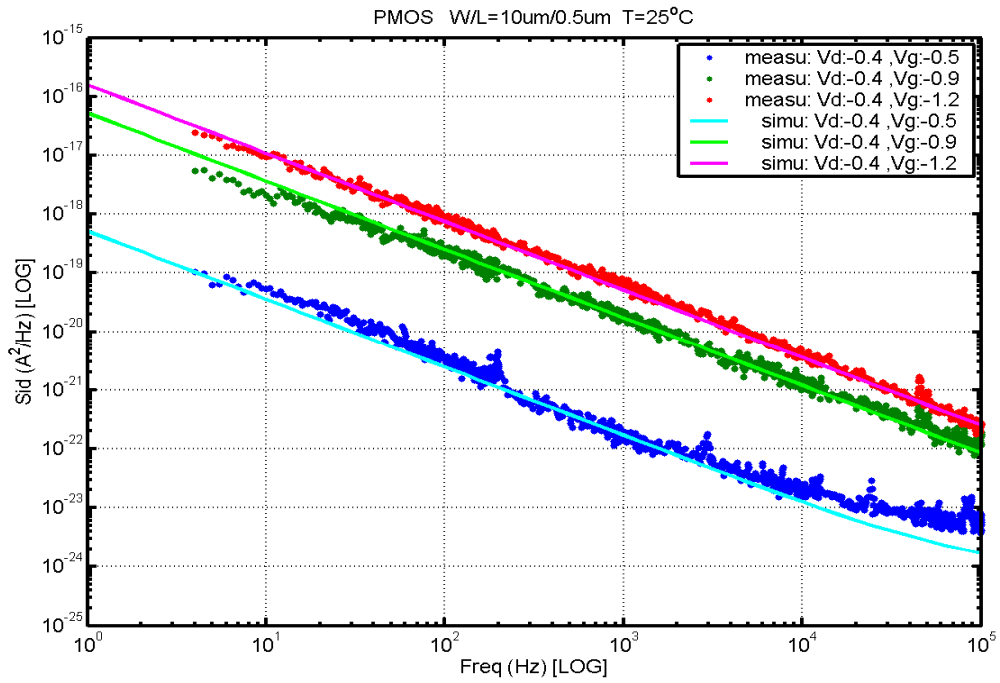


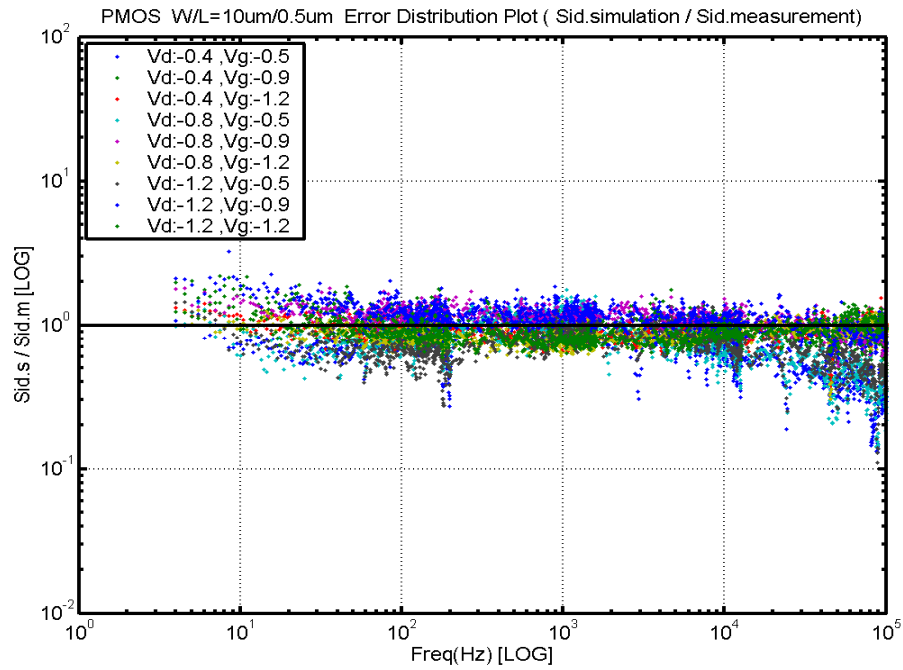
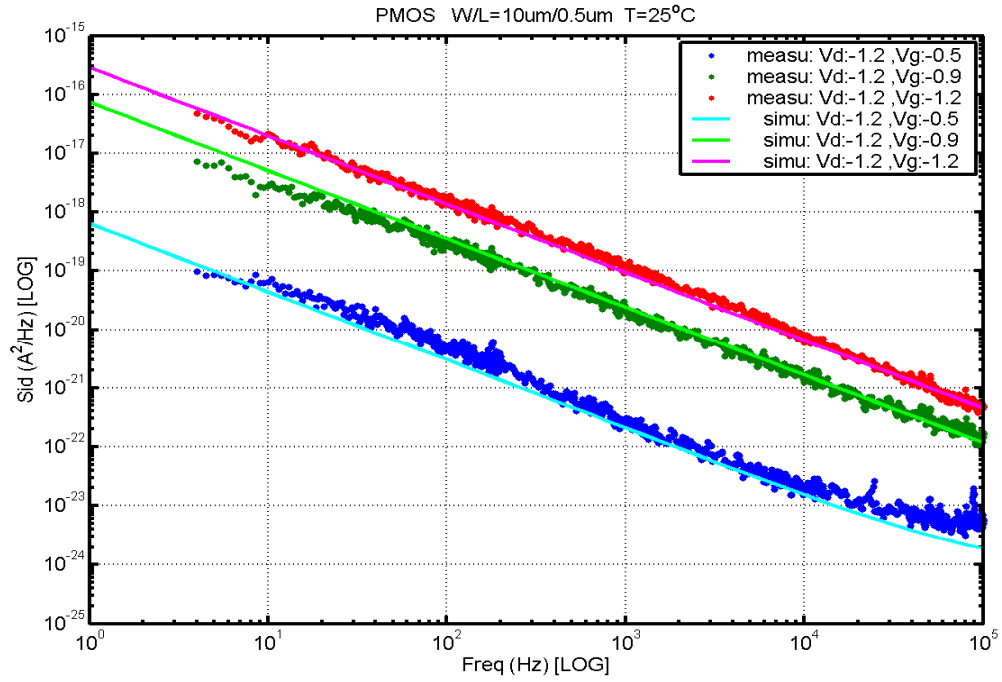
PMOS, W=10 μ m L=0.24 μ m



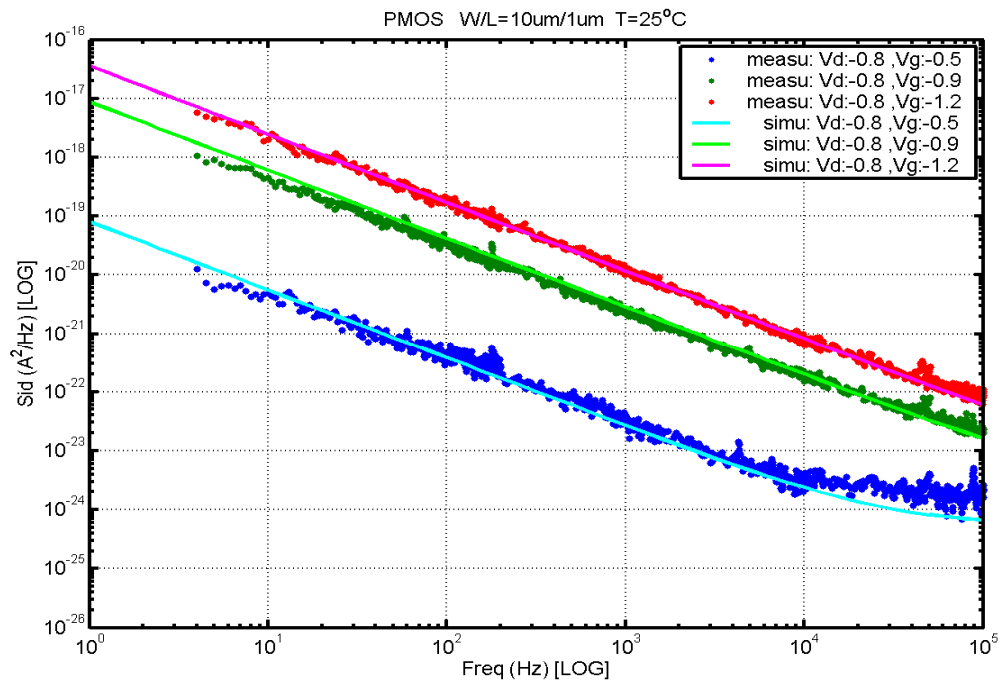
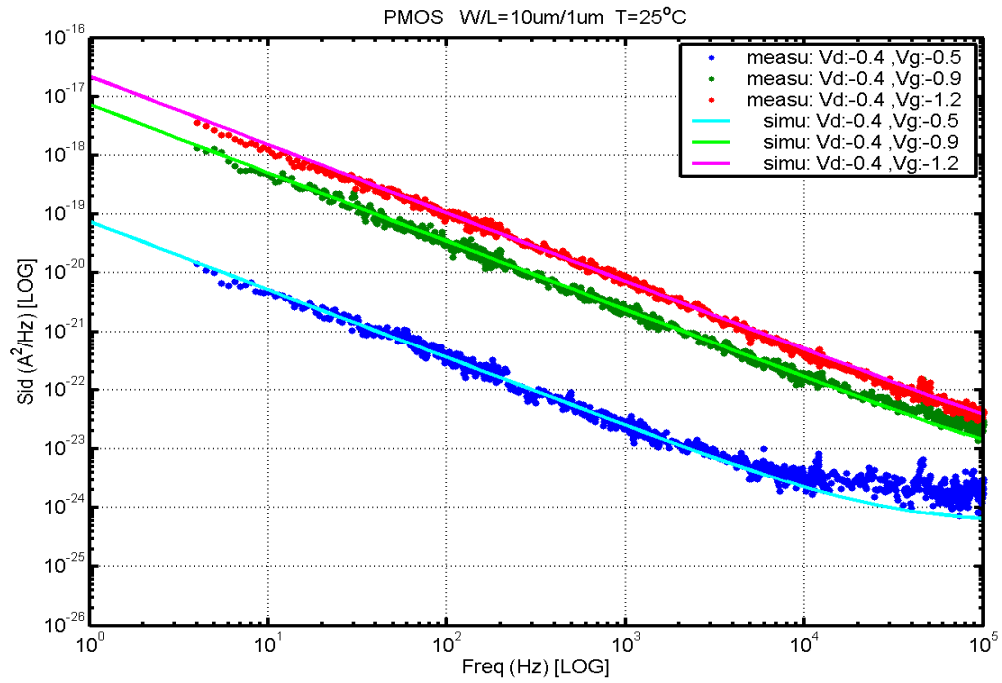


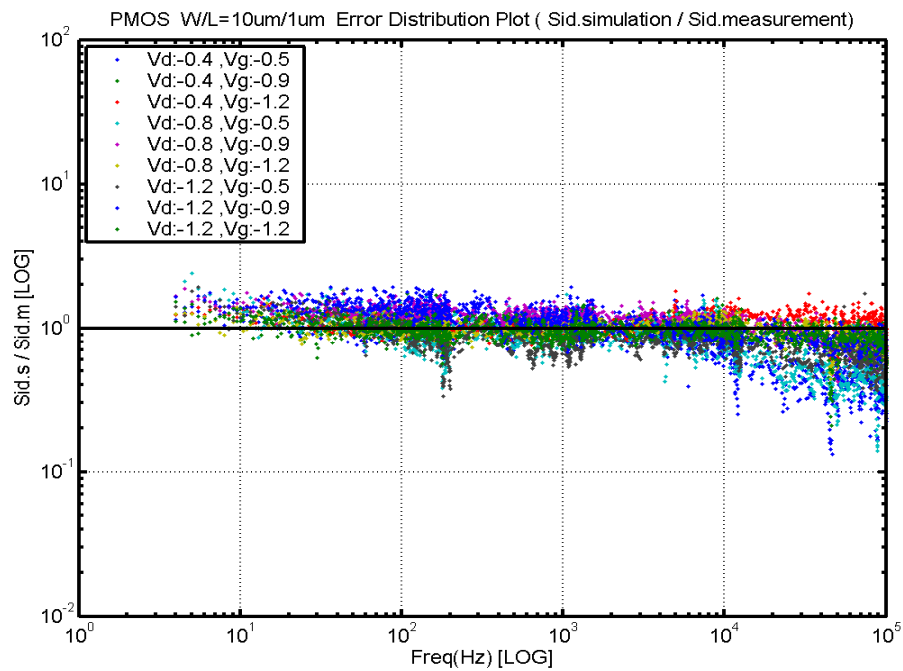
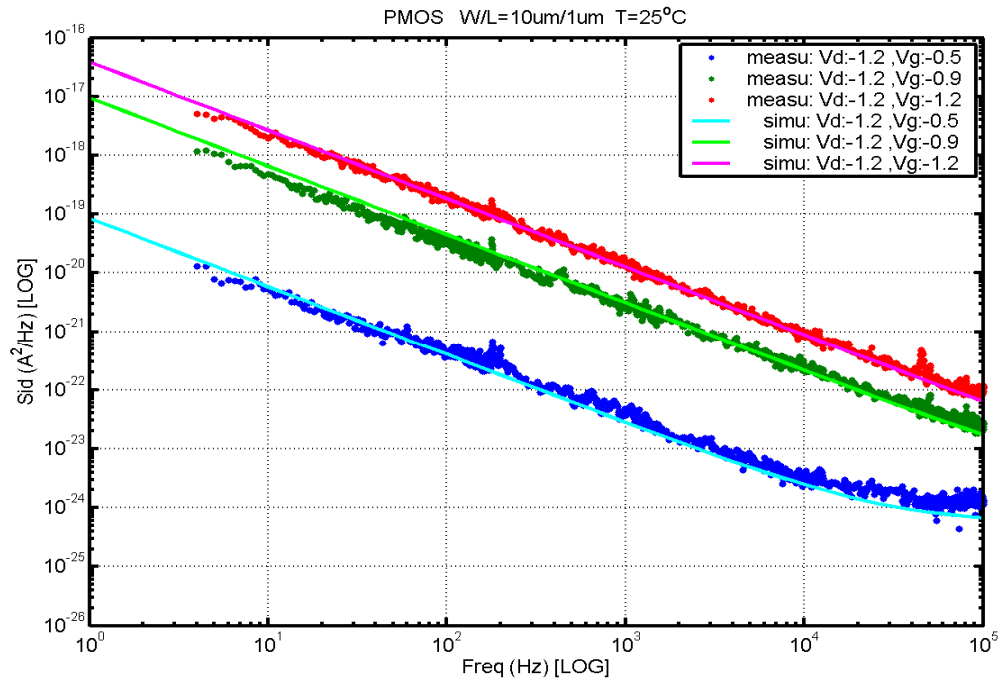
PMOS, $W=10\mu\text{m}$ $L=0.5\mu\text{m}$



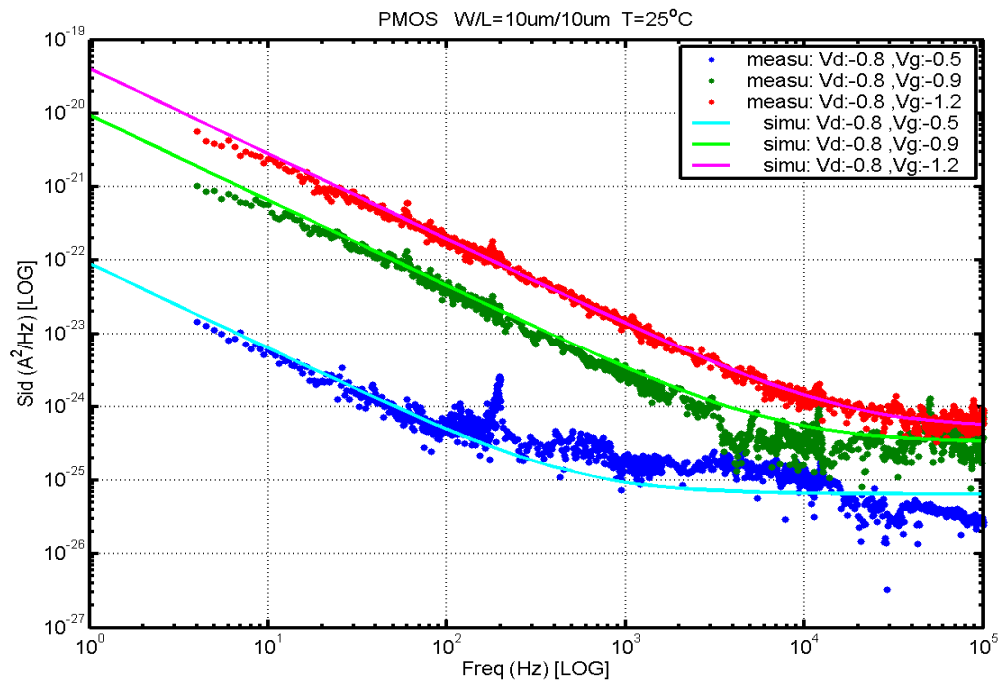
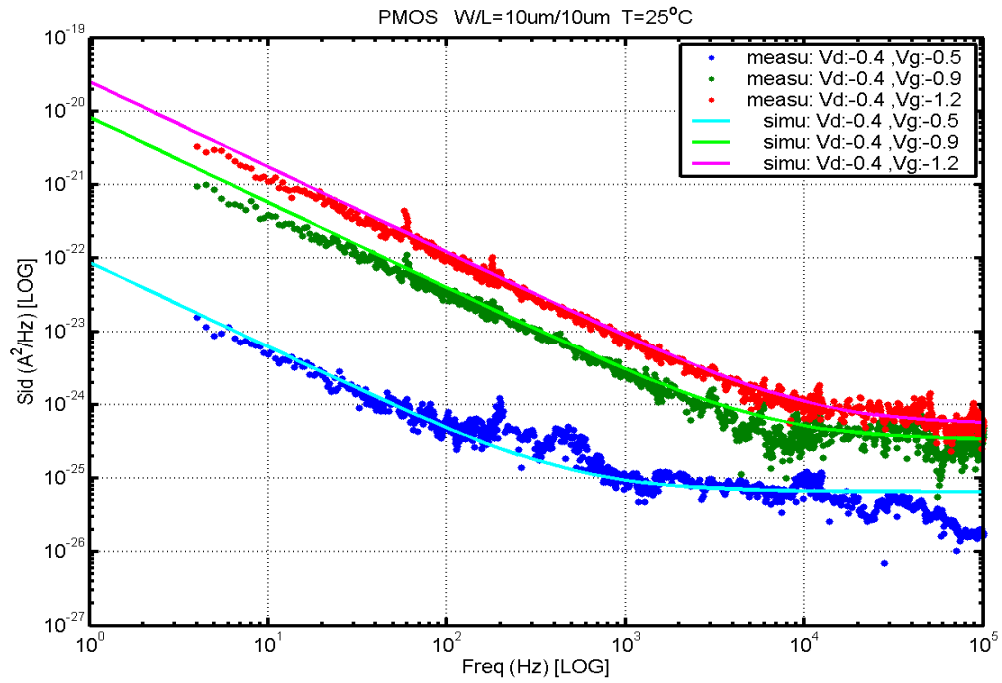


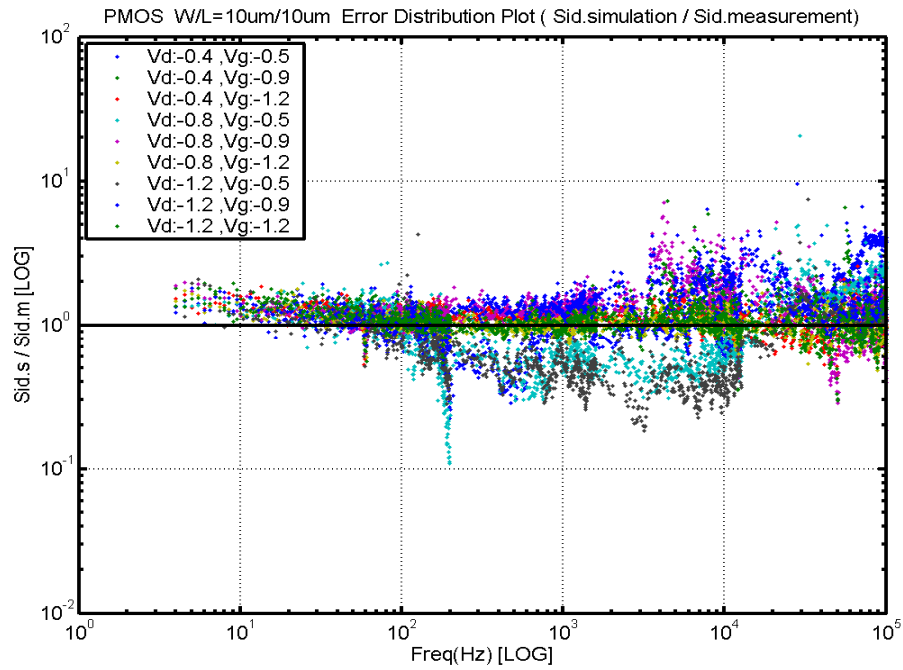
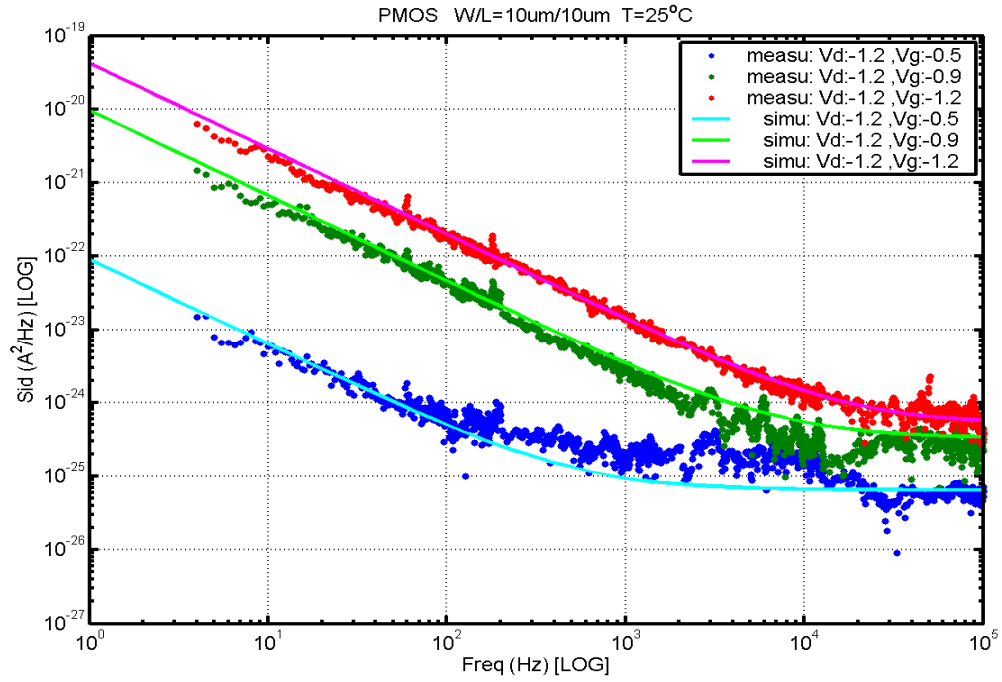
PMOS, $W=10\mu\text{m}$ $L=1\mu\text{m}$





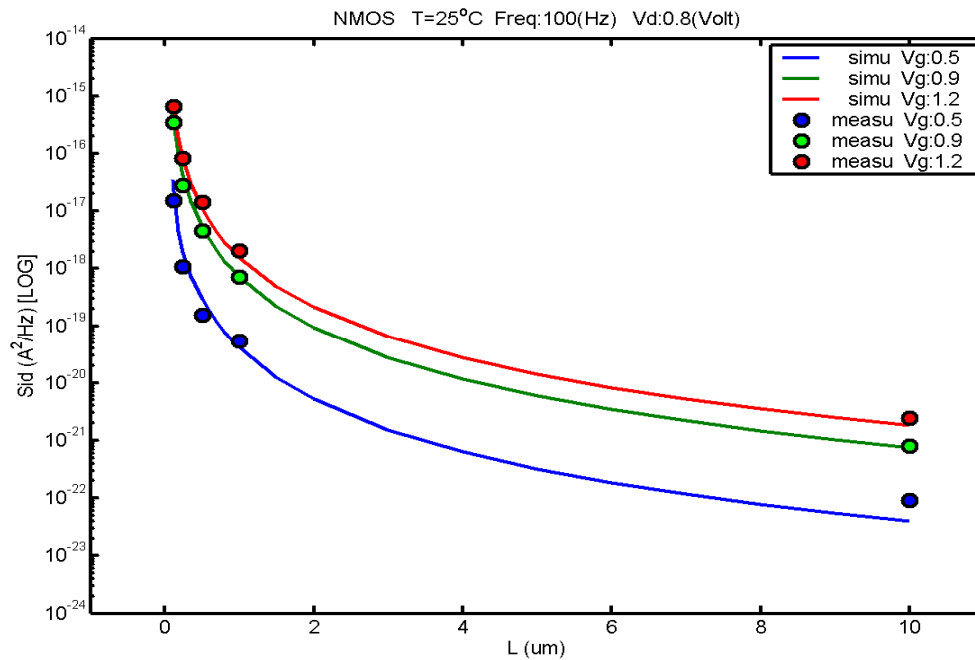
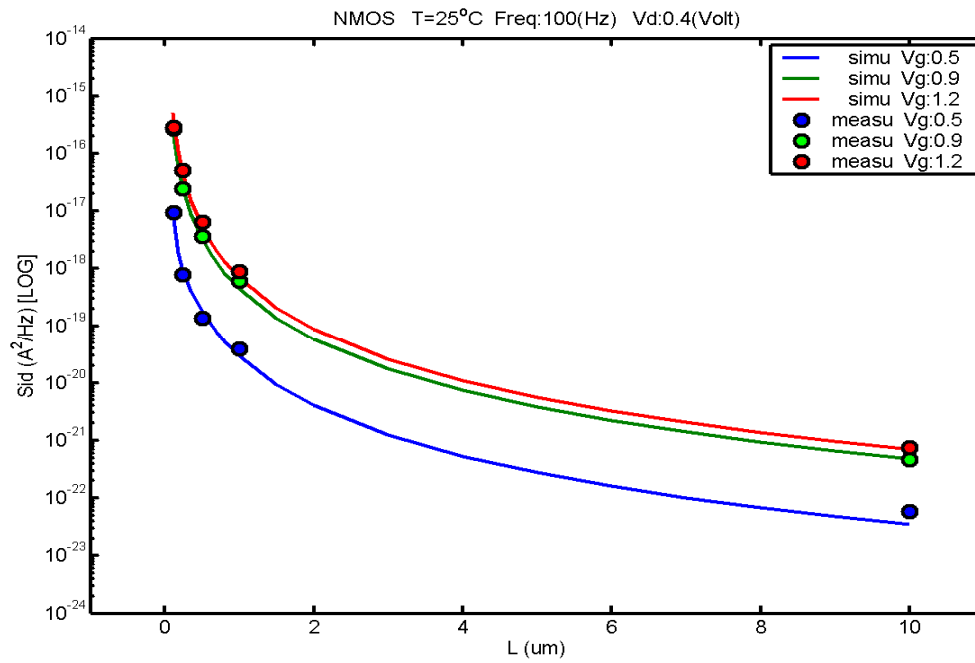
PMOS, $W=10\mu\text{m}$ $L=10\mu\text{m}$

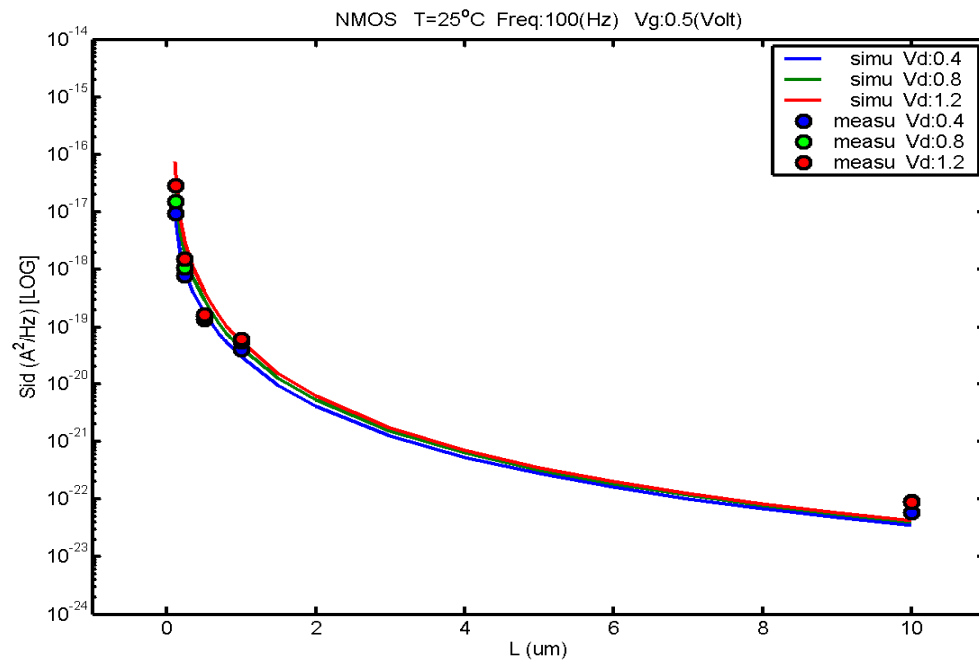
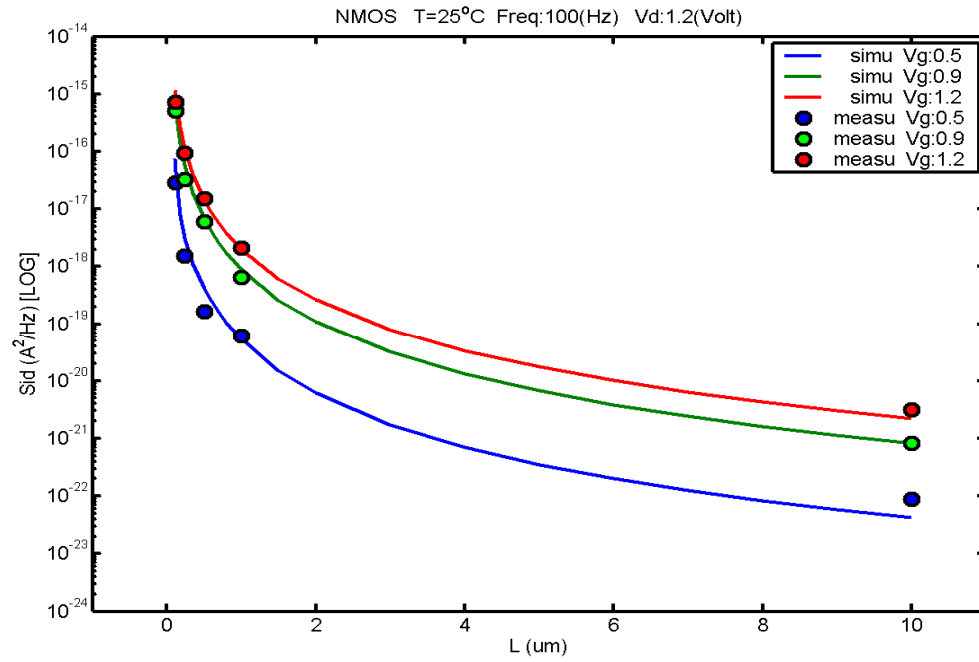


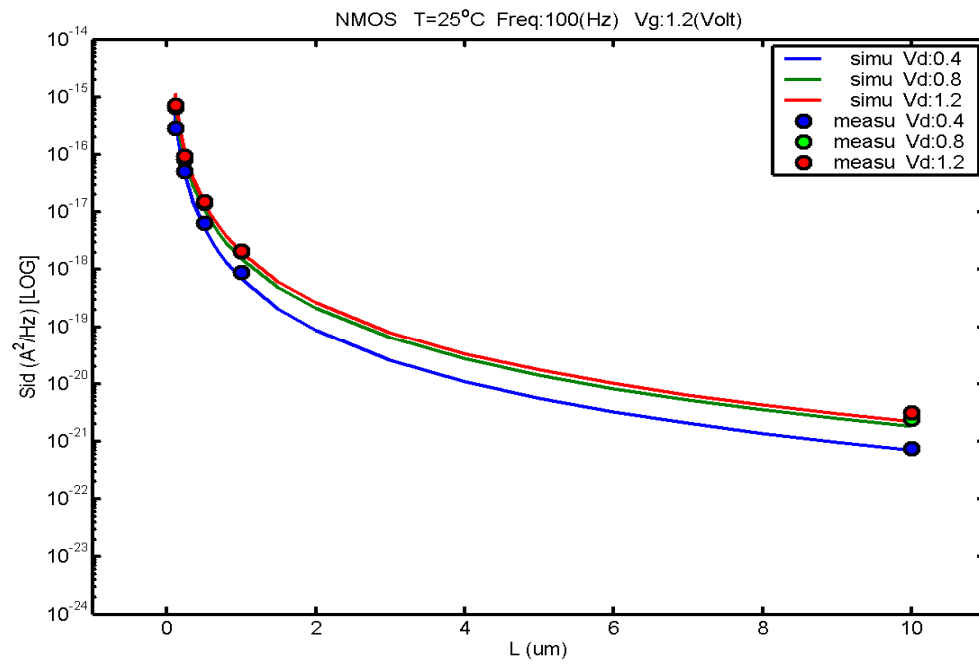
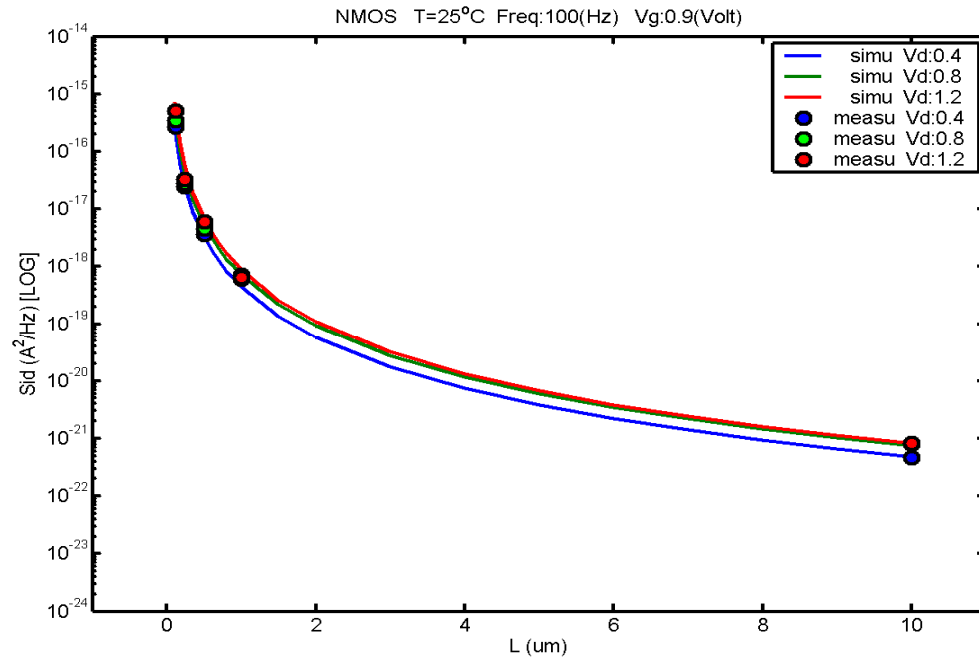


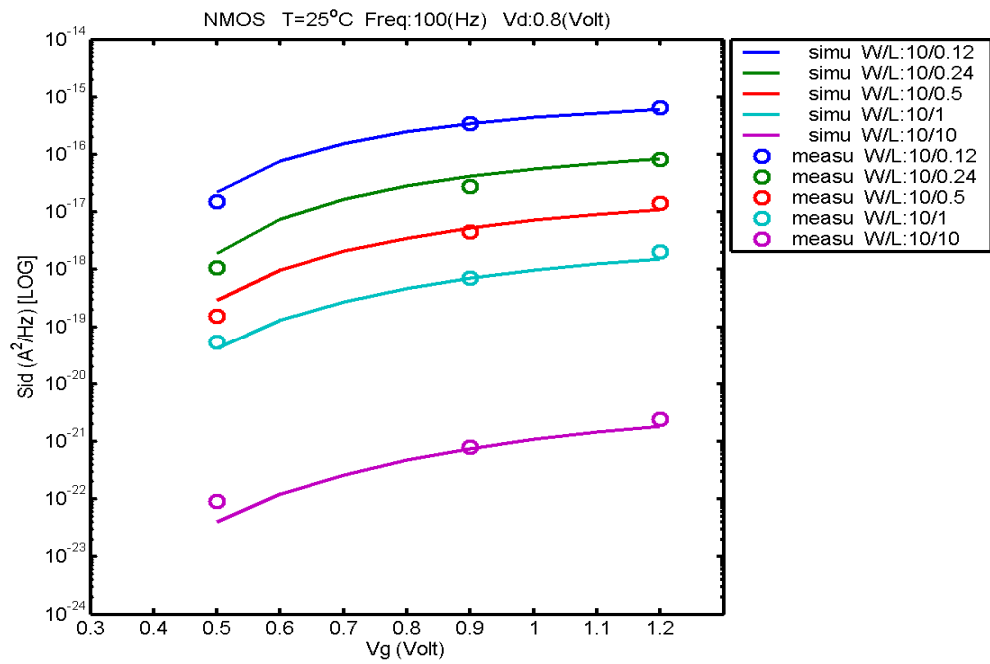
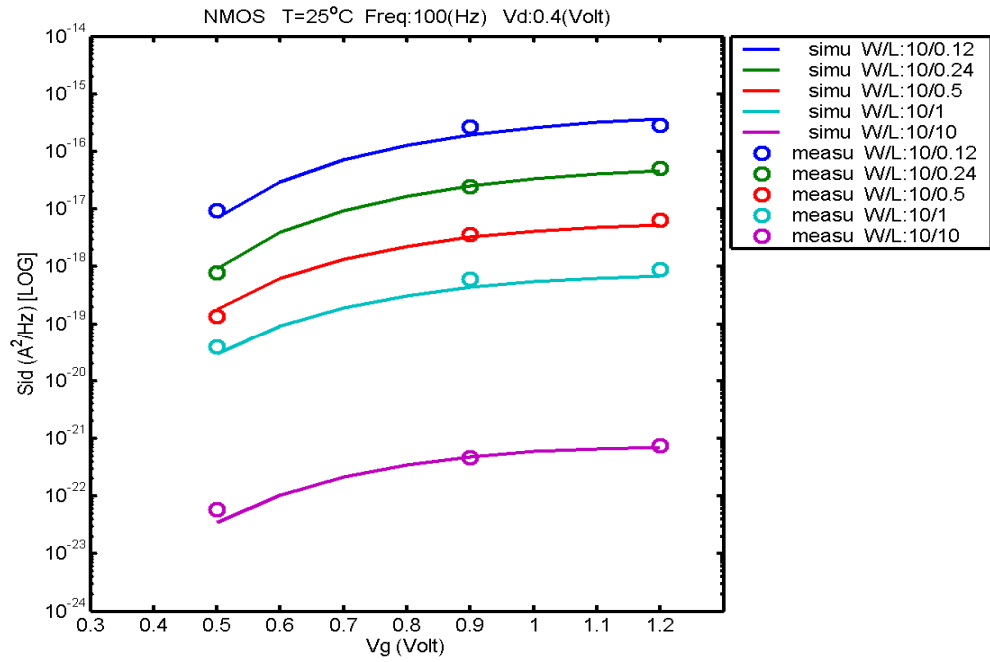
7.2 Trend plots at frequency=100Hz

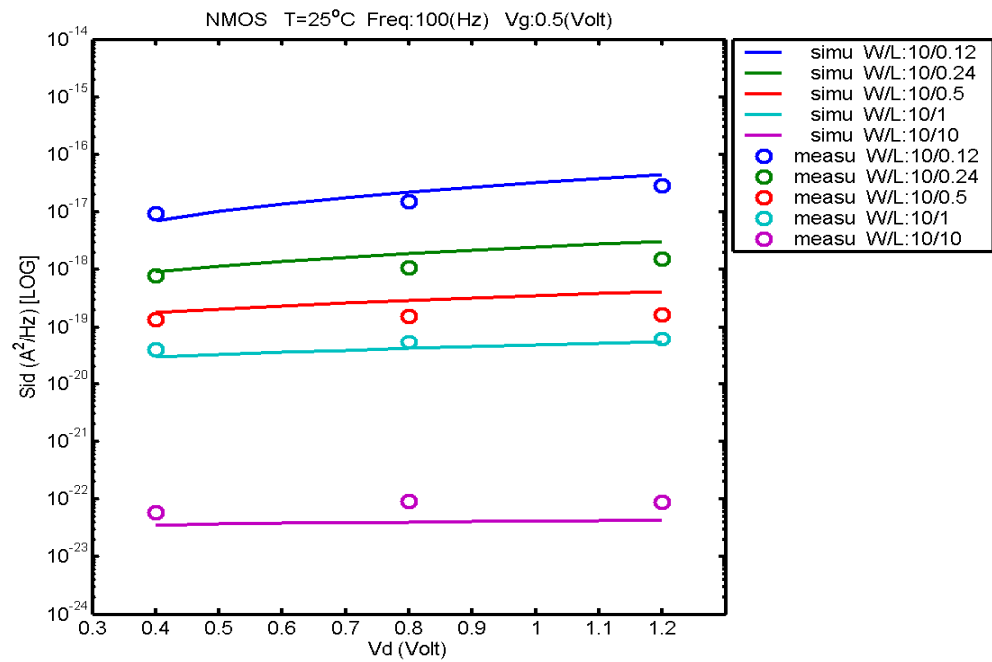
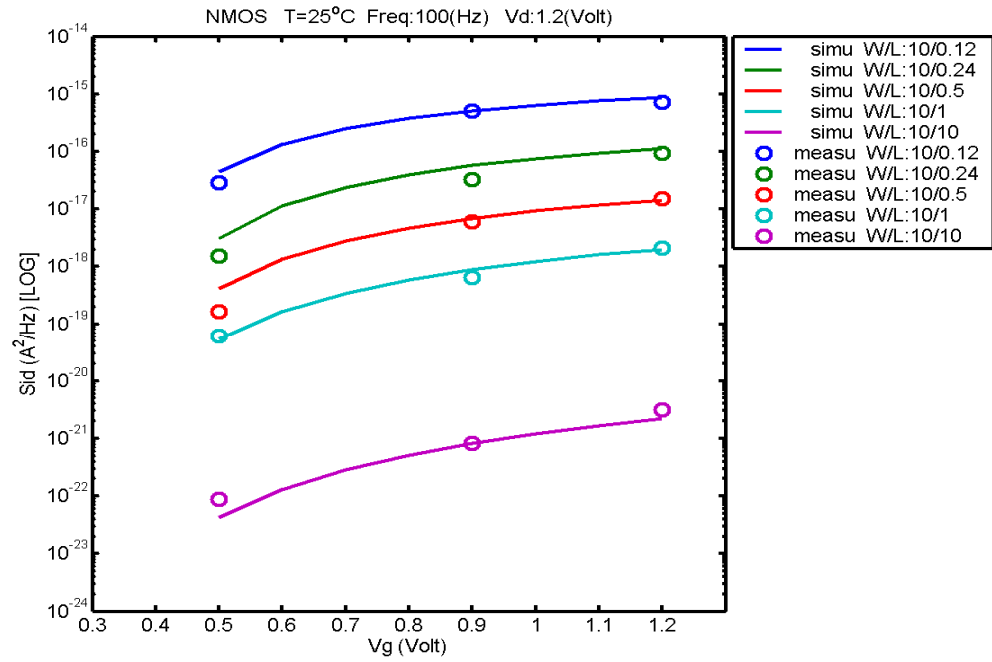
A. NMOS

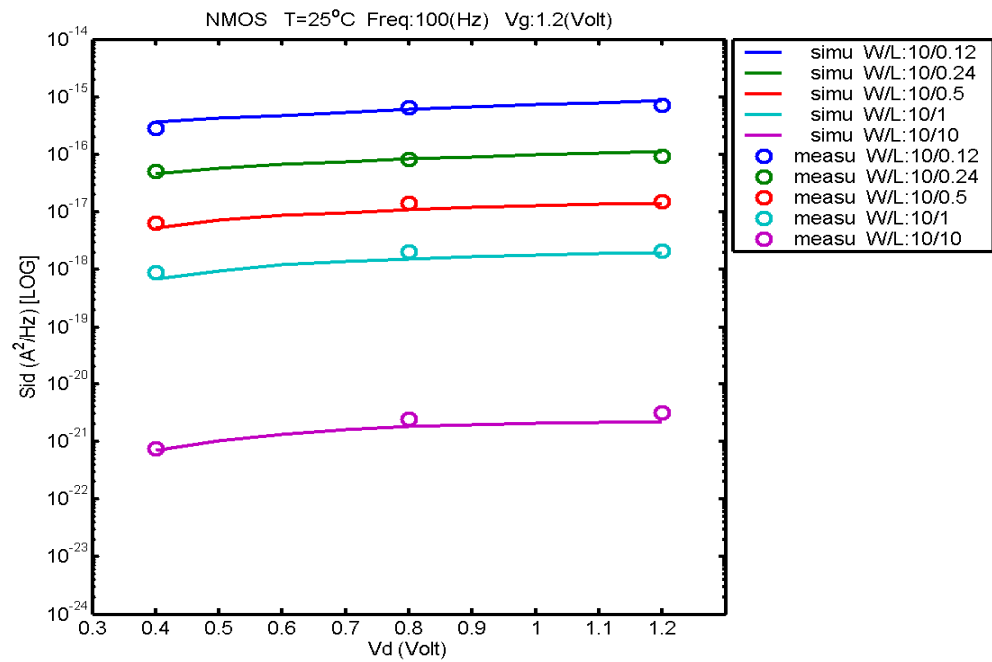
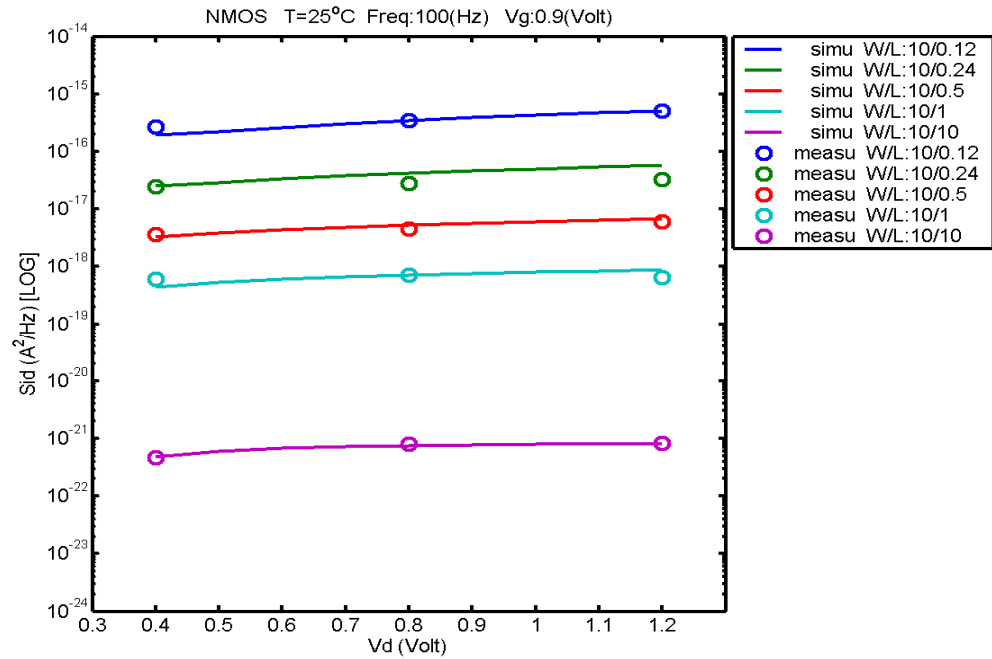




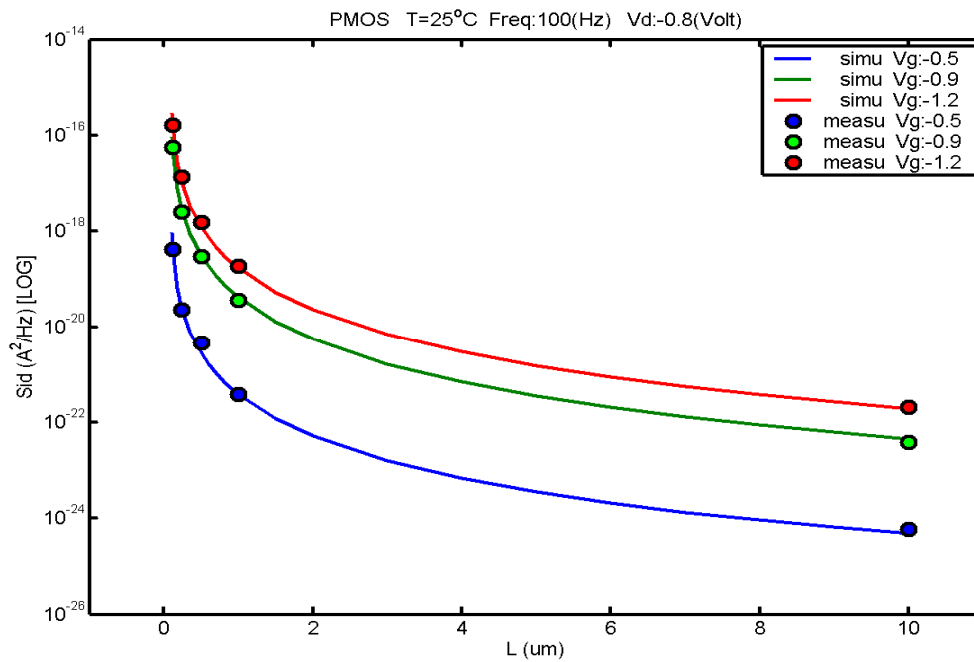
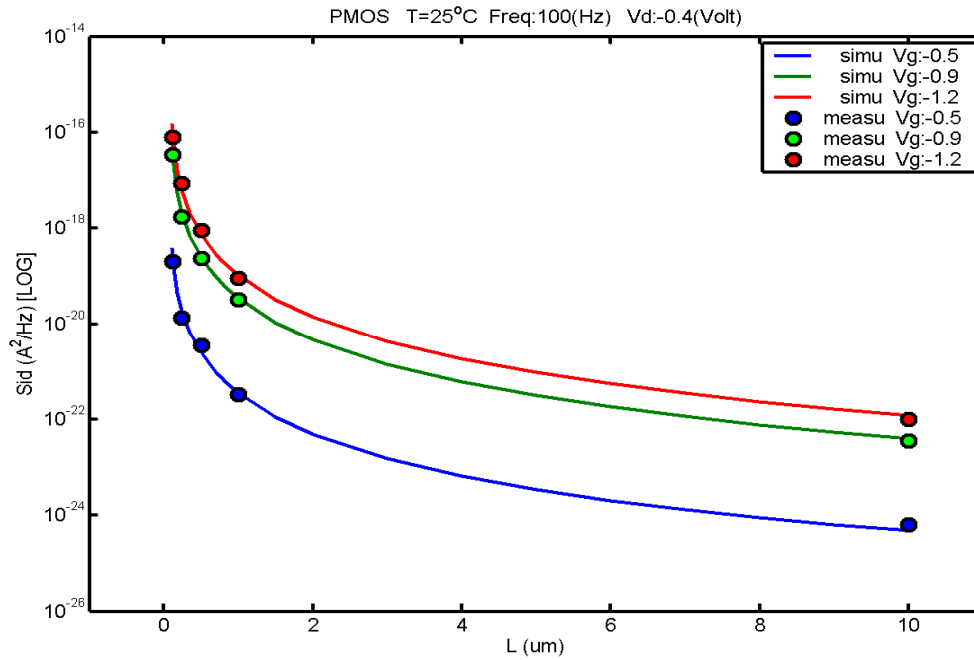


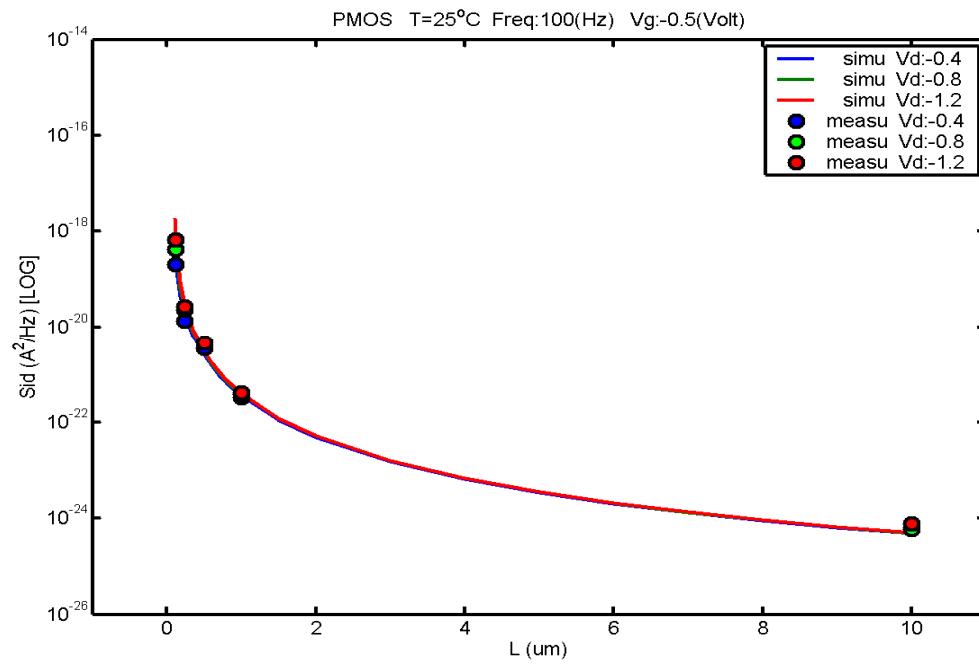
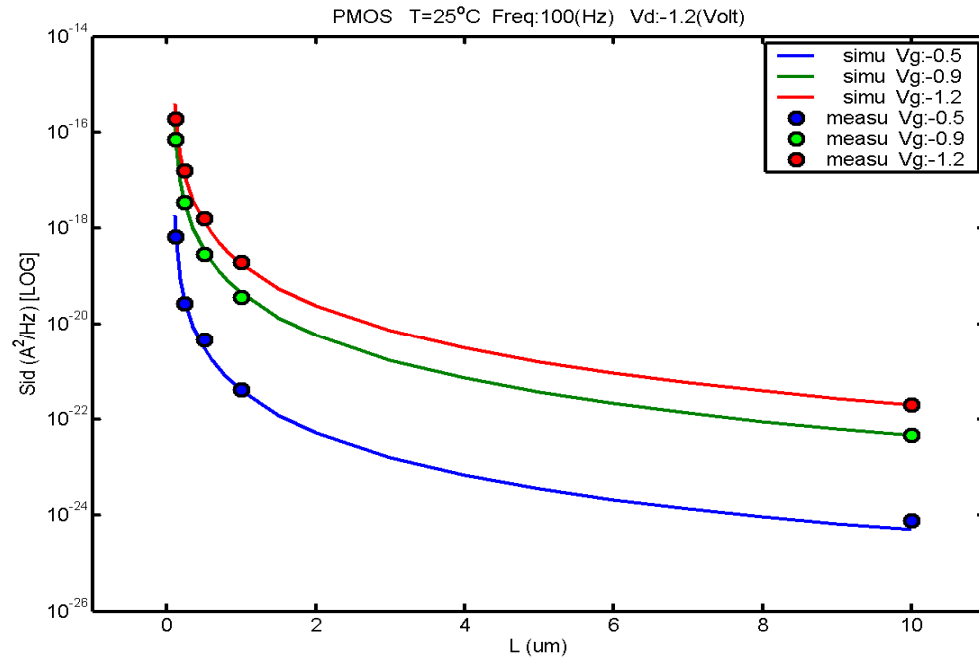


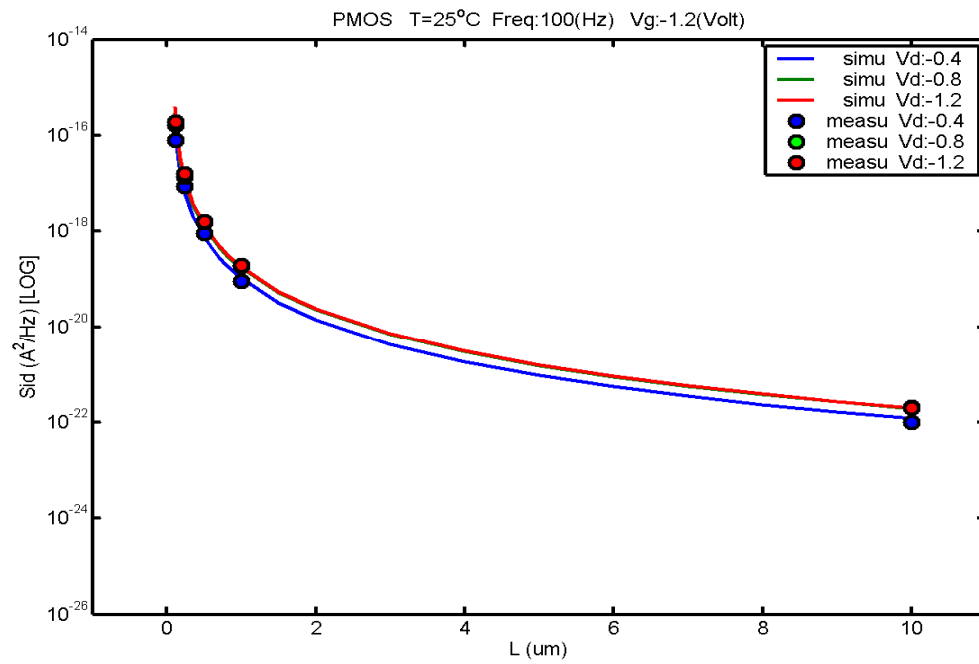
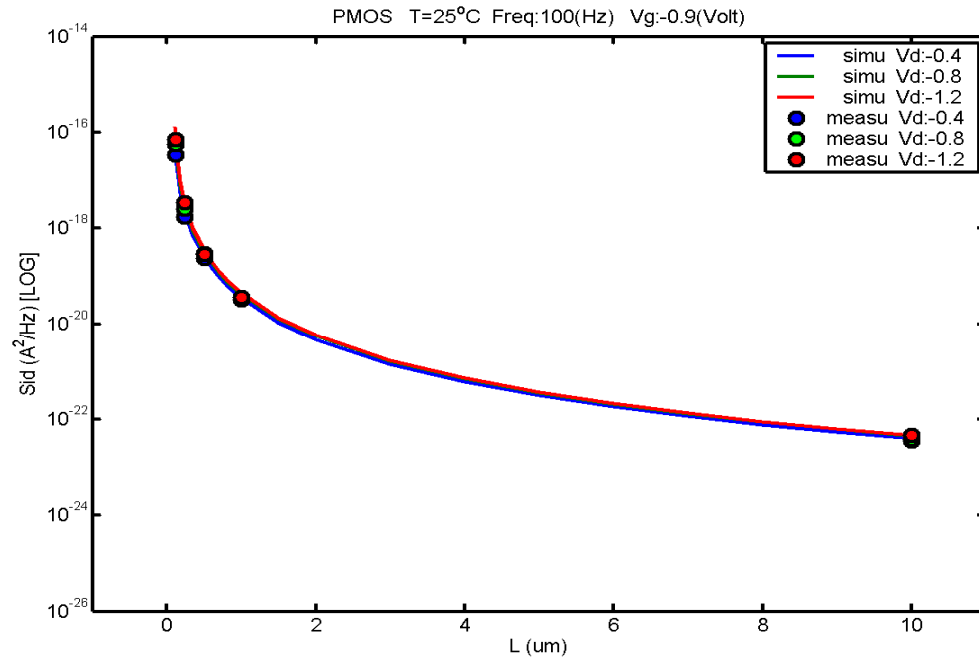


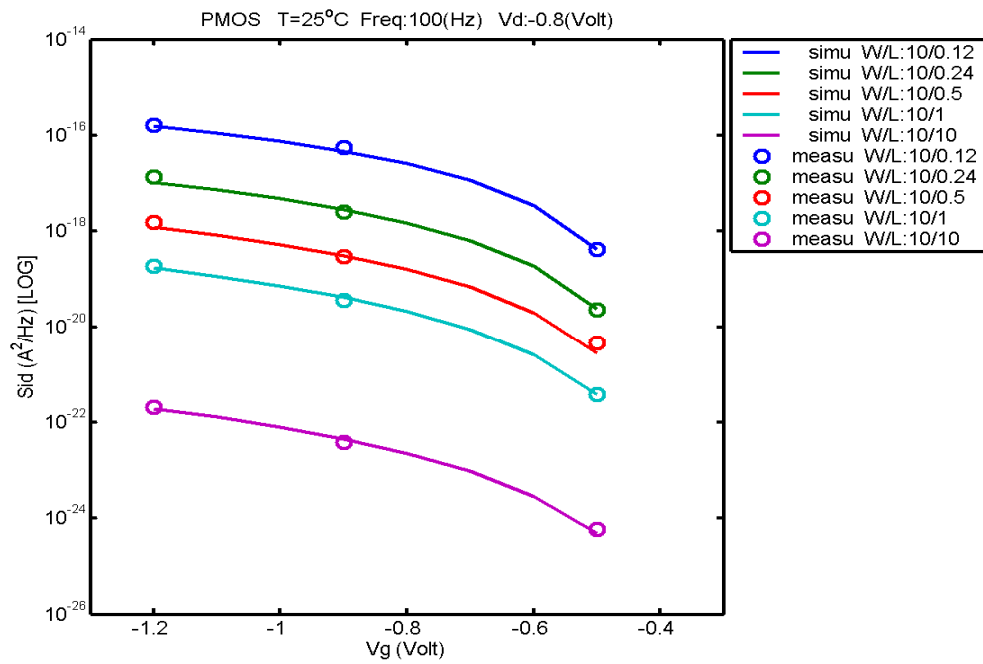
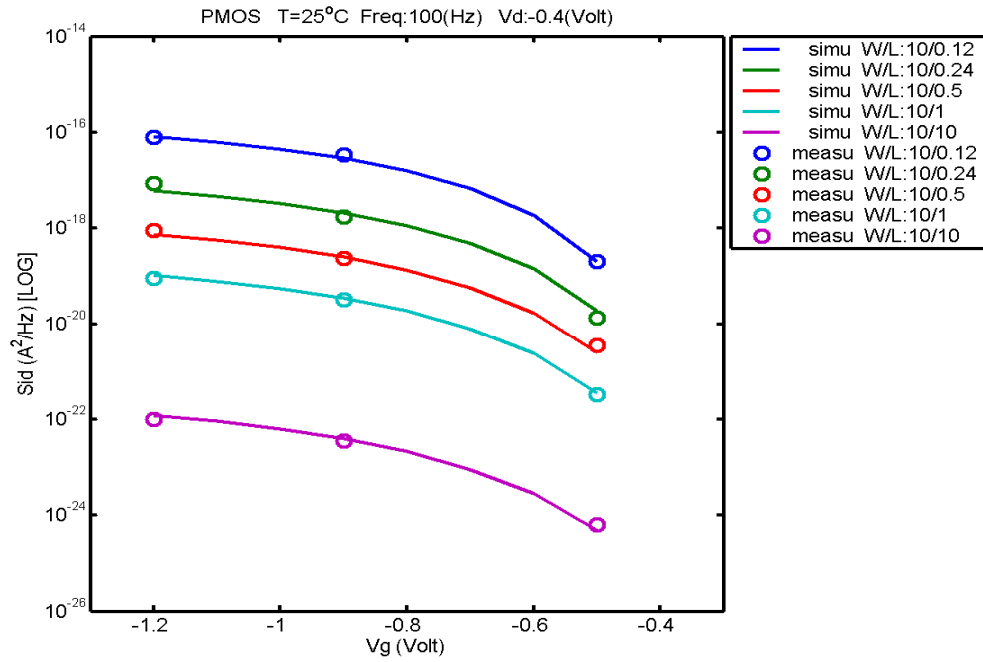


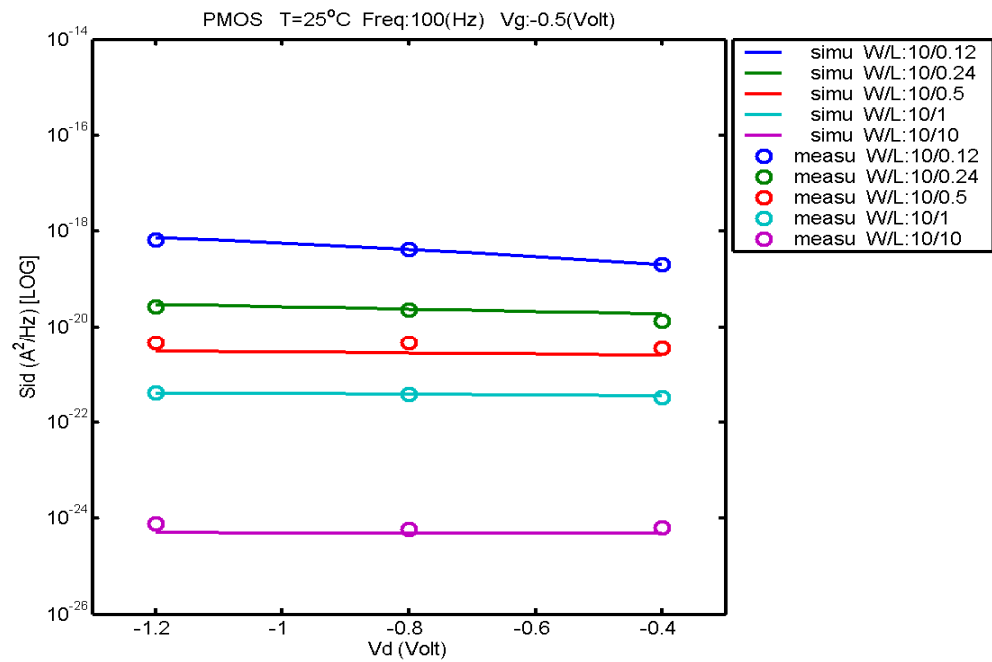
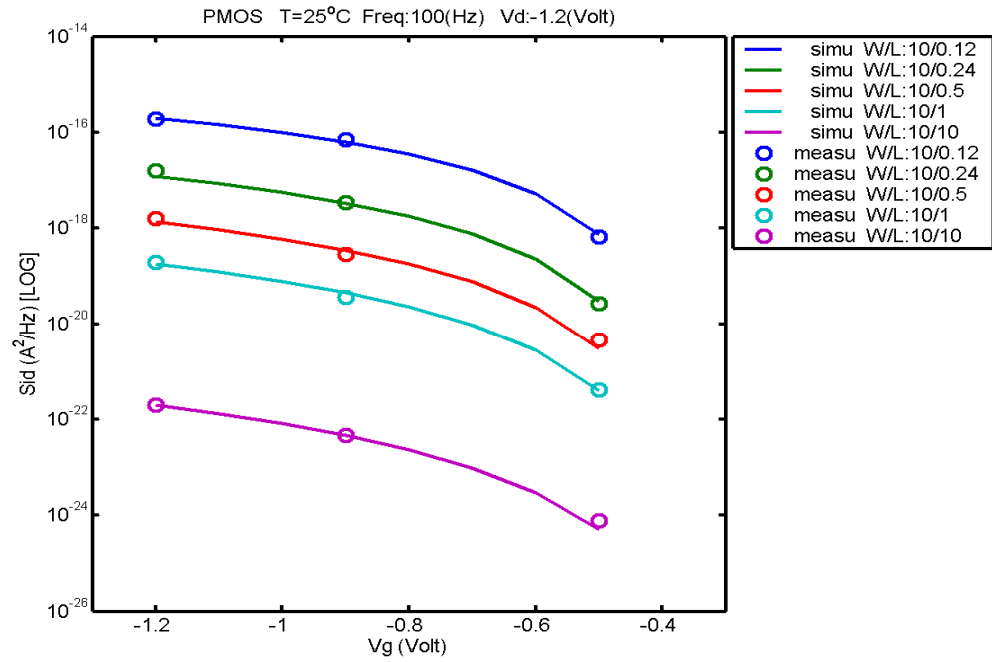
B. PMOS

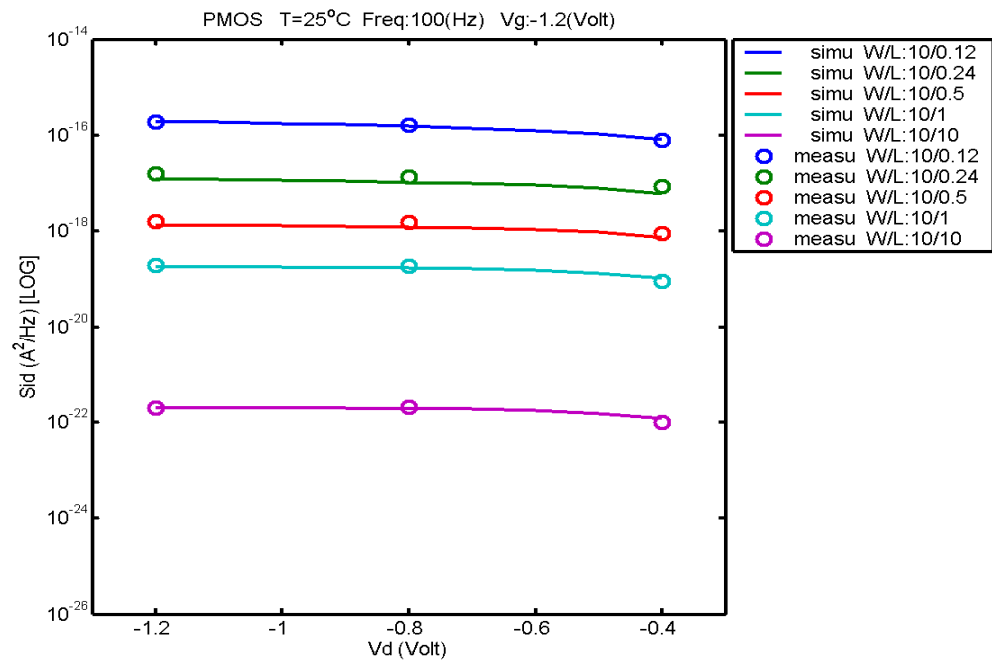
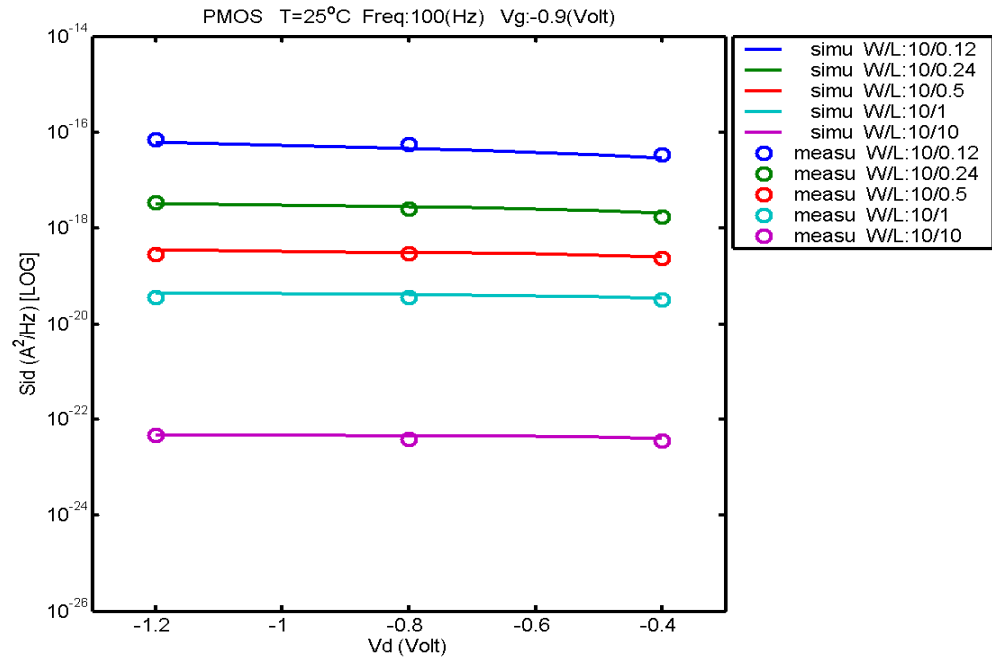












8 Appendix

8.1 Verify corner plot.

Table 8-1 Verify wafer information

Mask ID	A11001	A11001	A11001
Lot ID	D60K6	D5N0S	D63LJ
Wafer Number	#1 and #3	#1 and #2	#8 and #9

Threshold voltage in linear region V_{TLN}

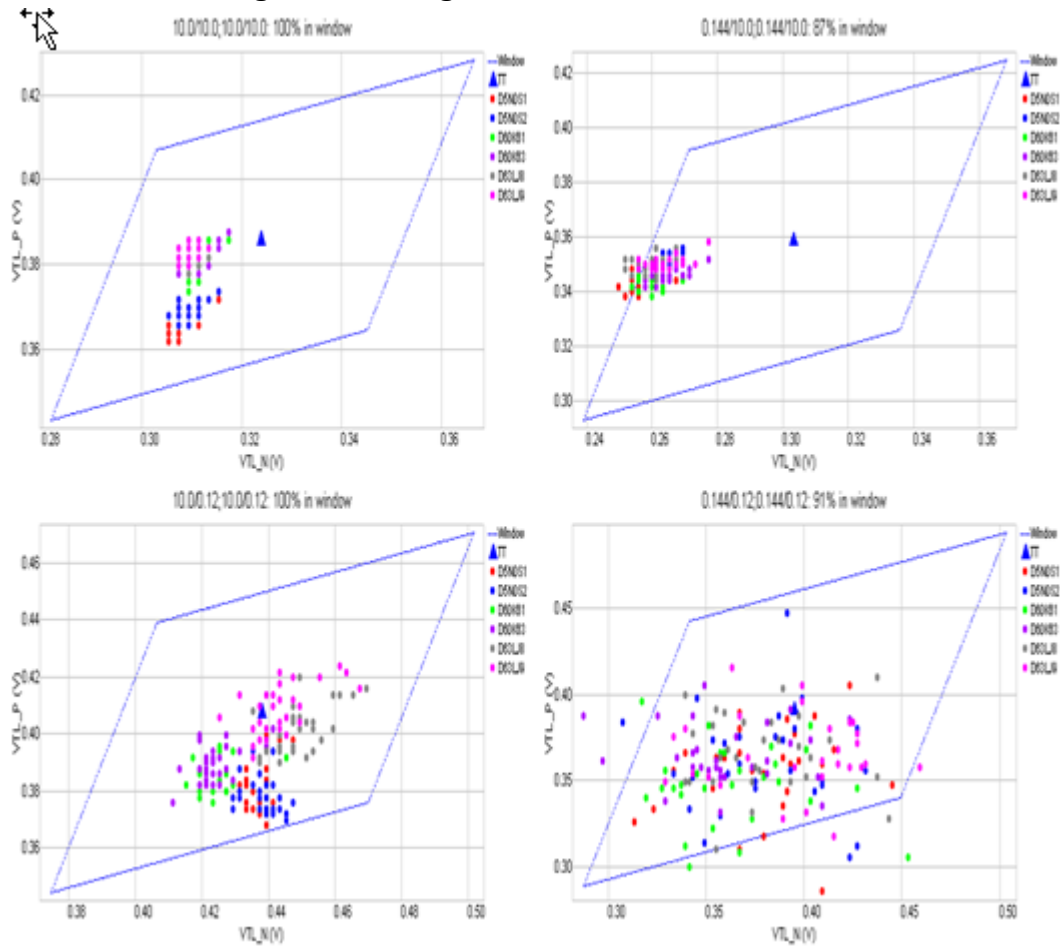


Figure 8-1 Corner graph of threshold voltage at low drain bias

Threshold voltage in Saturation region V_{Tsat}

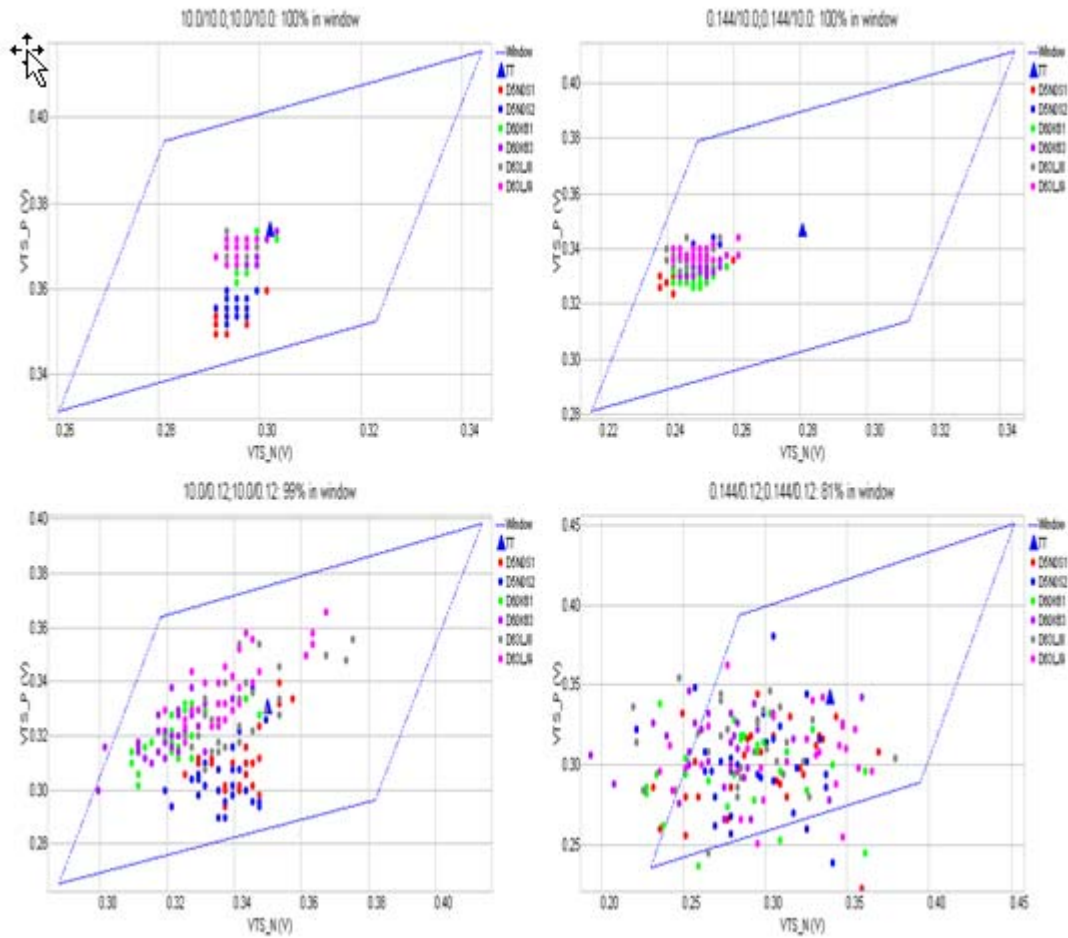


Figure 8-2 Corner graph of threshold voltage at high drain bias

Linear Current I_{LIN}

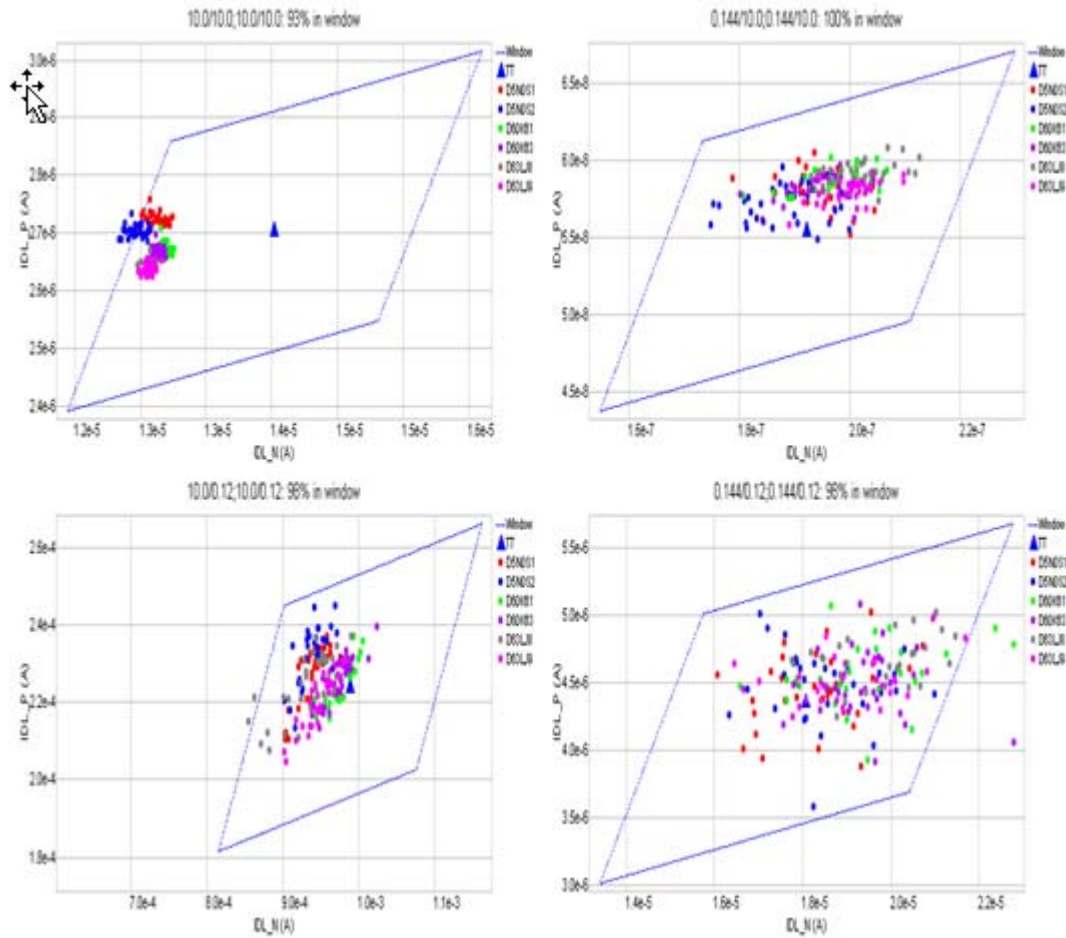


Figure 8-3 Corner graph of linear current

Saturation Current I_{ON}

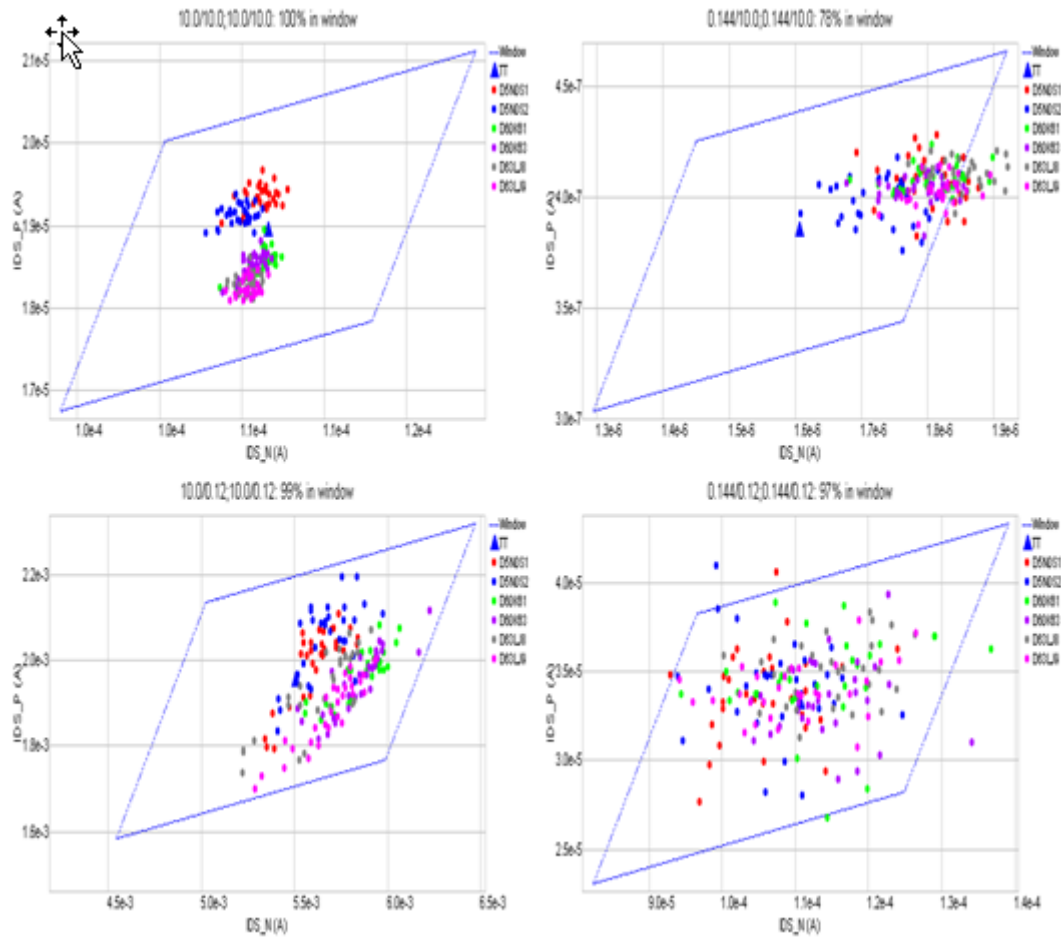


Figure 8-4 Corner graph of on-current

8.2 HSPICE Warning Message

Table 8-2 Hspice Warning Message

Item	Warning Message	Explanation
1	Acde = 0.283 is too small.	A small value of parameter ACDE is necessary to get good fitting in accumulation region of CV curve but would cause a harmless warning in HSPICE.

8.3 ELDO Warning Message

Table 8-3 Eldo Warning Message

Item	Warning Message	Explanation
1	ACDE = 0.283 is too small.	A small value of parameter ACDE is necessary to get good fitting in accumulation region of CV curve but would cause a harmless warning in ELDO.

8.4 SPECTRE Warning Message

Table 8-4 Spectre Warning Message

Item	Warning Message	Explanation
1	`Acde' = 2.83000e-01 is unusually small.	A small value of parameter ACDE is necessary to get good fitting in accumulation region of CV curve but would cause a harmless warning in SPECTRE.